

Airline Satisfaction Analysis

Joan Acero, Anna Monsó, Walter Troiani

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1. Introduction

To complement this report, an ANNEX document is included, containing all the results, visualizations, and plots necessary to interpret and validate the findings. This additional resource provides a thorough understanding of the analysis and facilitates the reproducibility of the study.

The primary objective of this project is to predict customer satisfaction probabilities using the training and test samples from the Airline Passengers Satisfaction dataset available on Kaggle. Additionally, we aim to interpret the final binary outcome model to identify which variables most significantly influence customer satisfaction decisions.

The project begins with a data preparation and exploratory analysis phase, which focuses on ensuring data quality. This includes generating a detailed Data Quality Report, identifying and imputing missing values, and examining variable distributions and relationships. A univariate descriptive analysis is conducted to assess the characteristics of individual variables.

In the modeling phase, we utilize the training dataset to build and refine predictive models, followed by an evaluation of model quality and predictive performance on the test dataset. The steps in this process include:

- Profiling and Feature Selection: Identifying key variables that contribute to the predictive power of the model.
- Residual Analysis: Detecting and filtering unusual and influential data points.
- Numerical Variable Transformations: Applying transformations to improve model fit and address data skewness.
- Factor Effects and Interactions: Adding main effects and second-order interactions involving categorical variables to enhance the model.
- Iterative Residual Analysis: Reassessing residuals and influential data iteratively to refine the model.
- Goodness of Fit and Model Interpretation: Evaluating model performance and interpretability using both the training and test datasets.

2. Data Preparation

2.1 Data loading

2.2 Univariate Descriptive Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the data set, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

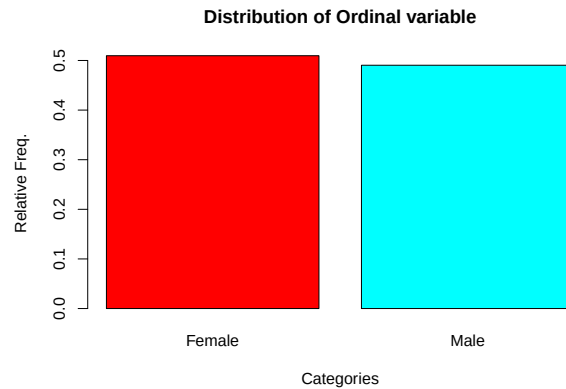
Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test data set. In this initial phase, we analyze all variables, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones. Not only that but also ordinal numerical variables like ratings will be converted into ordered factors. Subsequently, new induced factor variables like age-brackets will also be introduced in this section.

2.2.1 Variable Gender

Gender is a binary categorical variable representing the passenger's gender. The distribution is nearly balanced, with a slightly higher proportion of females compared to males, and there are no missing values.

```
summarize_categorical_data(train$Gender)
```

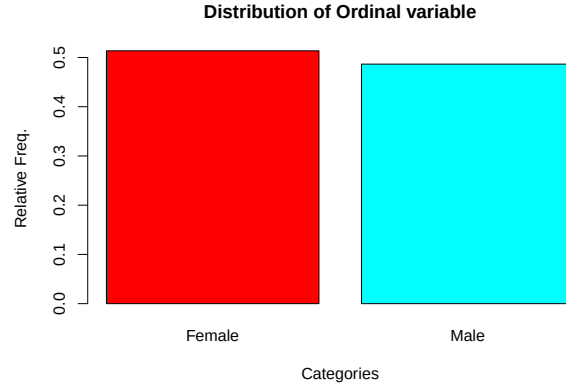
```
## [1] "Proportions: "  
## data  
## Female   Male  
## 0.5096 0.4904
```



```
## [1] "Number of nulls: "  
## [1] 0
```

```
summarize_categorical_data(test$Gender) # 0 NAs
```

```
## [1] "Proportions: "  
## data  
## Female   Male  
## 0.5135 0.4865
```



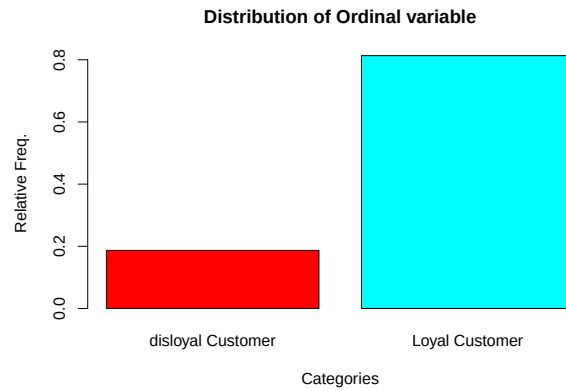
```
## [1] "Number of nulls: "  
## [1] 0
```

2.2.2 Variable Customer.Type

Customer.Type is a binary categorical variable indicating customer loyalty status (Loyal or Disloyal). The distribution is highly imbalanced, with 81% classified as Loyal and 19% as Disloyal. No missing values are present.

```
summarize_categorical_data(train$Customer.Type)
```

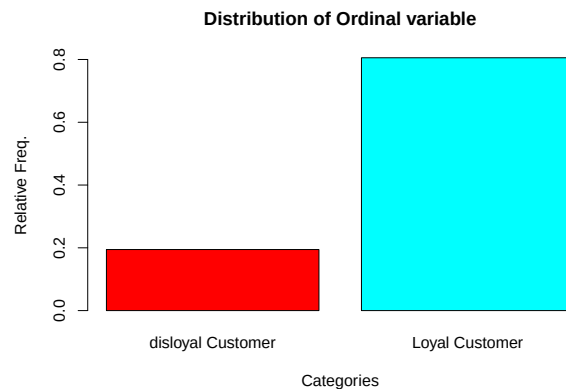
```
## [1] "Proportions: "
## data
## disloyal Customer    Loyal Customer
##           0.1868           0.8132
```



```
## [1] "Number of nulls: "
## [1] 0
```

```
summarize_categorical_data(test$Customer.Type) # 0 NAs
```

```
## [1] "Proportions: "
## data
## disloyal Customer    Loyal Customer
##           0.1945           0.8055
```



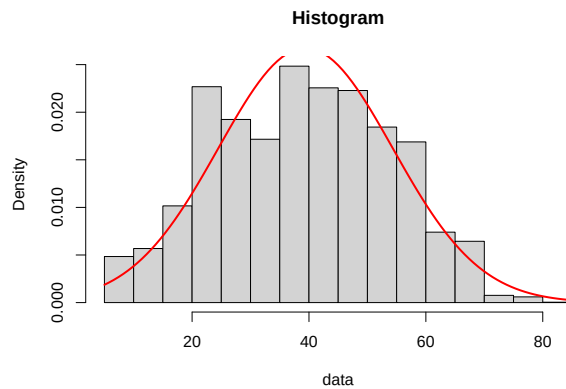
```
## [1] "Number of nulls: "
## [1] 0
```

2.2.3 Variable Age

Age is a numerical variable representing the age of customers, ranging from 7 to 85 years, with a mean of 39.41 and a standard deviation of 14.92. Despite the large sample size, both visual inspections (histogram and boxplot) and statistical tests strongly indicate no normality. No missing values or outliers are present in the variable.

```
summary_and_normality_check(train$Age,bins=25)
```

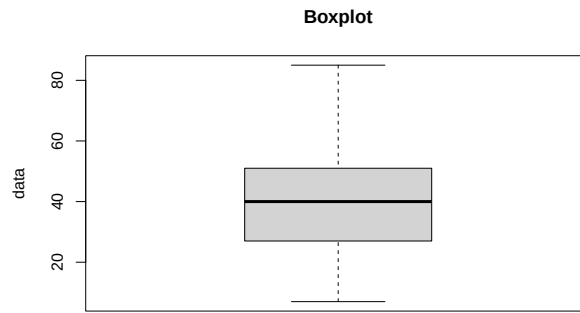
```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      7.00  27.00  40.00  39.41  51.00  85.00  
## [1] "Mean"  
## [1] 39.4126  
## [1] "Standard deviation"  
## [1] 14.92196
```



```
## [1] "Skewness"  
## [1] 0.0100164
```

```
## [1] "Num Outliers:"  
## [1] 0  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro-Wilk test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.98893, p-value < 2.2e-16  
##  
## [1] "Kolmogorov-Smirnov test: "
```

```
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =  
## sd(data)): ties should not be present for the Kolmogorov-Smirnov test
```

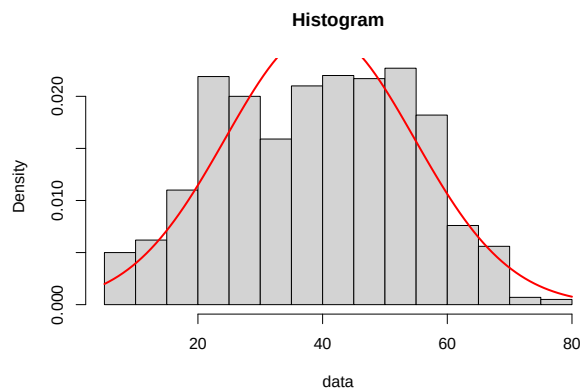


```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.055434, p-value = 9.026e-14
## alternative hypothesis: two-sided

## integer(0)
```

```
summary_and_normality_check(test$Age,bins=25) # 0 NAs
```

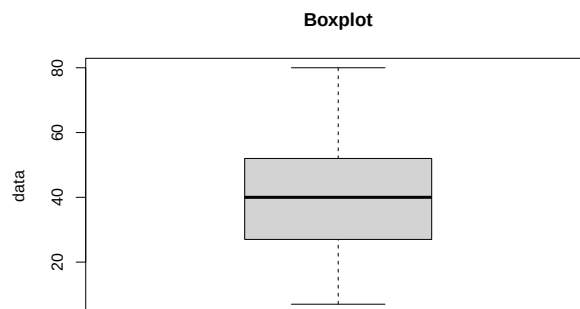
```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   7.00  27.00   40.00   39.55  52.00   80.00
## [1] "Mean"
## [1] 39.554
## [1] "Standard deviation"
## [1] 15.20068
```



```
## [1] "Skewness"
## [1] -0.07369344
```

```
## [1] "Num Outliers:"
## [1] 0
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9837, p-value = 2.474e-14
##
## [1] "Kolmogorov-Smirnov test: "

## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
## sd(data)): ties should not be present for the Kolmogorov-Smirnov test
```



```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.063065, p-value = 2.466e-07
## alternative hypothesis: two-sided

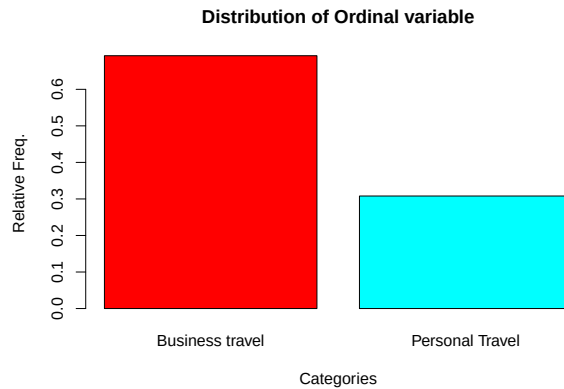
## integer(0)
```

2.2.4 Variable Type.of.Travel

Type.of.Travel is a categorical binary variable that represents the kind of flight of the observation (Business or Personal). The proportions are skewed towards business travel (0.692 vs 0.308) and there are no null values.

```
summarize_categorical_data(train$Type.of.Travel)
```

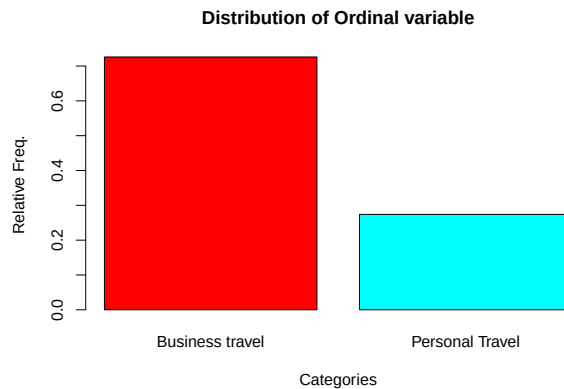
```
## [1] "Proportions: "
## data
## Business travel Personal Travel
##           0.692           0.308
```



```
## [1] "Number of nulls: "
## [1] 0
```

```
summarize_categorical_data(test$Type.of.Travel) # 0 NAs
```

```
## [1] "Proportions: "
## data
## Business travel Personal Travel
##          0.726          0.274
```



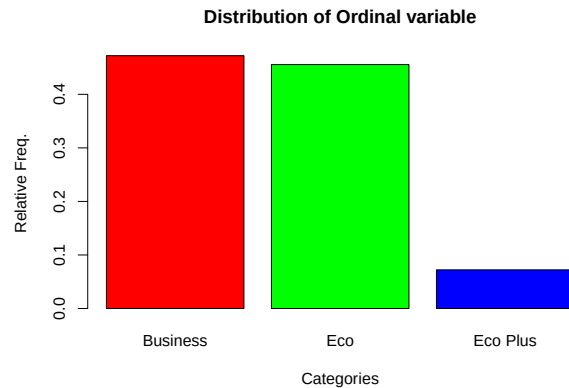
```
## [1] "Number of nulls: "
## [1] 0
```

2.2.5 Variable Class

Class categorical ternary variable that represents the type of cabin travel class of the observations (Business, Eco or Eco Plus). Vast majority of observations belong to Business or Eco evenly, and Eco plus has belongs to 7.22% of the observations. There are no null values.

```
summarize_categorical_data(train$Class)
```

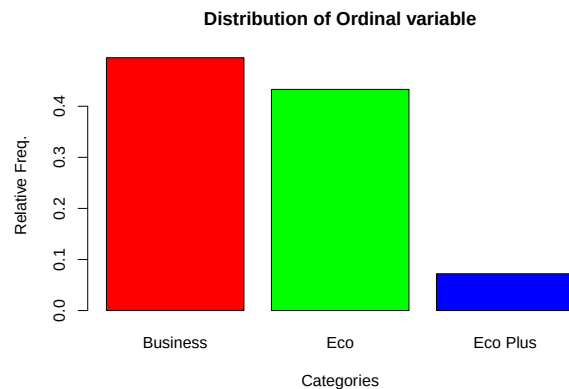
```
## [1] "Proportions: "
## data
## Business      Eco Eco Plus
##    0.4722    0.4556    0.0722
```



```
## [1] "Number of nulls: "
## [1] 0
```

```
summarize_categorical_data(test$Class) # 0 NAs
```

```
## [1] "Proportions: "
## data
## Business      Eco Eco Plus
##    0.495     0.433    0.072
```



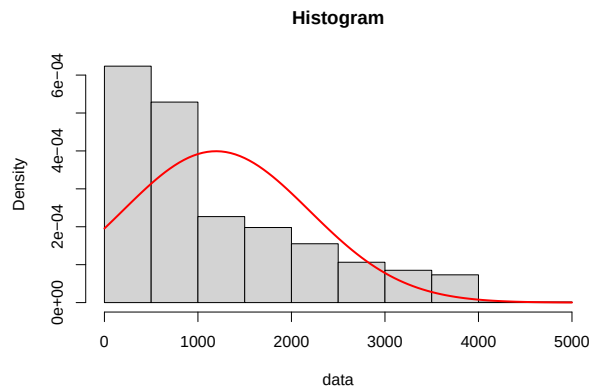
```
## [1] "Number of nulls: "
## [1] 0
```

2.2.6 Variable Flight.Distance

Flight.Distance is a discrete finite numerical variable that represents the distance of the given flight in a range between [67, 4983] with a mean value of 1194 and 999 std. Its distribution is obviously not normal given the almost exponential shape that it has and both inferential tests reject with tiny p-values. There are 95 mild outliers, none of them severe, and there is no metadata about the units. There are no null values.

```
outliers_indexes <- summary_and_normality_check(train$Flight.Distance, bins=10)
```

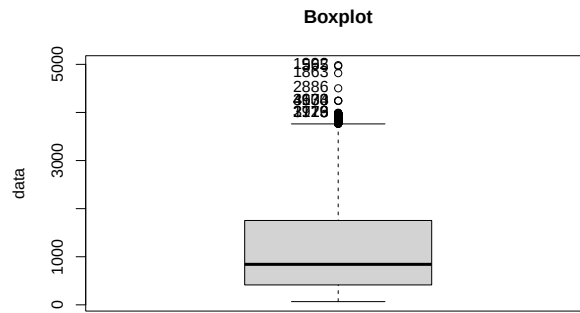
```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##       67    414     842   1194   1754   4983  
## [1] "Mean"  
## [1] 1194.205  
## [1] "Standard deviation"  
## [1] 999.2656
```



```
## [1] "Skewness"  
## [1] 1.087412
```

```
## [1] "Num Outliers:"  
## [1] 95  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro-Wilk test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.86754, p-value < 2.2e-16  
##  
## [1] "Kolmogorov-Smirnov test: "
```

```
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =  
## sd(data)): ties should not be present for the Kolmogorov-Smirnov test
```



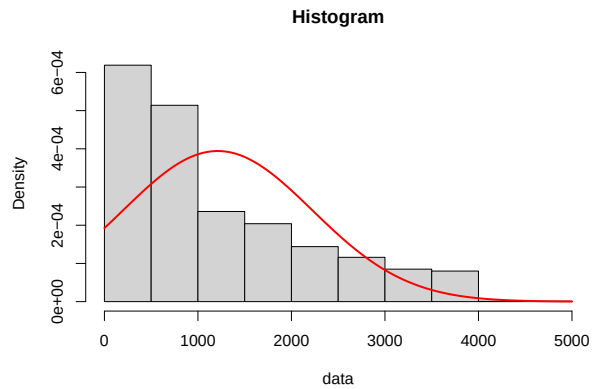
```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.15624, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

```
outliers_indexes
```

```
## [1] 205 206 221 379 388 458 523 533 565 622 692 720 825 878 880
## [16] 928 1090 1170 1272 1286 1428 1460 1470 1520 1579 1606 1758 1764 1825 1863
## [31] 1871 1887 1926 1992 2043 2101 2115 2295 2421 2474 2590 2624 2633 2720 2773
## [46] 2825 2859 2880 2886 2888 2974 2980 2985 3001 3119 3174 3215 3312 3333 3353
## [61] 3390 3533 3547 3617 3635 3680 3714 3744 3786 3821 3827 3837 3890 3916 3921
## [76] 3965 4020 4034 4183 4270 4278 4398 4553 4570 4587 4600 4639 4680 4689 4712
## [91] 4911 4918 4927 4933 4939
```

```
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes] + 1
summary_and_normality_check(test$Flight.Distance, bins=10) # 0 NAs
```

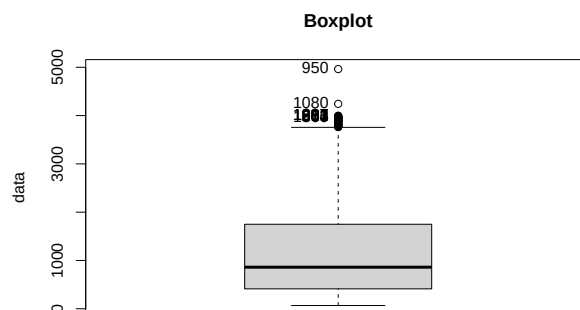
```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   67.0  412.5   861.0 1211.6 1750.0 4963.0
## [1] "Mean"
## [1] 1211.571
## [1] "Standard deviation"
## [1] 1011.707
```



```
## [1] "Skewness"
## [1] 1.071731

## [1] "Num Outliers:"
## [1] 42
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.86788, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
```

Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
sd(data)): ties should not be present for the Kolmogorov-Smirnov test



```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.1551, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

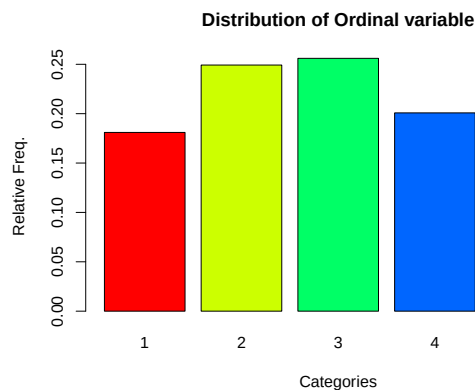
```
## [1] 19 49 208 226 283 285 487 540 611 673 729 804 807 838 950
## [16] 960 1033 1080 1082 1291 1327 1360 1401 1430 1442 1443 1484 1542 1609 1662
## [31] 1668 1713 1748 1790 1792 1818 1834 1878 1910 1922 1987 1993
```

2.2.7 Variable Inflight.wifi.service

Inflight.wifi.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The majority of values are grouped between 2 and 3, with a mean of 2.7, median of 3 and deviation of 1.26, producing a uni modal bell-shaped distribution centered at 2-3. There are 149 null values in the training set.

```
train$Inflight.wifi.service <- summarize_ordinal_data(train$Inflight.wifi.service)
```

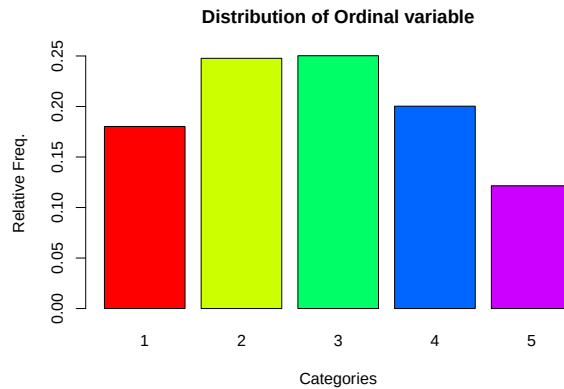
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1809936 0.2492270 0.2560297 0.2007833 0.1129664
```



```
## [1] "Mean: "
## [1] 2.815502
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.2618
## [1] "Number of nulls: "
## [1] 149
```

```
test$Inflight.wifi.service <- summarize_ordinal_data(test$Inflight.wifi.service) # 58 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1802266 0.2476828 0.2502575 0.2003090 0.1215242
```



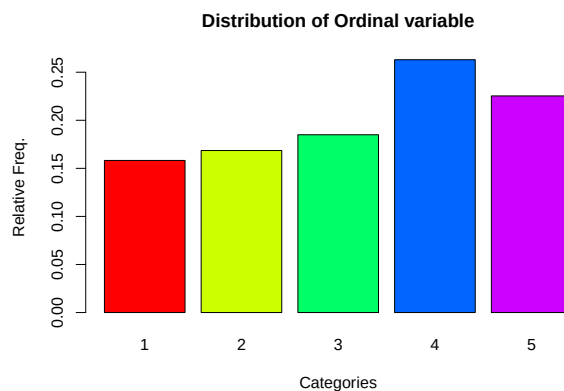
```
## [1] "Mean: "
## [1] 2.835221
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.276198
## [1] "Number of nulls: "
## [1] 58
```

2.2.8 Variable `Departure.Arrival.time.convenient`

`Departure.Arrival.time.convenient` is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The majority of values are grouped between 4 and 5, with a mean of 3.22, median of 3 and deviation of 1.38. The distribution is right-skewed and has a heavy left tail. There are 247 values equal to 0, we consider them as Not Applicable (NA).

```
train$Departure.Arrival.time.convenient <-
  summarize_ordinal_data(train$Departure.Arrival.time.convenient)
```

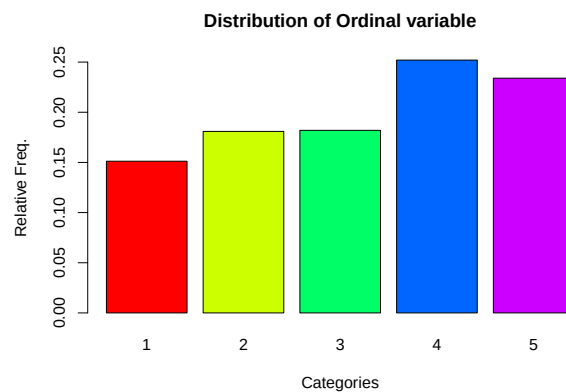
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1582159 0.1685251 0.1849358 0.2629918 0.2253314
```



```
## [1] "Mean: "
## [1] 3.228698
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.383404
## [1] "Number of nulls: "
## [1] 247
```

```
test$Departure.Arrival.time.convenient<-
  summarize_ordinal_data(test$Departure.Arrival.time.convenient) # 115 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1511936 0.1809019 0.1819629 0.2519894 0.2339523
```



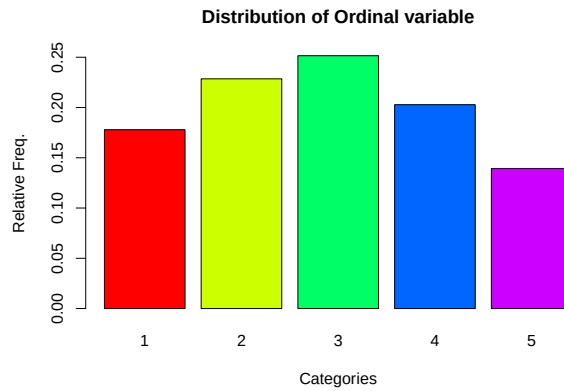
```
## [1] "Mean: "
## [1] 3.236605
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.385103
## [1] "Number of nulls: "
## [1] 115
```

2.2.9 Variable Ease.of.Online.booking

Ease.of.Online.booking is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution of ratings is somewhat bell-shaped and unimodal, with a mean of 2.89, median 3 and deviation of 1.29. There are 127 null values.

```
train$Ease.of.Online.booking <-
  summarize_ordinal_data(train$Ease.of.Online.booking)
```

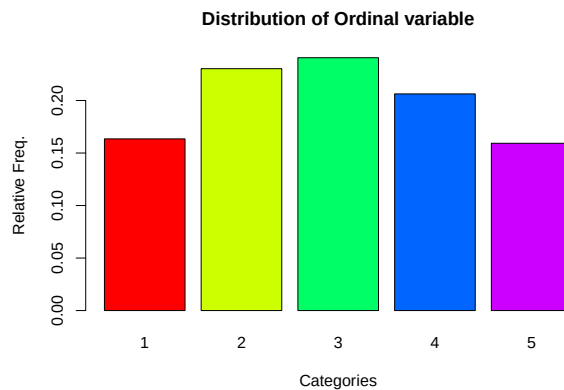
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1779218 0.2285177 0.2515158 0.2028016 0.1392432
```



```
## [1] "Mean: "
## [1] 2.896927
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.299888
## [1] "Number of nulls: "
## [1] 217
```

```
test$Ease.of.Online.booking <-
  summarize_ordinal_data(test$Ease.of.Online.booking) # 85 NAs
```

```
## [1] "Proportions: "
## data
##           1           2           3           4           5
## 0.1634465 0.2302872 0.2407311 0.2062663 0.1592689
```



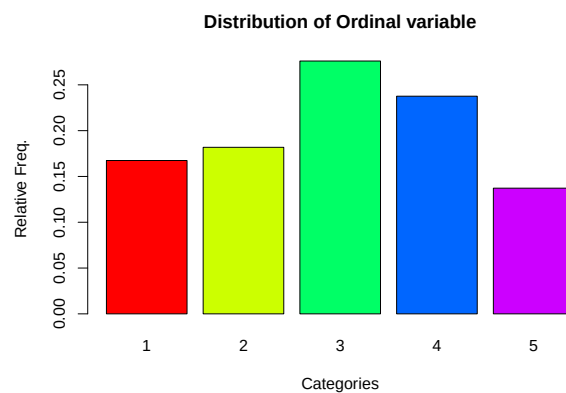
```
## [1] "Mean: "
## [1] 2.967624
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.314256
## [1] "Number of nulls: "
## [1] 85
```

2.2.10 Variable Gate.location

Gate.location is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution is somewhat uniform with a median of 3, mean of 3 and deviation of 1.28. There are no null values.

```
train$Gate.location <- summarize_ordinal_data(train$Gate.location)
```

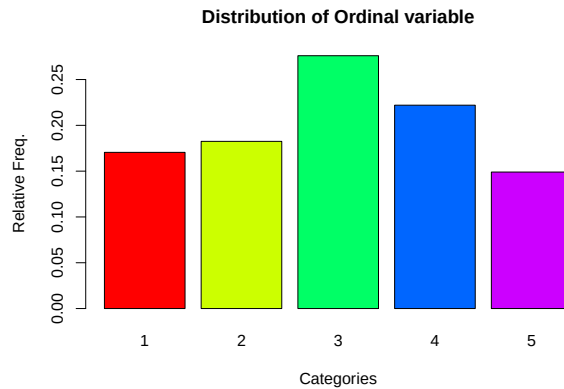
```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1674 0.1818 0.2760 0.2376 0.1372
```



```
## [1] "Mean: "  
## [1] 2.9954  
## [1] "Median: "  
## [1] 3  
## [1] "Standard Deviation: "  
## [1] 1.279885  
## [1] "Number of nulls: "  
## [1] 0
```

```
test$Gate.location <- summarize_ordinal_data(test$Gate.location) # 0 NAs
```

```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1705 0.1825 0.2760 0.2220 0.1490
```



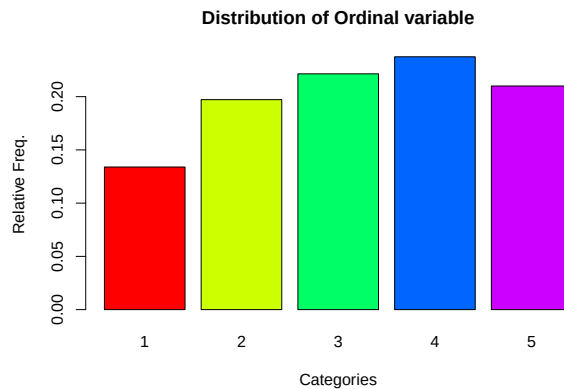
```
## [1] "Mean: "
## [1] 2.9965
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.297432
## [1] "Number of nulls: "
## [1] 0
```

2.2.11 Variable Food.and.drink

Food.and.drink is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution is uniform and right skewed, with a median of 3, mean of 3.19 and deviation of 1.33. There are 5 null values.

```
train$Food.and.drink <- summarize_ordinal_data(train$Food.and.drink)
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1339339 0.1971972 0.2214214 0.2374374 0.2100100
```

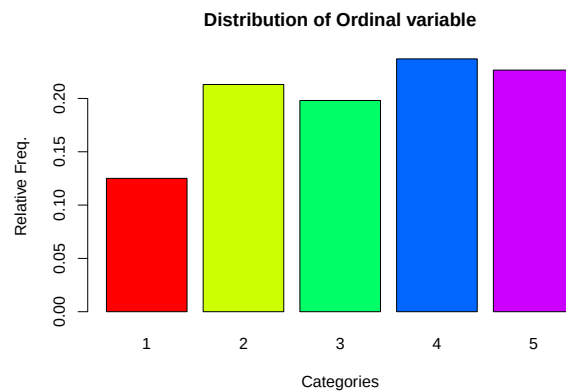


```
## [1] "Mean: "
```

```
## [1] 3.192392
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.331822
## [1] "Number of nulls: "
## [1] 5
```

```
test$Food.and.drink <- summarize_ordinal_data(test$Food.and.drink) # 1 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1250625 0.2131066 0.1980990 0.2371186 0.2266133
```



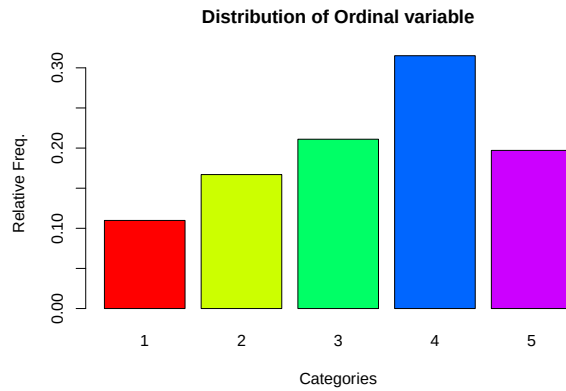
```
## [1] "Mean: "
## [1] 3.227114
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.343969
## [1] "Number of nulls: "
## [1] 1
```

2.2.12 Variable Online.boarding

Online.boarding is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The barplot showcases a fairly unimodal right skewed distribution, with a median of 4, mean of 3.32 and deviation of 1.26. There are 119 null values.

```
train$Online.boarding <- summarize_ordinal_data(train$Online.boarding)
```

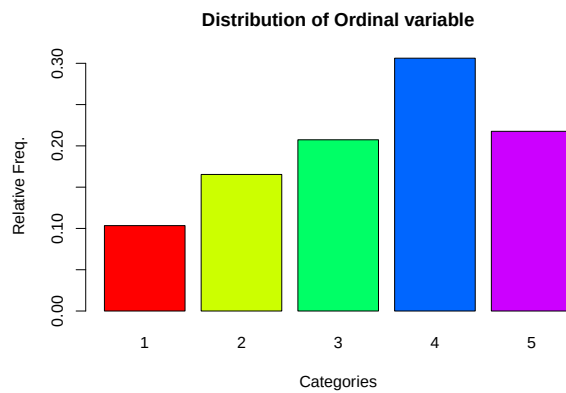
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1098136 0.1669740 0.2110223 0.3150994 0.1970908
```



```
## [1] "Mean: "
## [1] 3.32268
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.26724
## [1] "Number of nulls: "
## [1] 119
```

```
test$Online.boarding <- summarize_ordinal_data(test$Online.boarding) # 47 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1034306 0.1653866 0.2073733 0.3061956 0.2176139
```



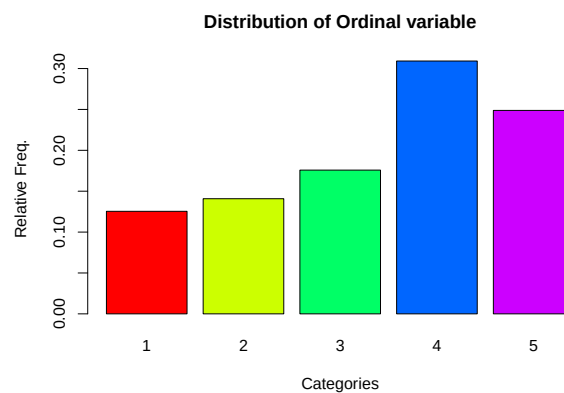
```
## [1] "Mean: "
## [1] 3.369176
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.27291
## [1] "Number of nulls: "
## [1] 47
```

2.2.13 Variable Seat.comfort

Seat.comfort is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is right skewed with a heavy left tail, and has a median of 4, mean of 3.41 and a deviation of 1.33. There are no null values.

```
train$Seat.comfort <- summarize_ordinal_data(train$Seat.comfort)
```

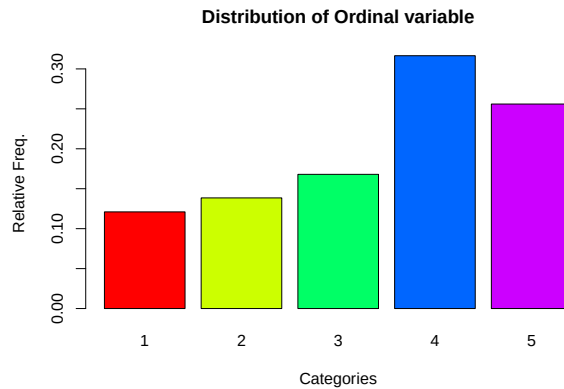
```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1254 0.1408 0.1758 0.3092 0.2488
```



```
## [1] "Mean: "  
## [1] 3.4152  
## [1] "Median: "  
## [1] 4  
## [1] "Standard Deviation: "  
## [1] 1.332203  
## [1] "Number of nulls: "  
## [1] 0
```

```
test$Seat.comfort <- summarize_ordinal_data(test$Seat.comfort) # 0 NAs
```

```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1210 0.1385 0.1680 0.3165 0.2560
```



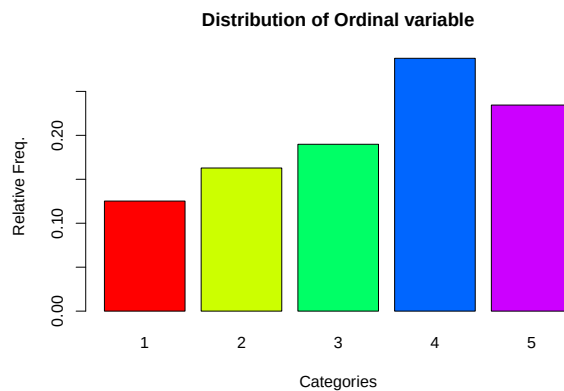
```
## [1] "Mean: "
## [1] 3.448
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.327847
## [1] "Number of nulls: "
## [1] 0
```

2.2.14 Variable Inflight.entertainment

Inflight.entertainment is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and slightly right-skewed, with a median of 4, mean 3.34 and deviation of 1.33. There is 1 null value.

```
train$Inflight.entertainment <-
  summarize_ordinal_data(train$Inflight.entertainment)
```

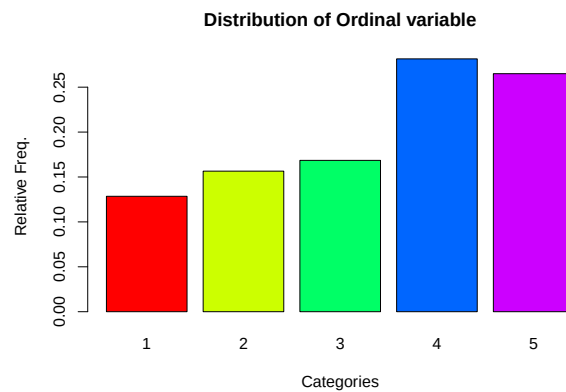
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1252250 0.1628326 0.1898380 0.2876575 0.2344469
```



```
## [1] "Mean: "
## [1] 3.343269
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.331052
## [1] "Number of nulls: "
## [1] 1
```

```
test$Inflight.entertainment <-
  summarize_ordinal_data(test$Inflight.entertainment) # 0 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1285 0.1565 0.1685 0.2815 0.2650
```



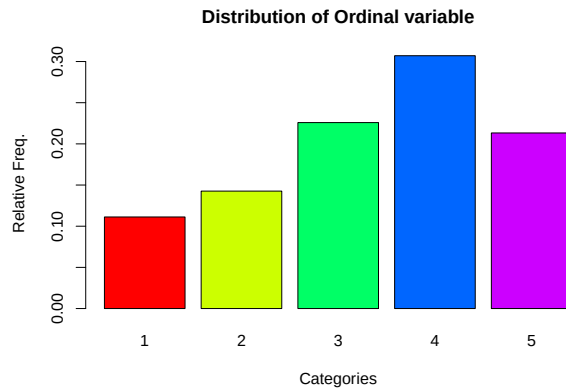
```
## [1] "Mean: "
## [1] 3.398
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.361809
## [1] "Number of nulls: "
## [1] 0
```

2.2.15 Variable On.board.service

On.board.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Similarly, its distribution is unimodal and right skewed with a median of 4, mean of 3.37 and deviation of 1.27. There is one null value.

```
train$On.board.service <- summarize_ordinal_data(train$On.board.service)
```

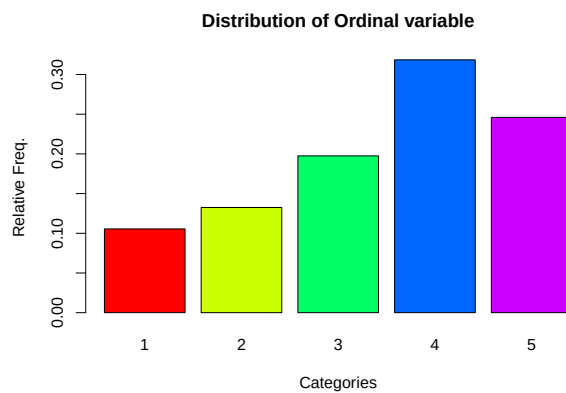
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1112222 0.1426285 0.2258452 0.3070614 0.2132426
```



```
## [1] "Mean: "
## [1] 3.368474
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.269685
## [1] "Number of nulls: "
## [1] 1
```

```
test$On.board.service<- summarize_ordinal_data(test$On.board.service) # 0 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1055 0.1325 0.1975 0.3185 0.2460
```



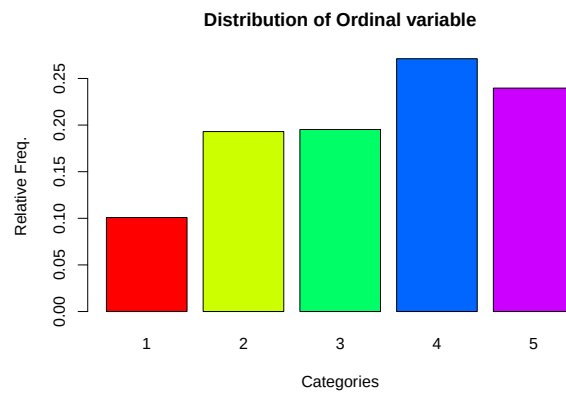
```
## [1] "Mean: "
## [1] 3.467
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.28052
## [1] "Number of nulls: "
## [1] 0
```

2.2.16 Variable Leg.room.service

Leg.room.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is more uniform than previous variables but its also somewhat right skewed, with a median of 4 and mean of 3.35, deviation 1.30. There are 22 null values.

```
train$Leg.room.service <- summarize_ordinal_data(train$Leg.room.service)
```

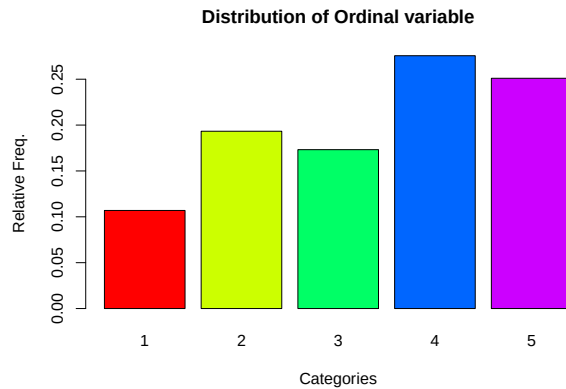
```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1008437 0.1930494 0.1952591 0.2711933 0.2396545
```



```
## [1] "Mean: "  
## [1] 3.355765  
## [1] "Median: "  
## [1] 4  
## [1] "Standard Deviation: "  
## [1] 1.303844  
## [1] "Number of nulls: "  
## [1] 22
```

```
test$Leg.room.service <- summarize_ordinal_data(test$Leg.room.service) # 8 NAs
```

```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1069277 0.1932731 0.1731928 0.2756024 0.2510040
```



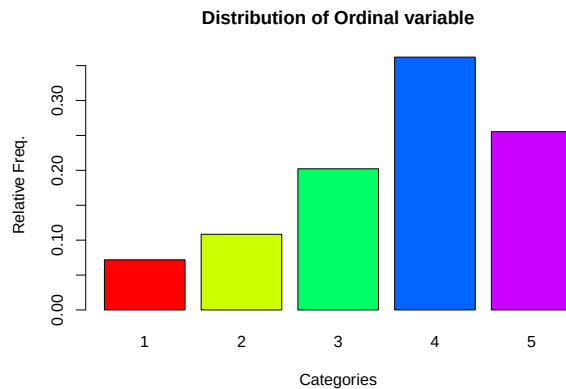
```
## [1] "Mean: "
## [1] 3.370482
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.328244
## [1] "Number of nulls: "
## [1] 8
```

2.2.17 Variable Baggage.handling

Baggage.handling is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and highly right-skewed, having a median of 4, mean of 3.62 and small deviation of 1.18. There are no null values.

```
train$Baggage.handling <- summarize_ordinal_data(train$Baggage.handling)
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.0718 0.1084 0.2022 0.3622 0.2554
```

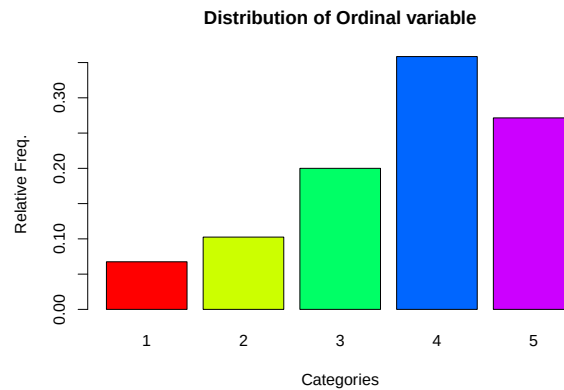


```
## [1] "Mean: "
```

```
## [1] 3.621
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.180694
## [1] "Number of nulls: "
## [1] 0
```

```
test$Baggage.handling <- summarize_ordinal_data(test$Baggage.handling) # 0 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.0675 0.1025 0.2000 0.3585 0.2715
```



```
## [1] "Mean: "
## [1] 3.664
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.173368
## [1] "Number of nulls: "
## [1] 0
```

2.2.18 Variable Checkin.service

Checking.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. In contrast, this distribution seems more uniform with a huge mode between 3 and 4 (median of 3, mean of 3.27, deviation 1.26). There are no null values.

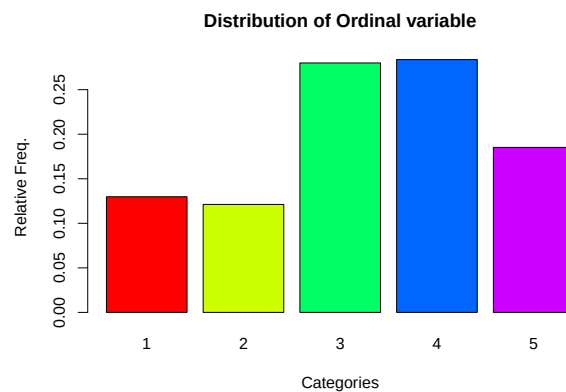
```
train$Checkin.service <- summarize_ordinal_data(train$Checkin.service)
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1298 0.1212 0.2800 0.2838 0.1852
```

```
## [1] "Mean: "
## [1] 3.2734
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.261178
## [1] "Number of nulls: "
## [1] 0
```

```
train$Checkin.service <- summarize_ordinal_data(train$Checkin.service) # 0 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1298 0.1212 0.2800 0.2838 0.1852
```



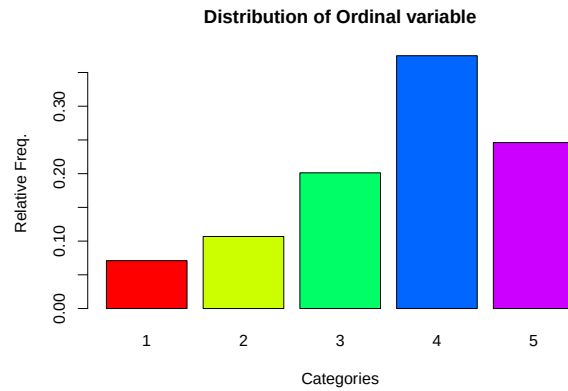
```
## [1] "Mean: "
## [1] 3.2734
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.261178
## [1] "Number of nulls: "
## [1] 0
```

2.2.19 Variable Inflight.service

Inflight.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and highly right-skewed, with a median of 4, mean of 3.61 and a small deviation 1.16. There is 1 null value.

```
train$Inflight.service <- summarize_ordinal_data(train$Inflight.service)
```

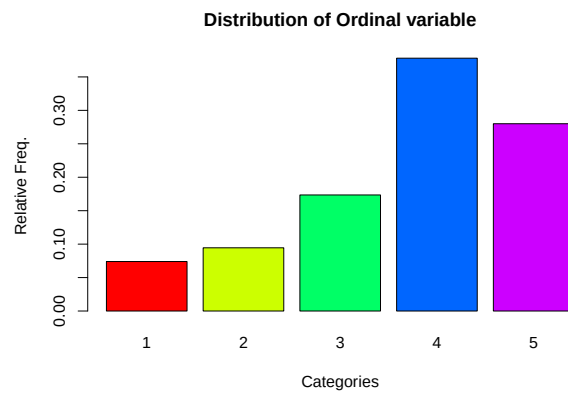
```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.0710142 0.1068214 0.2012402 0.3748750 0.2460492
```



```
## [1] "Mean: "
## [1] 3.618124
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.169678
## [1] "Number of nulls: "
## [1] 1
```

```
test$Inflight.service <- summarize_ordinal_data(test$Inflight.service) # 0 NAs
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.0740 0.0945 0.1735 0.3780 0.2800
```



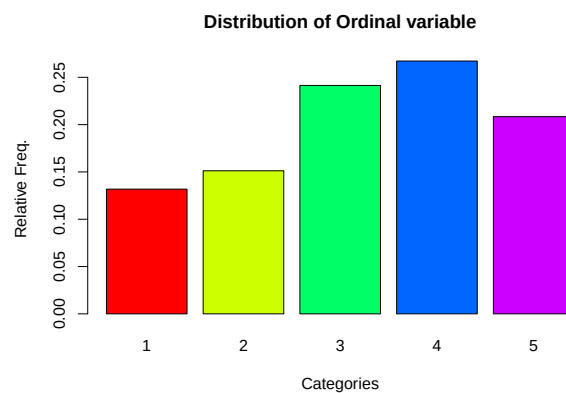
```
## [1] "Mean: "
## [1] 3.6955
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.18553
## [1] "Number of nulls: "
## [1] 0
```

2.2.20 Variable Cleanliness

Cleanliness is a self-describing ordinal categorical variable given by the customers in the range [1,5]. This variable has the most uniform distribution of them all, median 3, mean 3.26 and deviation 1.3, thus skewed slightly to the right. Again, there are no null values.

```
train$Cleanliness <- summarize_ordinal_data(train$Cleanliness)
```

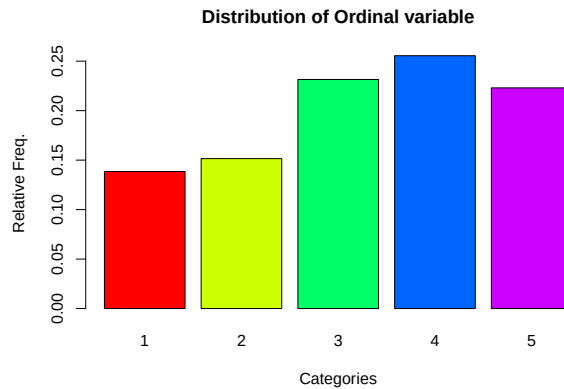
```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1318 0.1512 0.2414 0.2672 0.2084
```



```
## [1] "Mean: "  
## [1] 3.2692  
## [1] "Median: "  
## [1] 3  
## [1] "Standard Deviation: "  
## [1] 1.30655  
## [1] "Number of nulls: "  
## [1] 0
```

```
test$Cleanliness <- summarize_ordinal_data(test$Cleanliness) # 0 NAs
```

```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1385 0.1515 0.2315 0.2555 0.2230
```



```
## [1] "Mean: "
## [1] 3.273
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.333927
## [1] "Number of nulls: "
## [1] 0
```

2.2.21 Variable Departure.Delay.in.Minutes

Departure.Delay.in.Minutes is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled departure. Its values range from 0 to 692, with a mean of 15, median of 0 and huge deviation of 38.45. There are 675 mild outliers and 413 severe ones. Its distribution looks astonishingly exponential, with a skewness coefficient of 5.99. Both inferential tests reject the hypothesis of normality.

```
outliers_indexes_dep_delay <-
  summary_and_normality_check(train$Departure.Delay.in.Minutes)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.00   0.00   0.00   15.08   13.00   692.00
## [1] "Mean"
## [1] 15.081
## [1] "Standard deviation"
## [1] 38.45726
```

A histogram showing the distribution of 'data'. The x-axis is labeled 'data' and ranges from 0 to 700. The y-axis is labeled 'Density' and ranges from 0.000 to 0.015. The histogram bars are grey, and a red density curve is overlaid. The distribution is right-skewed, with a peak density of approximately 0.016 at a data value of about 25.[illegible]

32

```
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes_dep_delay] + 1
```

```
## Warning in all_outliers[outliers_indexes] <-  
## all_outliers[outliers_indexes_dep_delay] + : número de items para para  
## sustituir no es un múltiplo de la longitud del reemplazo
```

```
table(all_outliers)
```

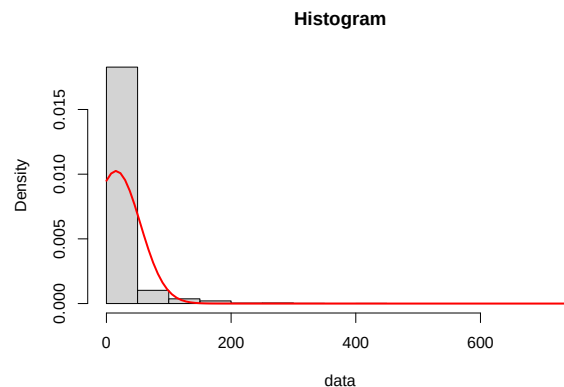
```
## all_outliers  
##      0      1      2  
## 4905    92      3
```

2.2.22 Variable Arrival.Delay.in.Minutes

Arrival.Delay.in.minutes is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled arrival. Similarly to departure delay, its distribution has an astonishingly exponential shape, and a huge coefficient of skewness of 6.06. Has a mean of 15.28 but a median of 0, with values ranging between $[0, 702]$ minutes, 12 null values, 669 mild outliers and 410 severe ones. Both inferential tests reject the hypothesis of normality.

```
outliers_indexes_arr_delay <-  
  summary_and_normality_check(train$Arrival.Delay.in.Minutes)
```

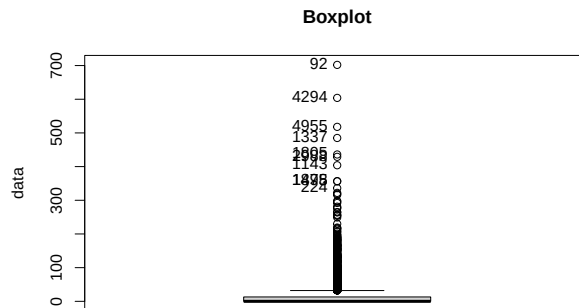
```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's  
##      0.00   0.00    0.00   15.28   13.00   702.00      12  
## [1] "Mean"  
## [1] 15.27506  
## [1] "Standard deviation"  
## [1] 38.91399
```



```
## [1] "Skewness"  
## [1] 6.060705  
  
## [1] "Num Outliers:"
```

```
## [1] 669
## [1] "Num. severe outliers:"
## [1] 410
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.42739, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
```

Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
sd(data)): ties should not be present for the Kolmogorov-Smirnov test



```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = NA, p-value = NA
## alternative hypothesis: two-sided
```

```
all_outliers[outliers_indexes] <- all_outliers[outliers_indexes_arr_delay] + 1
```

```
## Warning in all_outliers[outliers_indexes] <-
## all_outliers[outliers_indexes_arr_delay] + : número de items para para
## sustituir no es un múltiplo de la longitud del reemplazo
```

```
table(all_outliers)
```

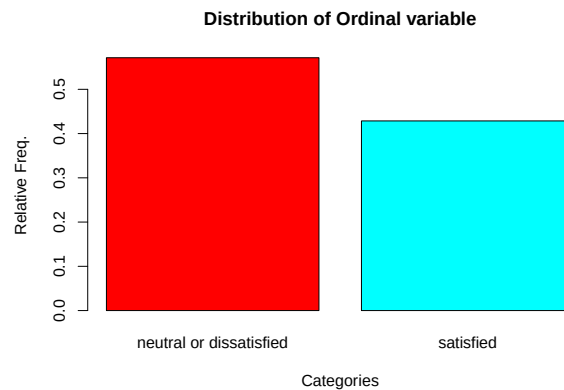
```
## all_outliers
##      0      1      2
## 4905   92      3
```

2.2.23 Variable satisfaction

satisfaction is a target binary categorical variable that represents whether the customer is satisfied or not/neutral with its overall experience. The dataset is somewhat balanced with a slight bias towards neutral/dissatisfied (57.14% vs 42.86%). There are no null values.

```
summarize_categorical_data(train$satisfaction)
```

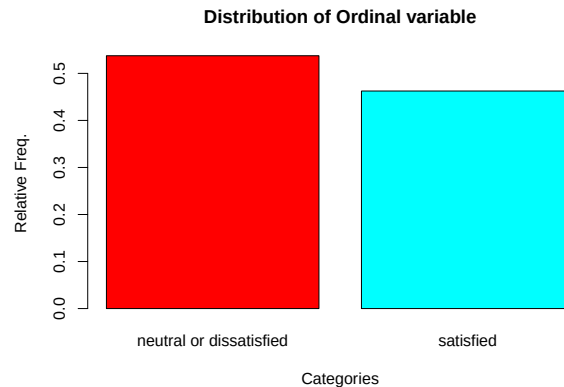
```
## [1] "Proportions: "  
## data  
## neutral or dissatisfied      satisfied  
##                0.5714                0.4286
```



```
## [1] "Number of nulls: "  
## [1] 0
```

```
summarize_categorical_data(test$satisfaction)
```

```
## [1] "Proportions: "  
## data  
## neutral or dissatisfied      satisfied  
##                0.5375                0.4625
```



```
## [1] "Number of nulls: "  
## [1] 0
```

Transform variables

Here, we transform variable types too their correct ones.

```
str(train)
```

```
## 'data.frame': 5000 obs. of 23 variables:
## $ Gender : chr "Female" "Female" "Male" "Female" ...
## $ Customer.Type : chr "Loyal Customer" "Loyal Customer" "Loyal Customer" "Loyal
## $ Age : int 40 39 58 19 42 30 47 56 46 77 ...
## $ Type.of.Travel : chr "Business travel" "Business travel" "Business travel" "Bu
## $ Class : chr "Business" "Business" "Business" "Business" ...
## $ Flight.Distance : int 1471 3447 3084 2756 793 516 1119 980 565 2288 ...
## $ Inflight.wifi.service : int 1 4 1 2 1 3 2 1 2 3 ...
## $ Departure.Arrival.time.convenient: int 1 1 1 5 1 5 1 1 4 1 ...
## $ Ease.of.Online.booking : int 1 4 1 5 1 3 1 1 2 1 ...
## $ Gate.location : int 1 4 1 5 3 2 1 1 1 1 ...
## $ Food.and.drink : int 5 3 3 2 3 5 3 2 4 4 ...
## $ Online.boarding : int 4 5 4 2 1 3 4 5 4 3 ...
## $ Seat.comfort : int 5 4 5 2 3 4 4 5 5 2 ...
## $ Inflight.entertainment : int 5 5 3 2 3 5 2 5 5 3 ...
## $ On.board.service : int 5 5 3 3 5 3 2 5 5 3 ...
## $ Leg.room.service : int 5 5 4 3 3 3 2 5 2 3 ...
## $ Baggage.handling : int 5 5 3 4 5 5 2 5 5 3 ...
## $ Checkin.service : int 5 4 4 3 5 3 2 4 5 3 ...
## $ Inflight.service : int 5 5 3 4 4 4 2 5 5 3 ...
## $ Cleanliness : int 4 5 5 2 3 5 2 5 4 3 ...
## $ Departure.Delay.in.Minutes : int 0 0 16 12 0 0 0 15 25 47 ...
## $ Arrival.Delay.in.Minutes : int 0 0 12 19 0 0 0 4 52 28 ...
## $ satisfaction : chr "satisfied" "satisfied" "satisfied" "neutral or dissatisf
```

```
# factor variables
train[, c(1,2,4,5,23)] <- lapply(train[, c(1,2,4,5,23)], as.factor)
test[, c(1,2,4,5,23)] <- lapply(test[, c(1,2,4,5,23)], as.factor)

# numerical variables
train[, c(3,6,21,22)] <- lapply(train[, c(3,6,21,22)], as.numeric)
test[, c(3,6,21,22)] <- lapply(test[, c(3,6,21,22)], as.numeric)

# ordinal variables
train[, c(7:20)] <- lapply(train[, c(7:20)],
                           function(x) factor(x, levels = c(1, 2, 3, 4, 5),
                                                ordered = TRUE))
test[, c(7:20)] <- lapply(test[, c(7:20)],
                           function(x) factor(x, levels = c(1, 2, 3, 4, 5),
                                                ordered = TRUE))

str(train)
```

```
## 'data.frame': 5000 obs. of 23 variables:
## $ Gender : Factor w/ 2 levels "Female","Male": 1 1 2 1 2 2 1 1 1 1 ...
## $ Customer.Type : Factor w/ 2 levels "disloyal Customer",...: 2 2 2 2 1 2 2 2 2 2
## $ Age : num 40 39 58 19 42 30 47 56 46 77 ...
```

```
## $ Type.of.Travel      : Factor w/ 2 levels "Business travel",...: 1 1 1 1 1 2 1 1 2 1 .
## $ Class              : Factor w/ 3 levels "Business","Eco",...: 1 1 1 1 1 1 3 1 2 2 ..
## $ Flight.Distance    : num  1471 3447 3084 2756 793 ...
## $ Inflight.wifi.service : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 1 4 1 2 1 3 2 1 2 3
## $ Departure.Arrival.time.convenient: Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 1 1 1 5 1 5 1 1 4 1
## $ Ease.of.Online.booking : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 1 4 1 5 1 3 1 1 2 1
## $ Gate.location      : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 1 4 1 5 3 2 1 1 1 1
## $ Food.and.drink     : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 3 3 2 3 5 3 2 4 4
## $ Online.boarding    : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 4 5 4 2 1 3 4 5 4 3
## $ Seat.comfort       : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 4 5 2 3 4 4 5 5 2
## $ Inflight.entertainment : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 5 3 2 3 5 2 5 5 3
## $ On.board.service   : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 5 3 3 5 3 2 5 5 3
## $ Leg.room.service   : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 5 4 3 3 3 2 5 2 3
## $ Baggage.handling   : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 5 3 4 5 5 2 5 5 3
## $ Checkin.service    : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 4 4 3 5 3 2 4 5 3
## $ Inflight.service   : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 5 5 3 4 4 4 2 5 5 3
## $ Cleanliness        : Ord.factor w/ 5 levels "1"<"2"<"3"<"4"<...: 4 5 5 2 3 5 2 5 4 3
## $ Departure.Delay.in.Minutes : num  0 0 16 12 0 0 0 15 25 47 ...
## $ Arrival.Delay.in.Minutes : num  0 0 12 19 0 0 0 4 52 28 ...
## $ satisfaction       : Factor w/ 2 levels "neutral or dissatisfied",...: 2 2 2 1 1 1 1 1
```

2.2.23 New variables definition

We defined 4 new variables that we think that might be improving our model since can provide information on the relation between variables and the satisfaction of an individual. These consist of:

- AgeGroup: Divide ages in 4 different groups
- f.FlightDist: Divide distances of flights in 3 different groups
- DelayedArrival: If the arrival time is delayed
- DelayedDeparture: If the departure is delayed.

```
# age factor
train$AgeGroup <- cut(train$Age, breaks = c(-Inf, 18, 35, 60, Inf),
                      labels = c("Under 18", "18-35", "36-60", "Over 60"))
test$AgeGroup <- cut(test$Age, breaks = c(-Inf, 18, 35, 60, Inf),
                    labels = c("Under 18", "18-35", "36-60", "Over 60"))

# distance factor
quant <- quantile(train$Flight.Distance, probs = seq(0, 1, 0.333), na.rm = TRUE)
train$f.FlightDist <- cut(train$Flight.Distance,
                         breaks = quant,
                         labels = c("Short Distance", "Medium Distance",
                                   "Long Distance"),
                         include.lowest = TRUE)

quant2 <- quantile(test$Flight.Distance, probs = seq(0, 1, 0.333), na.rm = TRUE)
test$f.FlightDist <- cut(test$Flight.Distance,
                       breaks = quant2,
                       labels = c("Short Distance", "Medium Distance",
                                 "Long Distance"),
                       include.lowest = TRUE)
```

```
# delay arrival (binary)
train$DelayedArrival <- as.factor(ifelse(train$Arrival.Delay.in.Minutes > 0,
                                         "DelayedArrival", "OK"))
test$DelayedArrival <- as.factor(ifelse(test$Arrival.Delay.in.Minutes > 0,
                                         "DelayedArrival", "OK"))
summary(train$DelayedArrival)
```

```
## DelayedArrival      OK      NA's
##           2206      2782      12
```

```
train$DelayedDeparture <- as.factor(ifelse(train$Departure.Delay.in.Minutes > 0,
                                           "DelayedDeparture", "OK"))
test$DelayedDeparture <- as.factor(ifelse(test$Departure.Delay.in.Minutes > 0,
                                           "DelayedDeparture", "OK"))
summary(train$DelayedDeparture)
```

```
## DelayedDeparture      OK
##           2170      2830
```

2.3 Data quality report

This section provides a comprehensive overview of the dataset's quality, complementing the prior identification of univariate outliers and missing values with a visual summary to highlight key patterns and anomalies. The data quality report is structured into two parts: an analysis regarding the variables and a review of individual records (stating the number of outliers per individual and identifying multivariate outliers).

2.3.1 Variables

This first part of the data quality report assesses the quality of the data regarding the variables.

2.3.1.1 Missing values and Errors We used the `md.pattern` function from the `mice` library to analyze the distribution of missing values across the dataset. This function provides a visual representation of missing data patterns, allowing us to identify the extent and location of missingness.

Note that we considered errors (and thus set to NA) observations with value 0 in an ordinal variable with range 1-5.

Based on the results, we can conclude that the dataset is relatively clean, with 4590 observations (91.8%) containing complete data. However, certain variables exhibit notable levels of missing values, particularly those on the right side of the plot. `Departure.Arrival.time.convenient` and `Ease.of.Online.Booking` have the highest number of missing values, with 247 and 217, respectively. `Inflight.wifi.service` has 149 missing values, while `Online.boarding` has 119.

In addition, eight variables have fewer than 25 missing values, and 14 variables are completely free of missing data.

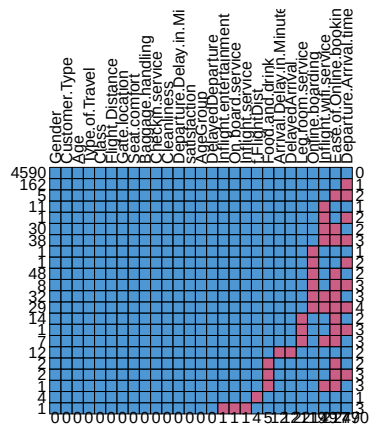
To ensure the integrity of the dataset and support robust and unbiased modeling, all missing and erroneous values will be carefully imputed in a way that preserves the dataset's original characteristics and patterns.

```
summary(train)
```

```
##      Gender      Customer.Type      Age      Type.of.Travel
## Female:2548  disloyal Customer: 934  Min.   : 7.00  Business travel:3460
## Male  :2452  Loyal Customer   :4066  1st Qu.:27.00  Personal Travel:1540
##                                     Median :40.00
##                                     Mean   :39.41
##                                     3rd Qu.:51.00
##                                     Max.   :85.00
##
##      Class      Flight.Distance Inflight.wifi.service
## Business:2361  Min.   : 67      1 : 878
## Eco          :2278  1st Qu.: 414    2 :1209
## Eco Plus: 361  Median : 842    3 :1242
##                                     Mean   :1194  4 : 974
##                                     3rd Qu.:1754  5 : 548
##                                     Max.   :4983  NA's: 149
##
## Departure.Arrival.time.convenient Ease.of.Online.booking Gate.location
## 1 : 752                                1 : 851                                1: 837
## 2 : 801                                2 :1093                                2: 909
## 3 : 879                                3 :1203                                3:1380
## 4 :1250                                4 : 970                                4:1188
## 5 :1071                                5 : 666                                5: 686
## NA's: 247                            NA's: 217
##
## Food.and.drink Online.boarding Seat.comfort Inflight.entertainment
## 1 : 669          1 : 536          1: 627          1 : 626
## 2 : 985          2 : 815          2: 704          2 : 814
## 3 :1106          3 :1030          3: 879          3 : 949
## 4 :1186          4 :1538          4:1546          4 :1438
## 5 :1049          5 : 962          5:1244          5 :1172
## NA's: 5          NA's: 119          NA's: 1
##
## On.board.service Leg.room.service Baggage.handling Checkin.service
## 1 : 556          1 : 502          1: 359          1: 649
## 2 : 713          2 : 961          2: 542          2: 606
## 3 :1129          3 : 972          3:1011          3:1400
## 4 :1535          4 :1350          4:1811          4:1419
## 5 :1066          5 :1193          5:1277          5: 926
## NA's: 1          NA's: 22
##
## Inflight.service Cleanliness Departure.Delay.in.Minutes
## 1 : 355          1: 659          Min.   : 0.00
## 2 : 534          2: 756          1st Qu.: 0.00
## 3 :1006          3:1207          Median : 0.00
## 4 :1874          4:1336          Mean   : 15.08
## 5 :1230          5:1042          3rd Qu.: 13.00
## NA's: 1          Max.   :692.00
##
## Arrival.Delay.in.Minutes      satisfaction      AgeGroup
## Min.   : 0.00      neutral or dissatisfied:2857  Under 18: 392
## 1st Qu.: 0.00      satisfied      :2143  18-35 :1602
```

```
## Median : 0.00
## Mean : 15.28
## 3rd Qu.: 13.00
## Max. :702.00
## NA's :12
## f.FlightDist DelayedArrival DelayedDeparture
## Short Distance :1666 DelayedArrival:2206 DelayedDeparture:2170
## Medium Distance:1664 OK :2782 OK :2830
## Long Distance :1666 NA's : 12
## NA's : 4
##
##
##
```

```
md.pattern(train, rotate.names = T)
```



```
## Gender Customer.Type Age Type.of.Travel Class Flight.Distance
## 4590 1 1 1 1 1 1
## 162 1 1 1 1 1 1
## 5 1 1 1 1 1 1
## 11 1 1 1 1 1 1
## 1 1 1 1 1 1 1
## 30 1 1 1 1 1 1
## 38 1 1 1 1 1 1
## 1 1 1 1 1 1 1
## 1 1 1 1 1 1 1
## 48 1 1 1 1 1 1 1
## 8 1 1 1 1 1 1 1
## 32 1 1 1 1 1 1 1
## 29 1 1 1 1 1 1 1
## 14 1 1 1 1 1 1 1
## 1 1 1 1 1 1 1
## 7 1 1 1 1 1 1 1
## 12 1 1 1 1 1 1 1
## 2 1 1 1 1 1 1 1
## 2 1 1 1 1 1 1 1
## 1 1 1 1 1 1 1 1
## 4 1 1 1 1 1 1 1
## 1 1 1 1 1 1 1 1
```

##	0	0	0	0	0	0
##	Gate.location	Seat.comfort	Baggage.handling	Checkin.service	Cleanliness	
## 4590	1	1	1	1	1	1
## 162	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 11	1	1	1	1	1	1
## 1	1	1	1	1	1	1
## 30	1	1	1	1	1	1
## 38	1	1	1	1	1	1
## 1	1	1	1	1	1	1
## 1	1	1	1	1	1	1
## 48	1	1	1	1	1	1
## 8	1	1	1	1	1	1
## 32	1	1	1	1	1	1
## 29	1	1	1	1	1	1
## 14	1	1	1	1	1	1
## 1	1	1	1	1	1	1
## 7	1	1	1	1	1	1
## 12	1	1	1	1	1	1
## 2	1	1	1	1	1	1
## 2	1	1	1	1	1	1
## 1	1	1	1	1	1	1
## 4	1	1	1	1	1	1
## 1	1	1	1	1	1	1
##	0	0	0	0	0	0
##	Departure.Delay.in.Minutes	satisfaction	AgeGroup	DelayedDeparture		
## 4590		1	1	1		1
## 162		1	1	1		1
## 5		1	1	1		1
## 11		1	1	1		1
## 1		1	1	1		1
## 30		1	1	1		1
## 38		1	1	1		1
## 1		1	1	1		1
## 1		1	1	1		1
## 48		1	1	1		1
## 8		1	1	1		1
## 32		1	1	1		1
## 29		1	1	1		1
## 14		1	1	1		1
## 1		1	1	1		1
## 7		1	1	1		1
## 12		1	1	1		1
## 2		1	1	1		1
## 2		1	1	1		1
## 1		1	1	1		1
## 4		1	1	1		1
## 1		1	1	1		1
##		0	0	0		0
##	Inflight.entertainment	On.board.service	Inflight.service	f.FlightDist		
## 4590		1	1	1		1
## 162		1	1	1		1
## 5		1	1	1		1
## 11		1	1	1		1

## 1	1	1	1	1
## 30	1	1	1	1
## 38	1	1	1	1
## 1	1	1	1	1
## 1	1	1	1	1
## 48	1	1	1	1
## 8	1	1	1	1
## 32	1	1	1	1
## 29	1	1	1	1
## 14	1	1	1	1
## 1	1	1	1	1
## 7	1	1	1	1
## 12	1	1	1	1
## 2	1	1	1	1
## 2	1	1	1	1
## 1	1	1	1	1
## 4	1	1	1	0
## 1	0	0	0	1
##	1	1	1	4
##	Food.and.drink	Arrival.Delay.in.Minutes	DelayedArrival	Leg.room.service
## 4590	1	1	1	1
## 162	1	1	1	1
## 5	1	1	1	1
## 11	1	1	1	1
## 1	1	1	1	1
## 30	1	1	1	1
## 38	1	1	1	1
## 1	1	1	1	1
## 1	1	1	1	1
## 48	1	1	1	1
## 8	1	1	1	1
## 32	1	1	1	1
## 29	1	1	1	1
## 14	1	1	1	0
## 1	1	1	1	0
## 7	1	1	1	0
## 12	1	0	0	1
## 2	0	1	1	1
## 2	0	1	1	1
## 1	0	1	1	1
## 4	1	1	1	1
## 1	1	1	1	1
##	5	12	12	22
##	Online.boarding	Inflight.wifi.service	Ease.of.Online.booking	
## 4590	1	1	1	
## 162	1	1	1	
## 5	1	1	0	
## 11	1	0	1	
## 1	1	0	1	
## 30	1	0	0	
## 38	1	0	0	
## 1	0	1	1	
## 1	0	1	1	
## 48	0	1	0	

## 8	0	1	0
## 32	0	0	0
## 29	0	0	0
## 14	1	1	0
## 1	1	1	0
## 7	1	0	0
## 12	1	1	1
## 2	1	1	0
## 2	1	1	0
## 1	1	0	0
## 4	1	1	1
## 1	1	1	1
##	119	149	217
##	Departure.Arrival.time.convenient		
## 4590		1 0	
## 162		0 1	
## 5		0 2	
## 11		1 1	
## 1		0 2	
## 30		1 2	
## 38		0 3	
## 1		1 1	
## 1		0 2	
## 48		1 2	
## 8		0 3	
## 32		1 3	
## 29		0 4	
## 14		1 2	
## 1		0 3	
## 7		1 3	
## 12		1 2	
## 2		1 2	
## 2		0 3	
## 1		1 3	
## 4		1 1	
## 1		1 3	
##		247 790	

2.3.1.2 Outliers In this section, we analyze the presence of outliers in the dataset, focusing on the four numerical variables: Age, Flight.Distance, Departure.Delay.in.Minutes, and Arrival.Delay.in.Minutes.

Age: No outliers were detected in the Age variable, indicating that all recorded values fall within a reasonable and expected range for this dataset.

Flight.Distance: A total of 95 mild outliers were identified in Flight.Distance, but no severe outliers were detected. This indicates that some flights are moderately longer than the majority, which is expected given variations in airline routes. These values are not concerning and reflect natural variability.

Departure.Delay.in.Minutes: Shows 675 mild outliers and 413 severe outliers, indicating that a significant number of flights experienced delays longer than the typical range.

Arrival.Delay.in.Minutes: Similarly, Arrival.Delay.in.Minutes has 669 mild outliers and 410 severe outliers, closely mirroring the outliers in Departure.Delay.in.Minutes. This similarity is expected, as the arrival delay is inherently related to the departure delay. A delayed departure often leads to a delayed arrival unless measures such as flight rescheduling or speed adjustments are taken.

2.3.2 Individuals

This second part of the data quality report assesses the number of outliers and missing values per individual and identifies which individuals are multivariate outliers.

2.3.2.1 Missing values and Errors Out of the total observations in the dataset, 4590 have no missing values across any variable. Additionally, 178 observations contain a missing value in one variable, 113 observations have missing values in two variables, 90 observations have missing values in three variables, and 29 individuals have missing values in 4 variables

```
na_counts <- rowSums(is.na(train))
table(na_counts)
```

```
## na_counts
##    0     1     2     3     4
## 4590  178  113   90   29
```

2.3.2.2 Outliers Since there are only four numerical variables in the dataset, and the Age variable does not have any outliers, the maximum number of outliers an observation can have is three. Based on the analysis, the distribution of outliers across observations is as follows: - 4140 observations have no outliers at all. - 297 observations have one outlier. - 547 observations have two outliers, and notably, 545 of these are outliers in both Departure.Delay.in.Minutes and Arrival.Delay.in.Minutes. This overlap is expected, as these variables are closely related and delays in departure often propagate to arrival delays. - 13 observations have outliers in all three variables.

This analysis highlights that most observations are free of outliers, while the relationship between Departure.Delay.in.Minutes and Arrival.Delay.in.Minutes contributes significantly to the overlapping outliers.

```
table(all_outliers)
```

```
## all_outliers
##    0     1     2
## 4905   92    3
```

```
common_indexes <- intersect(outliers_indexes_arr_delay,
                             outliers_indexes_dep_delay)
length(common_indexes) # 558
```

```
## [1] 558
```

2.3.2.3 Multivariate Outliers The *find_valid_moutlier* function identifies the largest combination of numeric variables without missing values that successfully executes the Moutlier function. By iterating through combinations of variables, starting with the largest possible subsets, this function ensures that the process for detecting multivariate outliers incorporates the maximum amount of information available.

Since the Moutlier function depends on some stochastic processes, one can get different results when executing it several times. Thus, we comment the results obtained in one single execution, which has been selected carefully to represent the average values.

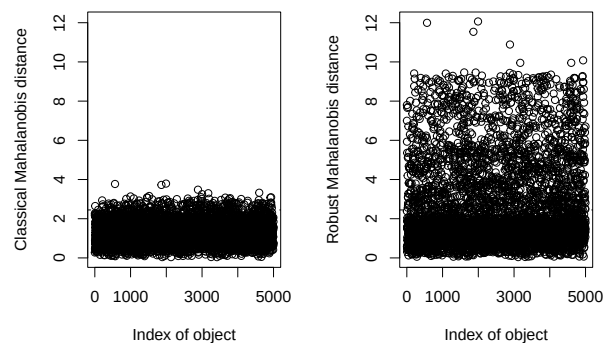
There are 213 multivariate outliers, which we proceed to characterize in the next part.

```
result <- find_valid_moutlier(train, quantile = 0.95)
```

```
## Total numeric variables without NAs: 3
## Testing combinations of size 3
## Total combinations of size 3 : 1
```

```
## Warning in covMcd(X): The 2502-th order statistic of the absolute deviation of variable 3 is
## zero.
## There are 2830 observations (in the entire dataset of 5000 obs.) lying
## on the plane with equation 0 (x_i1-m_1) + 0 (x_i2-m_2) + 1 (x_i3-m_3) =
## 0 with (m_1,m_2) the mean of these observations.
```

```
## Combination failed.
## Testing combinations of size 2
## Total combinations of size 2 : 3
```



```
## Valid combination found! Indexes: 3, 6
## Variable names: Age, Flight.Distance
## Number of multivariate outliers: 213
```

```
res.out <- result$res_mout;
mult.outliers <- result$multoutliers;
```

```
#Print number of outliers
length(mult.outliers)
```

```
## [1] 213
```

```
# Plot Multivariate Outliers
dev.off()
```

```
## null device
##          1
```

```
plot(res.out$md, res.out$rd)
abline(h=res.out$cutoff, col="red", lty=2)
abline(v=res.out$cutoff, col="red", lty=2)
```

Finally, we present the multivariate outlier profiling:

The chi-square test results suggest significant relationships between the multivariate outliers and several categorical variables, such as *Class*, *Type of Travel*, *Satisfaction*, *Customer Type*, *Flight Distance*, and *Seat Comfort*. Multivariate outliers are primarily associated with specific travel class types, such as “Business” and “Eco Plus,” as well as travel categories like “Business Travel” and “Personal Travel.” Additionally, multivariate outliers tend to have distinct satisfaction levels, with the “satisfied” and “neutral or dissatisfied” categories being most common. There are also strong links to online boarding status and comfort ratings, which help separate outlier from non-outlier categories.

From the analysis of quantitative variables, we see strong associations with Flight Distance and Age. Outliers are more likely to be associated with significantly higher or lower flight distances and age groups, with long-distance flights and older travelers being characteristic of the outlier group. Eta-squared values indicate that Flight Distance has the highest effect size (0.248), which suggests it is a key determinant in distinguishing multivariate outliers. Age, with a smaller effect size (0.0057), also plays a role, though its influence is less pronounced.

Profiling the outliers by cluster, we observe distinct patterns. For the “Not M.O.” (Non-Outlier) group, the mean values for Age and Flight Distance are relatively close to the overall means. In contrast, the “Yes M.O.” (Outlier) group shows significant deviations in both variables, with outliers being characterized by extremely high or low Flight Distance values and older Age profiles.

```
bi_mvout_df <- train
#Binary factor if its a multivariate outlier
bi_mvout_df$mout <- 0
bi_mvout_df[mult.outliers, "mout"] <- 1
bi_mvout_df$mout <- factor(bi_mvout_df$mout, labels=c("Not M.O", "Yes M.O"))
table(bi_mvout_df$mout)
```

```
##
## Not M.O Yes M.O
##      4787      213
```

```
summary(bi_mvout_df[, c("mout")]);
```

```
## Not M.O Yes M.O
##      4787      213
```

```
catdes(bi_mvout_df, num.var=28, proba=0.05)
```

```
##
## Link between the cluster variable and the categorical variables (chi-square test)
## =====
##
##                p.value df
## f.FlightDist    5.457652e-103  3
## Class           8.810722e-45  2
## Type.of.Travel   2.182700e-21  1
## satisfaction     9.780713e-17  1
## Customer.Type    4.092893e-10  1
```

```

## Online.boarding      3.242950e-05  5
## AgeGroup            1.101448e-03  3
## Seat.comfort        1.324376e-02  4
## Inflight.entertainment 3.998201e-02  5
##
## Description of each cluster by the categories
## =====
## $'Not M.O'
##
## Cla/Mod  Mod/Cla Global      p.value
## Class=Eco 99.69271 47.440986 45.56 2.103765e-46
## f.FlightDist=Medium Distance 99.93990 34.739921 33.28 4.117743e-37
## Type.of.Travel=Personal Travel 99.80519 32.107792 30.80 1.956186e-30
## f.FlightDist=Short Distance 99.63986 34.677251 33.32 9.577189e-30
## satisfaction=neutral or dissatisfied 97.79489 58.366409 57.14 1.200679e-16
## Customer.Type=disloyal Customer 99.46467 19.406727 18.68 8.912262e-14
## Class=Eco Plus 98.89197 7.457698 7.22 4.608857e-04
## Online.boarding=1 98.13433 10.988093 10.72 1.599517e-03
## Online.boarding=NA 100.00000 2.485899 2.38 5.278809e-03
## Online.boarding=2 97.42331 16.586589 16.30 6.406030e-03
## Seat.comfort=1 97.60766 12.784625 12.54 8.891521e-03
## Food.and.drink=1 97.45889 13.620221 13.38 1.308200e-02
## Inflight.entertainment=1 97.44409 12.742845 12.52 1.803467e-02
## Inflight.entertainment=2 97.05160 16.503029 16.28 3.694813e-02
## Inflight.service=2 94.00749 10.486735 10.68 4.445564e-02
## Departure.Arrival.time.convenient=1 94.28191 14.810946 15.04 3.798898e-02
## Inflight.wifi.service=NA 91.94631 2.861918 2.98 3.471342e-02
## Online.boarding=5 94.38669 18.968038 19.24 2.491566e-02
## Seat.comfort=4 94.56662 30.541049 30.92 7.081685e-03
## Online.boarding=4 94.53836 30.373929 30.76 6.035491e-03
## AgeGroup=Over 60 92.12598 7.332358 7.62 9.419008e-04
## f.FlightDist=NA 0.00000 0.000000 0.08 3.205225e-06
## Customer.Type=Loyal Customer 94.88441 80.593273 81.32 8.912262e-14
## satisfaction=satisfied 93.00047 41.633591 42.86 1.200679e-16
## Type.of.Travel=Business travel 93.93064 67.892208 69.20 1.956186e-30
## Class=Business 91.44430 45.101316 47.22 1.553273e-53
## f.FlightDist=Long Distance 87.87515 30.582828 33.32 1.073960e-84
##
## v.test
## Class=Eco 14.302735
## f.FlightDist=Medium Distance 12.728287
## Type.of.Travel=Personal Travel 11.465942
## f.FlightDist=Short Distance 11.327620
## satisfaction=neutral or dissatisfied 8.283042
## Customer.Type=disloyal Customer 7.456098
## Class=Eco Plus 3.502516
## Online.boarding=1 3.155995
## Online.boarding=NA 2.789509
## Online.boarding=2 2.726240
## Seat.comfort=1 2.616197
## Food.and.drink=1 2.481529
## Inflight.entertainment=1 2.364906
## Inflight.entertainment=2 2.086337
## Inflight.service=2 -2.009769
## Departure.Arrival.time.convenient=1 -2.074974
## Inflight.wifi.service=NA -2.111686

```

```

## Online.boarding=5 -2.242708
## Seat.comfort=4 -2.692978
## Online.boarding=4 -2.745847
## AgeGroup=Over 60 -3.307329
## f.FlightDist=NA -4.657211
## Customer.Type=Loyal Customer -7.456098
## satisfaction=satisfied -8.283042
## Type.of.Travel=Business travel -11.465942
## Class=Business -15.403383
## f.FlightDist=Long Distance -19.501124
##
## $'Yes M.O'
## Cla/Mod Mod/Cla Global
## f.FlightDist=Long Distance 12.12484994 94.8356808 33.32
## Class=Business 8.55569674 94.8356808 47.22
## Type.of.Travel=Business travel 6.06936416 98.5915493 69.20
## satisfaction=satisfied 6.99953336 70.4225352 42.86
## Customer.Type=Loyal Customer 5.11559272 97.6525822 81.32
## f.FlightDist=NA 100.00000000 1.8779343 0.08
## AgeGroup=Over 60 7.87401575 14.0845070 7.62
## Online.boarding=4 5.46163849 39.4366197 30.76
## Seat.comfort=4 5.43337646 39.4366197 30.92
## Online.boarding=5 5.61330561 25.3521127 19.24
## Inflight.wifi.service=NA 8.05369128 5.6338028 2.98
## Departure.Arrival.time.convenient=1 5.71808511 20.1877934 15.04
## Inflight.service=2 5.99250936 15.0234742 10.68
## Inflight.entertainment=2 2.94840295 11.2676056 16.28
## Inflight.entertainment=1 2.55591054 7.5117371 12.52
## Food.and.drink=1 2.54110613 7.9812207 13.38
## Seat.comfort=1 2.39234450 7.0422535 12.54
## Online.boarding=2 2.57668712 9.8591549 16.30
## Online.boarding=NA 0.00000000 0.0000000 2.38
## Online.boarding=1 1.86567164 4.6948357 10.72
## Class=Eco Plus 1.10803324 1.8779343 7.22
## Customer.Type=disloyal Customer 0.53533191 2.3474178 18.68
## satisfaction=neutral or dissatisfied 2.20511026 29.5774648 57.14
## f.FlightDist=Short Distance 0.36014406 2.8169014 33.32
## Type.of.Travel=Personal Travel 0.19480519 1.4084507 30.80
## f.FlightDist=Medium Distance 0.06009615 0.4694836 33.28
## Class=Eco 0.30728709 3.2863850 45.56
## p.value v.test
## f.FlightDist=Long Distance 1.073960e-84 19.501124
## Class=Business 1.553273e-53 15.403383
## Type.of.Travel=Business travel 1.956186e-30 11.465942
## satisfaction=satisfied 1.200679e-16 8.283042
## Customer.Type=Loyal Customer 8.912262e-14 7.456098
## f.FlightDist=NA 3.205225e-06 4.657211
## AgeGroup=Over 60 9.419008e-04 3.307329
## Online.boarding=4 6.035491e-03 2.745847
## Seat.comfort=4 7.081685e-03 2.692978
## Online.boarding=5 2.491566e-02 2.242708
## Inflight.wifi.service=NA 3.471342e-02 2.111686
## Departure.Arrival.time.convenient=1 3.798898e-02 2.074974
## Inflight.service=2 4.445564e-02 2.009769

```

```

## Inflight.entertainment=2          3.694813e-02 -2.086337
## Inflight.entertainment=1          1.803467e-02 -2.364906
## Food.and.drink=1                  1.308200e-02 -2.481529
## Seat.comfort=1                    8.891521e-03 -2.616197
## Online.boarding=2                  6.406030e-03 -2.726240
## Online.boarding=NA                 5.278809e-03 -2.789509
## Online.boarding=1                  1.599517e-03 -3.155995
## Class=Eco Plus                    4.608857e-04 -3.502516
## Customer.Type=disloyal Customer   8.912262e-14 -7.456098
## satisfaction=neutral or dissatisfied 1.200679e-16 -8.283042
## f.FlightDist=Short Distance        9.577189e-30 -11.327620
## Type.of.Travel=Personal Travel     1.956186e-30 -11.465942
## f.FlightDist=Medium Distance       4.117743e-37 -12.728287
## Class=Eco                         2.103765e-46 -14.302735
##
##
## Link between the cluster variable and the quantitative variables
## =====
##                               Eta2      P-value
## Flight.Distance 0.247963988 1.189649e-311
## Age              0.005686402 9.375160e-08
##
## Description of each cluster by quantitative variables
## =====
## $'Not M.O'
##               v.test Mean in category Overall mean sd in category
## Age           -5.331634          39.17527      39.4126    14.77567
## Flight.Distance -35.207556      1089.25360    1194.2054    872.29345
##               Overall sd      p.value
## Age           14.92047 9.733273e-08
## Flight.Distance 999.16562 1.532217e-271
##
## $'Yes M.O'
##               v.test Mean in category Overall mean sd in category
## Flight.Distance 35.207556      3552.91080    1194.2054    723.51316
## Age              5.331634       44.74648      39.4126    17.01591
##               Overall sd      p.value
## Flight.Distance 999.16562 1.532217e-271
## Age           14.92047 9.733273e-08

```

At the moment, we do not remove any observation. However, this are potential influential data for the model that we build as follows, so we will assess carefully which data biases too much the model and see if it is wise to remove it from the data set after computing each model version.

2.4 Imputation

In this section, all the missing values previously identified are addressed through an imputation process using mice function.

Note that values are imputed both in the training and test sets, as stated in the statement of the project.

```
summary(train)
```

```

##      Gender      Customer.Type      Age      Type.of.Travel
## Female:2548  disloyal Customer: 934  Min.   : 7.00  Business travel:3460
## Male  :2452  Loyal Customer   :4066  1st Qu.:27.00  Personal Travel:1540
##                                           Median :40.00
##                                           Mean   :39.41
##                                           3rd Qu.:51.00
##                                           Max.   :85.00
##
##      Class      Flight.Distance  Inflight.wifi.service
## Business:2361  Min.   : 67      1 : 878
## Eco          :2278  1st Qu.: 414    2 :1209
## Eco Plus: 361  Median : 842    3 :1242
##                                           Mean   :1194    4 : 974
##                                           3rd Qu.:1754    5 : 548
##                                           Max.   :4983    NA's: 149
##
## Departure.Arrival.time.convenient  Ease.of.Online.booking  Gate.location
## 1 : 752                             1 : 851                 1: 837
## 2 : 801                             2 :1093                 2: 909
## 3 : 879                             3 :1203                 3:1380
## 4 :1250                             4 : 970                 4:1188
## 5 :1071                             5 : 666                 5: 686
## NA's: 247                          NA's: 217
##
## Food.and.drink  Online.boarding  Seat.comfort  Inflight.entertainment
## 1 : 669          1 : 536          1: 627        1 : 626
## 2 : 985          2 : 815          2: 704        2 : 814
## 3 :1106          3 :1030          3: 879        3 : 949
## 4 :1186          4 :1538          4:1546        4 :1438
## 5 :1049          5 : 962          5:1244        5 :1172
## NA's: 5          NA's: 119          NA's: 1
##
## On.board.service  Leg.room.service  Baggage.handling  Checkin.service
## 1 : 556           1 : 502           1: 359           1: 649
## 2 : 713           2 : 961           2: 542           2: 606
## 3 :1129           3 : 972           3:1011           3:1400
## 4 :1535           4 :1350           4:1811           4:1419
## 5 :1066           5 :1193           5:1277           5: 926
## NA's: 1           NA's: 22
##
## Inflight.service  Cleanliness  Departure.Delay.in.Minutes
## 1 : 355            1: 659      Min.   : 0.00
## 2 : 534            2: 756      1st Qu.: 0.00
## 3 :1006            3:1207      Median : 0.00
## 4 :1874            4:1336      Mean   : 15.08
## 5 :1230            5:1042      3rd Qu.: 13.00
## NA's: 1            Max.   :692.00
##
## Arrival.Delay.in.Minutes      satisfaction      AgeGroup
## Min.   : 0.00      neutral or dissatisfied:2857  Under 18: 392
## 1st Qu.: 0.00      satisfied :2143      18-35 :1602
## Median : 0.00      Over 60 :2625
## Mean   : 15.28
## 3rd Qu.: 13.00

```

```
## Max. :702.00
## NA's :12
## f.FlightDist DelayedArrival DelayedDeparture
## Short Distance :1666 DelayedArrival:2206 DelayedDeparture:2170
## Medium Distance:1664 OK :2782 OK :2830
## Long Distance :1666 NA's : 12
## NA's : 4
##
##
##
```

```
imputed_data <- mice(train)
```

```
##
## iter imp variable
## 1 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
```

```
df_complete <- complete(imputed_data)
summary(df_complete)
```

```
## Gender Customer.Type Age Type.of.Travel
## Female:2548 disloyal Customer: 934 Min. : 7.00 Business travel:3460
## Male :2452 Loyal Customer :4066 1st Qu.:27.00 Personal Travel:1540
## Median :40.00
## Mean :39.41
## 3rd Qu.:51.00
## Max. :85.00
## Class Flight.Distance Inflight.wifi.service
## Business:2361 Min. : 67 1: 894
```

```

## Eco      :2278   1st Qu.: 414   2:1238
## Eco Plus: 361   Median : 842   3:1269
##           Mean   :1194   4:1017
##           3rd Qu.:1754   5: 582
##           Max.    :4983
## Departure.Arrival.time.convenient Ease.of.Online.booking Gate.location
## 1: 784                      1: 887                      1: 837
## 2: 853                      2:1137                      2: 909
## 3: 931                      3:1252                      3:1380
## 4:1322                      4:1023                      4:1188
## 5:1110                      5: 701                      5: 686
##
## Food.and.drink Online.boarding Seat.comfort Inflight.entertainment
## 1: 671          1: 556          1: 627          1: 626
## 2: 985          2: 827          2: 704          2: 814
## 3:1109         3:1062         3: 879          3: 950
## 4:1186         4:1569         4:1546         4:1438
## 5:1049         5: 986          5:1244         5:1172
##
## On.board.service Leg.room.service Baggage.handling Checkin.service
## 1: 557          1: 506          1: 359          1: 649
## 2: 713          2: 969          2: 542          2: 606
## 3:1129         3: 976          3:1011         3:1400
## 4:1535         4:1353         4:1811         4:1419
## 5:1066         5:1196         5:1277         5: 926
##
## Inflight.service Cleanliness Departure.Delay.in.Minutes
## 1: 355          1: 659          Min.   : 0.00
## 2: 535          2: 756          1st Qu.: 0.00
## 3:1006         3:1207          Median : 0.00
## 4:1874         4:1336          Mean   : 15.08
## 5:1230         5:1042          3rd Qu.: 13.00
##           Max.    :692.00
##
## Arrival.Delay.in.Minutes      satisfaction      AgeGroup
## Min.   : 0.00      neutral or dissatisfied:2857  Under 18: 392
## 1st Qu.: 0.00      satisfied      :2143    18-35   :1602
## Median : 0.00                                36-60   :2625
## Mean   : 15.37                                Over 60  : 381
## 3rd Qu.: 13.00
## Max.    :702.00
##
## f.FlightDist      DelayedArrival      DelayedDeparture
## Short Distance :1666 DelayedArrival:2216 DelayedDeparture:2170
## Medium Distance:1664 OK          :2784 OK          :2830
## Long Distance  :1670
##
##
##

```

Here, we check the correctness of imputed values.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:MASS':
##
##      select

## The following object is masked from 'package:car':
##
##      recode

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
```

```
# Identify columns with NAs
cols_with_na <- names(train)[sapply(train, function(x) any(is.na(x)))]
numeric_cols_with_na <- cols_with_na[sapply(train[cols_with_na], is.numeric)]

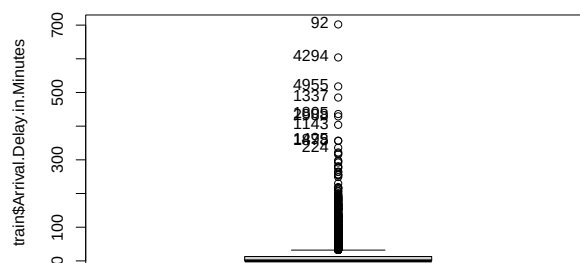
# Numeric variables
quantiles_original <- quantile(train$Arrival.Delay.in.Minutes,
                                probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_original
```

```
##      0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
##      0     0     0     0     0     0     2     9    20    44   702
```

```
quantiles_imputed <- quantile(df_complete$Arrival.Delay.in.Minutes,
                              probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imputed
```

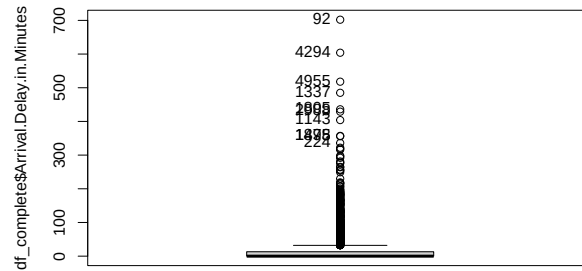
```
##      0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
##      0     0     0     0     0     0     3     9    20    44   702
```

```
Boxplot(train$Arrival.Delay.in.Minutes)
```



```
## [1] 92 4294 4955 1337 1805 2988 1143 1495 1878 224
```

```
Boxplot(df_complete$Arrival.Delay.in.Minutes)
```



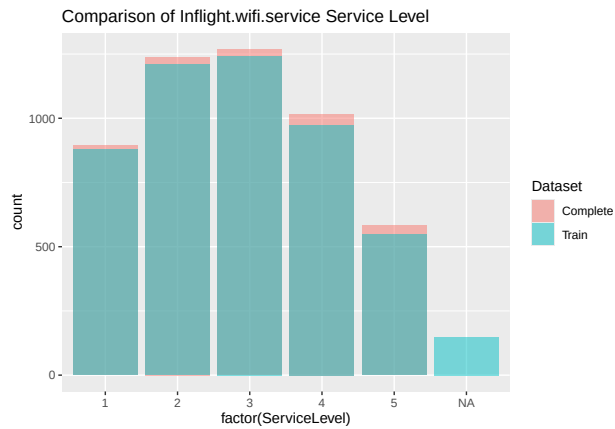
```
## [1] 92 4294 4955 1337 1805 2988 1143 1495 1878 224
```

```
#Categorical variables
categorical <- cols_with_na[!sapply(train[cols_with_na], is.numeric)]

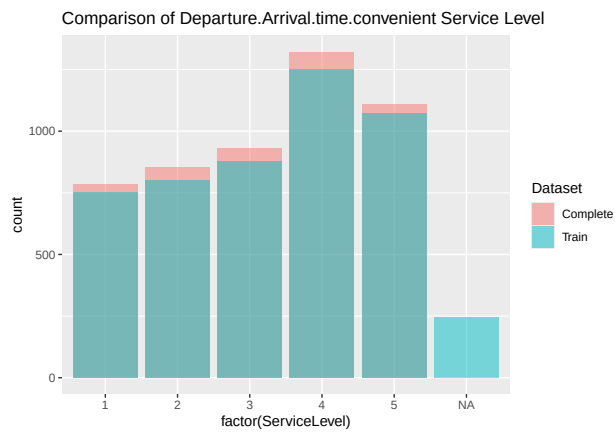
# Loop through each variable
for (var in categorical) {
  # Print the table comparison for both datasets
  print(list(Train = table(train[[var]]), Complete = table(df_complete[[var]])))

  # Create and print the superimposed bar plot
  print(
    ggplot(rbind(data.frame(ServiceLevel = train[[var]], Dataset = "Train"),
                  data.frame(ServiceLevel = df_complete[[var]],
                             Dataset = "Complete")),
            aes(x = factor(ServiceLevel), fill = Dataset)) +
    geom_bar(position = "identity", alpha = 0.5) +
    labs(title = paste("Comparison of", var, "Service Level"))
  )
}
```

```
## $Train
##
## 1 2 3 4 5
## 878 1209 1242 974 548
##
## $Complete
##
## 1 2 3 4 5
## 894 1238 1269 1017 582
```



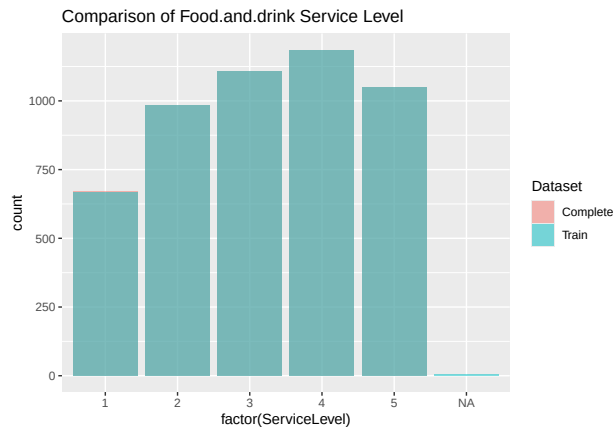
```
## $Train
##
##      1      2      3      4      5
## 752  801  879 1250 1071
##
## $Complete
##
##      1      2      3      4      5
## 784  853  931 1322 1110
```



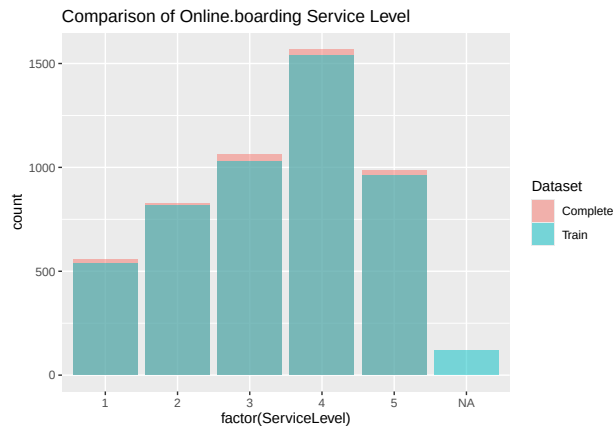
```
## $Train
##
##      1      2      3      4      5
## 851 1093 1203  970  666
##
## $Complete
##
##      1      2      3      4      5
## 887 1137 1252 1023  701
```



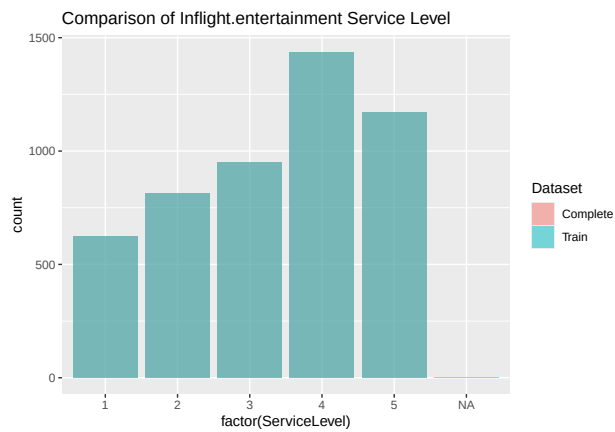
```
## $Train
##
##      1      2      3      4      5
## 669  985 1106 1186 1049
##
## $Complete
##
##      1      2      3      4      5
## 671  985 1109 1186 1049
```



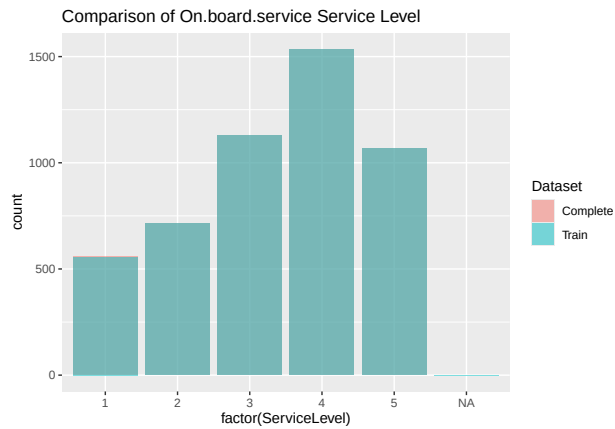
```
## $Train
##
##      1      2      3      4      5
## 536  815 1030 1538  962
##
## $Complete
##
##      1      2      3      4      5
## 556  827 1062 1569  986
```



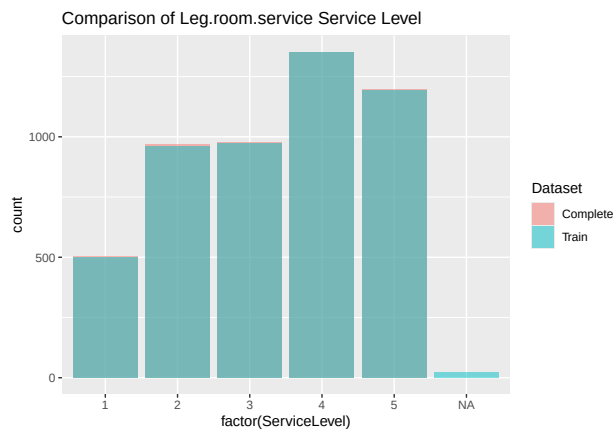
```
## $Train
##
##      1      2      3      4      5
## 626  814  949 1438 1172
##
## $Complete
##
##      1      2      3      4      5
## 626  814  950 1438 1172
```



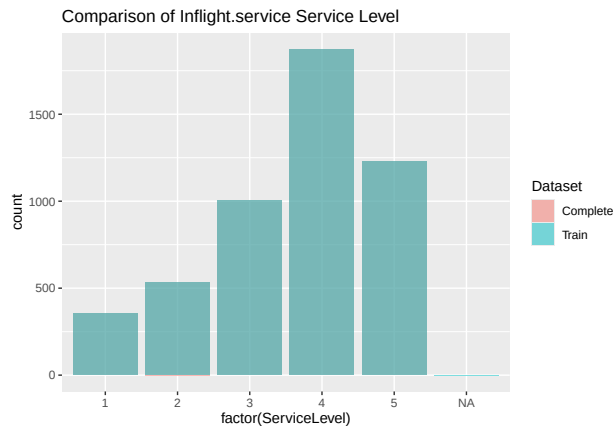
```
## $Train
##
##      1      2      3      4      5
## 556  713 1129 1535 1066
##
## $Complete
##
##      1      2      3      4      5
## 557  713 1129 1535 1066
```



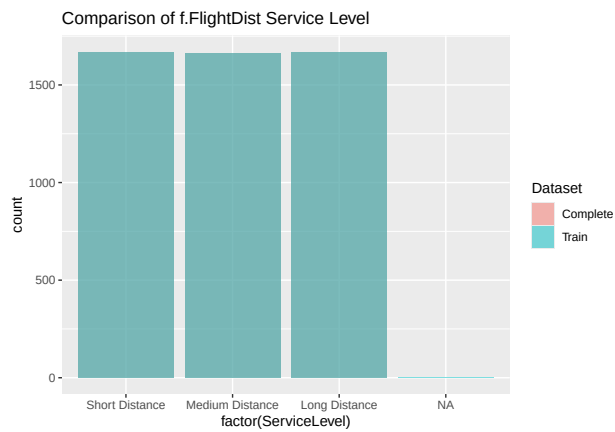
```
## $Train
##
##      1      2      3      4      5
## 502  961  972 1350 1193
##
## $Complete
##
##      1      2      3      4      5
## 506  969  976 1353 1196
```



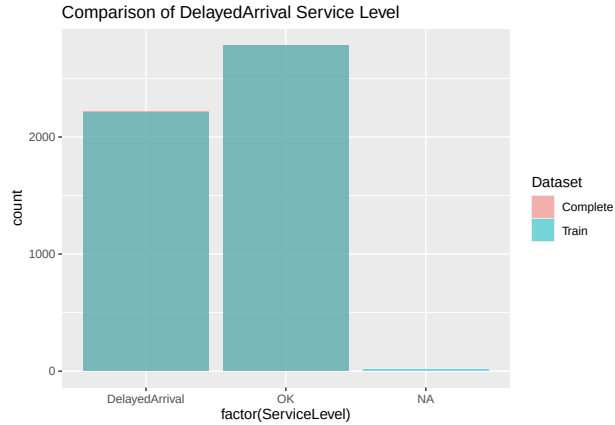
```
## $Train
##
##      1      2      3      4      5
## 355  534 1006 1874 1230
##
## $Complete
##
##      1      2      3      4      5
## 355  535 1006 1874 1230
```



```
## $Train
##
## Short Distance Medium Distance Long Distance
##      1666      1664      1666
##
## $Complete
##
## Short Distance Medium Distance Long Distance
##      1666      1664      1670
```



```
## $Train
##
## DelayedArrival OK
##      2206      2782
##
## $Complete
##
## DelayedArrival OK
##      2216      2784
```



Now that we have checked the validity of the imputed values, we assign them to the main data frame and impute values in the test set.

```
# Impute values to the original df
train <- df_complete

# Impute test data
imputed_data_test <- mice(test)
```

```
##
## iter imp variable
## 1 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 1 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 2 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 3 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 4 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 1 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 2 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 3 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 4 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
## 5 5 Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking Food.and.
```

```
df_complete_test <- complete(imputed_data_test)
test <- df_complete_test
```

2.5 Profiling analysis and correlations

The target binary variable **satisfaction** splits the data unevenly: 57.14% neutral or dissatisfied vs 42.86% satisfied. Moreover this variable contains no null values at all. To delve deeper into the main differences between these 2 groups, a clustering profiling will be performed. Just as a remainder, the dataset contains 18 categorical variables (14 ordinal, 3 binary, 1 ternary) and 4 numerical, besides the target variable.

Firstly, the **test.chi2** output points out the fact that almost all categorical variables are statistically significantly related to satisfaction, except for **Gender**. The statistical significance of the performed χ^2 tests is huge for all variables except for **Departure.Arrival.time.convenient**, which has the lowest p-value of them all, $1.53 \cdot 10^{-2}$.

Secondly, the **quanti.var** output points out that all the numerical variables of the dataset (Flight.Distance, Age, Arrival.Delay.in.Minutes and Departure.Delay.in.Minutes) are statistically significant after performing Anova/T-test, being Flight.Distance and Age the most significant variables.

Finally, taking a glance over **quanti** output we can gather numerical insights on the main difference between the 2 groups after performing numerous v-tests: - Those belonging to the “satisfied” group present a slightly above age, below average delays in departure and arrival and a significantly bigger Flight.Distance, compared to the overall averages. - Those belonging to the “neutral or dissatisfied” in contrast have a below average age of 37 and slightly above average delays in both departure and arrival but significantly below average Flight.Distance, compared to the overall averages.

```
res_cat <- catdes(train, proba=0.05, num.var=23)

#Find categorical variables that are significantly related to satisfaction
#Chi^2 test
res_cat$test.chi2
```

##	p.value	df
## Online.boarding	0.000000e+00	4
## Class	3.762928e-290	2
## Inflight.wifi.service	3.507407e-260	4
## Type.of.Travel	3.141061e-216	1
## Inflight.entertainment	3.213696e-189	4
## Seat.comfort	2.391451e-160	4
## On.board.service	2.620052e-123	4
## Leg.room.service	6.279765e-120	4
## Ease.of.Online.booking	6.818395e-103	4
## Cleanliness	1.661592e-102	4
## f.FlightDist	3.534811e-98	2
## AgeGroup	1.639188e-94	3
## Baggage.handling	4.589485e-87	4
## Inflight.service	8.651761e-83	4
## Checkin.service	1.310091e-59	4
## Food.and.drink	9.138131e-49	4
## Customer.Type	1.626909e-46	1
## Gate.location	2.329562e-25	4
## DelayedArrival	5.939692e-11	1
## DelayedDeparture	2.768182e-04	1
## Departure.Arrival.time.convenient	4.150319e-02	4

```
#Find numerical variables that are significantly related to satisfaction
# Anova/T test
res_cat$quanti.var
```

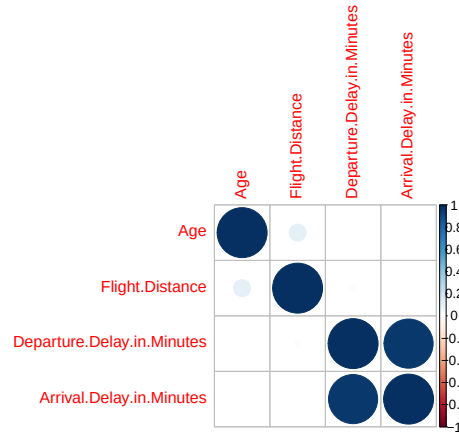
```
##                               Eta2      P-value
## Flight.Distance              0.090411953 5.327880e-105
## Age                          0.021854338 7.895448e-26
## Arrival.Delay.in.Minutes     0.001595345 4.732040e-03
## Departure.Delay.in.Minutes   0.001095035 1.928555e-02
```

```
#Compare the 2 clusters assessing numerical differences ---> v test
res_cat$quanti
```

```
## $'neutral or dissatisfied'
##                               v.test Mean in category Overall mean
## Arrival.Delay.in.Minutes      2.824027      16.72209      15.3730
## Departure.Delay.in.Minutes     2.339675      16.18306      15.0810
## Age                          -10.452265      37.50228      39.4126
## Flight.Distance               -21.259571     934.00595     1194.2054
##                               sd in category Overall sd      p.value
## Arrival.Delay.in.Minutes       41.50781     38.99927 4.742438e-03
## Departure.Delay.in.Minutes      40.88688     38.45342 1.930052e-02
## Age                           16.34893     14.92047 1.430676e-25
## Flight.Distance                809.05758    999.16562 2.688635e-100
##
## $satisfied
##                               v.test Mean in category Overall mean
## Flight.Distance               21.259571     1541.09753     1194.2054
## Age                          10.452265      41.95940      39.4126
## Departure.Delay.in.Minutes    -2.339675      13.61176      15.0810
## Arrival.Delay.in.Minutes     -2.824027      13.57443      15.3730
##                               sd in category Overall sd      p.value
## Flight.Distance              1116.25574    999.16562 2.688635e-100
## Age                          12.31746     14.92047 1.430676e-25
## Departure.Delay.in.Minutes     34.89252     38.45342 1.930052e-02
## Arrival.Delay.in.Minutes      35.29927     38.99927 4.742438e-03
```

Now, Looking at the relationship between numerical variables, an obvious perfect positive correlation is divided: Departure delay and Arrival delay. Other variables have negligible relations and variables Age and Flight.distance share a small positive correlation.

```
corrplot(cor(train[, num_vars]))
```



3. Modeling

First, note that we treat the model Nominally, as the response variable *Satisfaction* have no inherent order.

3.1 Modeling with numeric variables & Residual Analysis

To see which numeric variables provide value to the model, we first create the null model as m0.

In the case of the null model, the intercept estimate is $\hat{\beta}_0 = -0.28757$. This means that the log-odds of satisfaction when no explanatory variables are included is -0.28757. The corresponding probability $P(\text{satisfaction} = 1)$ can be calculated by converting the log-odds to a probability:

$$P(\text{satisfaction} = 1) = \frac{1}{1 + e^{-\hat{\beta}_0}} = \frac{1}{1 + e^{0.28757}} \approx 0.428$$

The AIC (Akaike Information Criterion) value is 6831.2.

```
# Set satisfaction to 0 or 1
train$satisfaction <- ifelse(train$satisfaction == "satisfied", 1, 0)
test$satisfaction <- ifelse(test$satisfaction == "satisfied", 1, 0)
```

```
# Verify the transformation
summary(as.factor(train$satisfaction))
```

```
##      0      1
## 2857 2143
```

```
summary(as.factor(test$satisfaction))
```

```
##      0      1
## 1075  925
```

```
#Null model
m0 <- glm(satisfaction ~ 1, family=binomial, data=train)
summary(m0)
```

```
##
## Call:
## glm(formula = satisfaction ~ 1, family = binomial, data = train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.28757    0.02858  -10.06  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6829.2  on 4999  degrees of freedom
## AIC: 6831.2
##
## Number of Fisher Scoring iterations: 4
```

```
AIC(m0)
```

```
## [1] 6831.163
```

Now, we aim to develop the best numerical variable model, balancing both model fit and complexity. It is clear that the *Age* variable contributes meaningfully to the model. Upon inspecting the residuals plot, we observe that the residuals do not exhibit a flat pattern, suggesting that a transformation of the variable might improve the model. Squaring the variable resolves this issue effectively.

On the other hand *Flight.Distance* also contributes meaningfully to the model. Incorporating the variable to the model it is clearly worth it, since we obtain better AIC and Residual deviance values and the residuals plot keeps showing a flat line. Transforming the variable is not worth, so we keep it plane.

After incorporating these two variables, we evaluate the potential addition of *Departure.Delay.in.Minutes* and *Arrival.Delay.in.Minutes*.

Following the AIC criteria, including *Departure.Delay.in.Minutes* to the existing model it is not worth it. Moreover, the Anova function (which provides insight into the net effect of adding variables to the model), shows that, at a 95% confidence level, including *Departure.Delay.in.Minutes* after *Arrival.Delay.in.Minutes* does not significantly improve the model.

Therefore, the optimal numerical variable model includes the variables *Age*, *Flight.Distance*, and *Arrival.Delay.in.Minutes*. The residuals vs fitted plot shows a flat line.

The model is :

$$Y = -0.9981 + (22.67)\text{poly}(\text{Age}, 2)_1 - (36.96)\text{poly}(\text{Age}, 2)_2 \\ + (5.877e - 04)\text{Flight.Distance} - (3.163e - 03)\text{Arrival.Delay.in.Minutes}$$

The model's fit statistics are as follows:

- **AIC:** 6064.3
- **Residual deviance:** 6054.3 (on 4995 degrees of freedom)

Additionally, the Variance Inflation Factor (VIF) analysis confirms the absence of significant multicollinearity among the predictor variables.

Residual and marginal model plots validate that all included variables are appropriate for the model.

After finalizing the model, we assessed influence in the glm models. By removing the severe outliers in the cooks distance metric, we remove the most influential points in the model. In the final model, we might repeat the process iteratively to remove all influential points.

```
#Numerical variables
numeric_vars <- names(which(sapply(train, is.numeric)==TRUE));numeric_vars
```

```
## [1] "Age" "Flight.Distance"
## [3] "Departure.Delay.in.Minutes" "Arrival.Delay.in.Minutes"
## [5] "satisfaction"
```

```
par(mfrow=c(2,2))
```

```
#####
#Including Age to the model#
#####
```

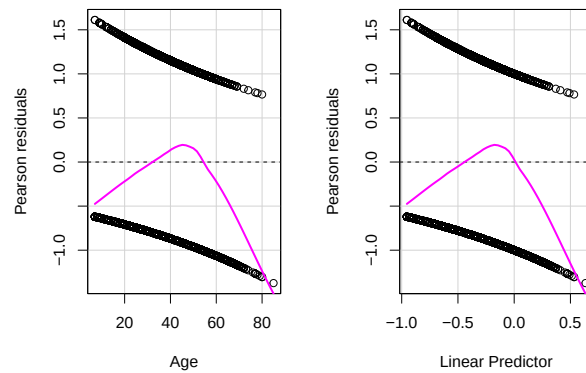
```
m1 <- glm(satisfaction ~ Age, family=binomial, data=train)
summary(m1)
```

```
##
## Call:
## glm(formula = satisfaction ~ Age, family = binomial, data = train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.096707   0.083822  -13.08  <2e-16 ***
## Age          0.020365   0.001965   10.36  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6719.0  on 4998  degrees of freedom
## AIC: 6723
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m1, test= "LR")
```

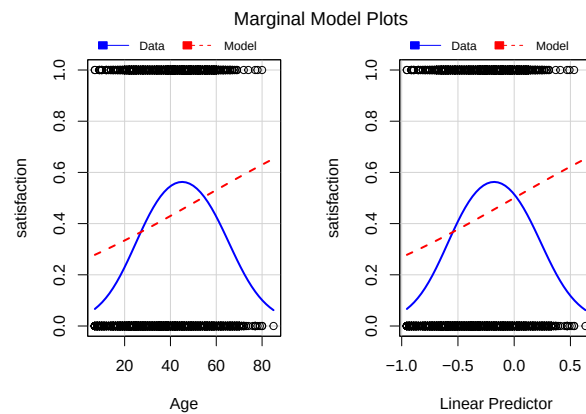
```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##      LR Chisq Df Pr(>Chisq)
## Age    110.2  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
residualPlots( m1 ) # Residual vs fitted plot not showing a flat line.
```



```
##      Test stat Pr(>|Test stat|)
## Age      291.04      < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
marginalModelPlots(m1)
```

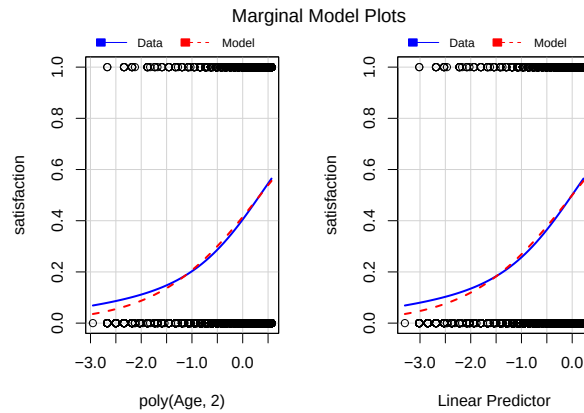


```
m1.2 <- glm(satisfaction ~ poly(Age,2), family=binomial, data=train)
anova(m1, m1.2, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ Age
## Model 2: satisfaction ~ poly(Age, 2)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4998      6719.0
## 2      4997      6427.9  1    291.04 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
marginalModelPlots(m1.2)
```

```
## Warning in mmps(...): Splines and/or polynomials replaced by a fitted linear
## combination
```

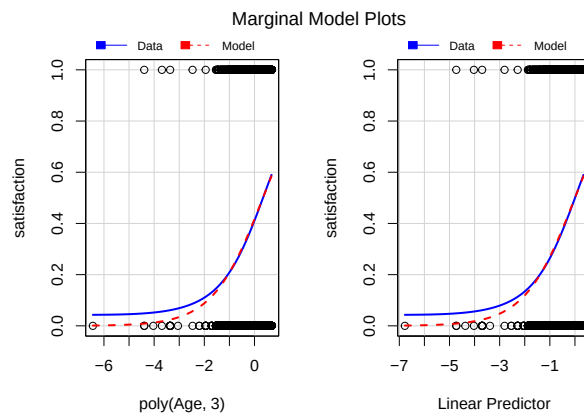


```
m1.3 <- glm(satisfaction ~ poly(Age,3), family=binomial, data=train)
anova(m1, m1.3, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ Age
## Model 2: satisfaction ~ poly(Age, 3)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4998      6719.0
## 2      4996      6390.4  2    328.61 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
marginalModelPlots(m1.3)
```

```
## Warning in mmps(...): Splines and/or polynomials replaced by a fitted linear
## combination
```

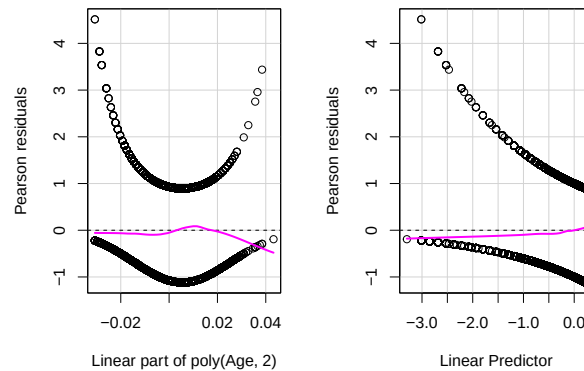


```
anova(m1.3, m1.2, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 3)
## Model 2: satisfaction ~ poly(Age, 2)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4996      6390.4
## 2      4997      6427.9 -1   -37.567 8.831e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

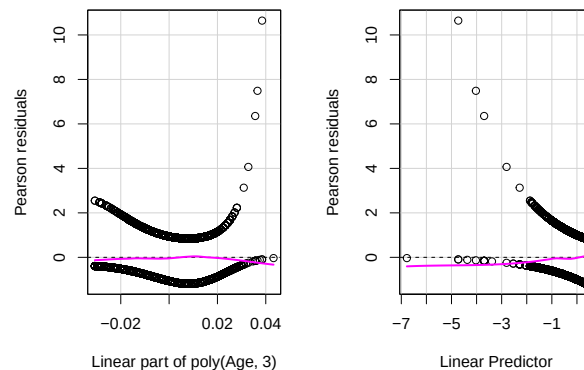
```
residualPlots( m1.2)
```

```
## Warning in residualPlots.default(model, ...): No possible lack-of-fit tests
```



```
residualPlots( m1.3) # residual vs fitted plot not showing a flat line.
```

```
## Warning in residualPlots.default(model, ...): No possible lack-of-fit tests
```



```
AIC(m1, m1.2, m1.3)
```

```
##      df      AIC
## m1    2 6722.964
## m1.2  3 6433.924
## m1.3  4 6398.357
```

```
m1 <- m1.2
```

```
#####
#Including Flight.Distance to the model#
#####
```

```
m2 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance, family=binomial,
          data=train)
summary(m2)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance,
##      family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.042e+00  4.944e-02 -21.066  <2e-16 ***
## poly(Age, 2)1    2.258e+01  2.437e+00   9.264  <2e-16 ***
## poly(Age, 2)2   -3.660e+01  2.627e+00 -13.936  <2e-16 ***
## Flight.Distance  5.848e-04  3.244e-05  18.028  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6068.6  on 4996  degrees of freedom
## AIC: 6076.6
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m2, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)    300.04  2  < 2.2e-16 ***
## Flight.Distance  359.30  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m1, m2, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2)
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4997      6427.9
## 2      4996      6068.6  1    359.3 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
residualPlots( m2, id.n=0 )
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in box(...): "id.n" is not a graphical parameter
```

```
## Warning in title(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in box(...): "id.n" is not a graphical parameter
```

```
## Warning in title(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

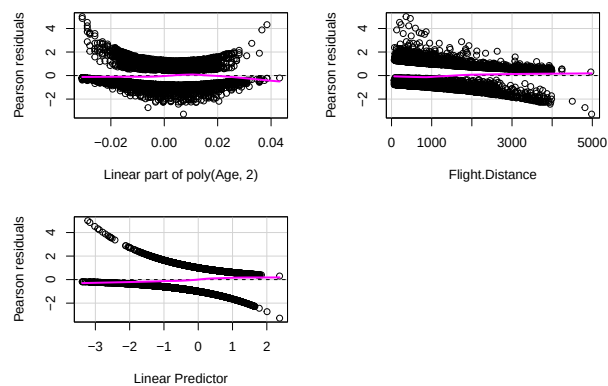
```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

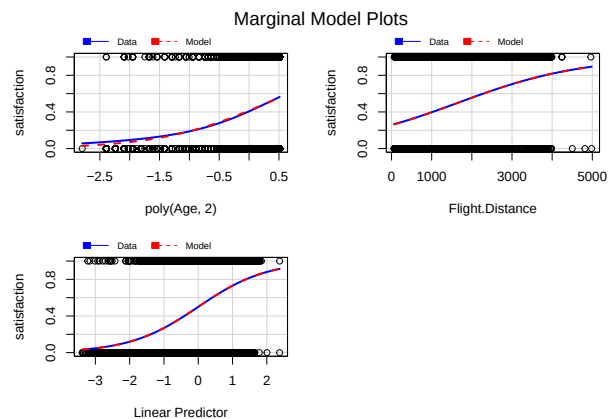
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```



```
## Test stat Pr(>|Test stat|)
## poly(Age, 2)
## Flight.Distance 0.6677 0.4139
```

```
marginalModelPlots(m2)
```

```
## Warning in mmpls(...): Splines and/or polynomials replaced by a fitted linear
## combination
```



```
m2.2 <- glm(satisfaction ~ poly(Age,2) + poly(Flight.Distance,2),
            family=binomial, data=train)
summary(m2.2)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + poly(Flight.Distance,
##      2), family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.34543    0.03166  -10.91  <2e-16 ***
## poly(Age, 2)1       22.65961    2.43907   9.29  <2e-16 ***
## poly(Age, 2)2      -36.66430    2.62769  -13.95  <2e-16 ***
## poly(Flight.Distance, 2)1  41.14040    2.28793  17.98  <2e-16 ***
## poly(Flight.Distance, 2)2  -1.87515    2.28745  -0.82    0.412
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6068.0  on 4995  degrees of freedom
## AIC: 6078
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m2.2, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      300.69  2  < 2.2e-16 ***
## poly(Flight.Distance, 2)  359.97  2  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2, m2.2, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance
## Model 2: satisfaction ~ poly(Age, 2) + poly(Flight.Distance, 2)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4996      6068.6
## 2      4995      6068.0  1  0.66771    0.4139
```

```
residualPlots( m2.2, id.n=0 )
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

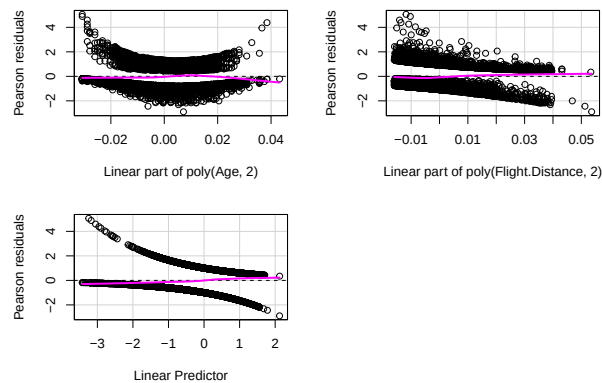
## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in residualPlots.default(model, ...): No possible lack-of-fit tests

```



```
#####
#Including Departure.Delay.in.Minutes to the model#
#####
```

```
m3 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
           Departure.Delay.in.Minutes, family=binomial, data=train)
summary(m3)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Departure.Delay.in.Minutes, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.005e+00  5.064e-02  -19.84 < 2e-16 ***
## poly(Age, 2)1     2.267e+01  2.440e+00   9.29 < 2e-16 ***
## poly(Age, 2)2    -3.686e+01  2.631e+00  -14.01 < 2e-16 ***
## Flight.Distance   5.873e-04  3.248e-05  18.08 < 2e-16 ***
## Departure.Delay.in.Minutes -2.730e-03  8.586e-04  -3.18  0.00147 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6057.9  on 4995  degrees of freedom
## AIC: 6067.9
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m3, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      303.03  2 < 2.2e-16 ***
```

```
## Flight.Distance          361.53  1  < 2.2e-16 ***
## Departure.Delay.in.Minutes  10.70  1   0.001074 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2, m3, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Departure.Delay.in.Minutes
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4996      6068.6
## 2      4995      6057.9  1   10.695 0.001074 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#####
#Including Arrival.Delay.in.Minutes to the model#
#####

m4 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Departure.Delay.in.Minutes + Arrival.Delay.in.Minutes,
          family=binomial, data=train)
summary(m4)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##   Departure.Delay.in.Minutes + Arrival.Delay.in.Minutes, family = binomial,
##   data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -9.997e-01  5.070e-02 -19.718  <2e-16 ***
## poly(Age, 2)1     2.265e+01  2.441e+00   9.279  <2e-16 ***
## poly(Age, 2)2    -3.704e+01  2.635e+00 -14.058  <2e-16 ***
## Flight.Distance     5.874e-04  3.251e-05  18.072  <2e-16 ***
## Departure.Delay.in.Minutes  4.963e-03  3.236e-03   1.534   0.1251
## Arrival.Delay.in.Minutes  -7.939e-03  3.226e-03  -2.461   0.0138 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##   Null deviance: 6829.2  on 4999  degrees of freedom
## Residual deviance: 6051.8  on 4994  degrees of freedom
## AIC: 6063.8
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m4, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##          LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      304.48  2    < 2e-16 ***
## Flight.Distance    361.20  1    < 2e-16 ***
## Departure.Delay.in.Minutes  2.36  1    0.12427
## Arrival.Delay.in.Minutes   6.13  1    0.01331 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2, m4, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Departure.Delay.in.Minutes +
##      Arrival.Delay.in.Minutes
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4996      6068.6
## 2      4994      6051.8  2    16.822 0.0002224 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#####
#Removing Departure.Delay.in.Minutes of the model --> FINAL MODEL #
#####
```

```
m5 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes, family=binomial, data=train)
summary(m5)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Arrival.Delay.in.Minutes, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -9.979e-01  5.068e-02 -19.692 < 2e-16 ***
## poly(Age, 2)1    2.267e+01  2.441e+00   9.289 < 2e-16 ***
## poly(Age, 2)2   -3.696e+01  2.633e+00 -14.039 < 2e-16 ***
## Flight.Distance    5.877e-04  3.250e-05  18.084 < 2e-16 ***
## Arrival.Delay.in.Minutes -3.182e-03  8.663e-04  -3.673  0.00024 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6829.2  on 4999  degrees of freedom
```

```
## Residual deviance: 6054.2 on 4995 degrees of freedom
## AIC: 6064.2
##
## Number of Fisher Scoring iterations: 4
```

```
Anova(m5, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##          LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      303.98  2 < 2.2e-16 ***
## Flight.Distance    361.75  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 14.46  1 0.0001432 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m2, m5, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4996      6068.6
## 2      4995      6054.2  1    14.46 0.0001432 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m5, m4, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Departure.Delay.in.Minutes +
##   Arrival.Delay.in.Minutes
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4995      6054.2
## 2      4994      6051.8  1    2.3626  0.1243
```

```
vif(m5)
```

```
##
##          GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      1.005848  2      1.001459
## Flight.Distance    1.005663  1      1.002828
## Arrival.Delay.in.Minutes 1.003710  1      1.001853
```

```
#####
#Asses final model residuals#
#####
par(mfrow=c(2,2))
residualPlots( m5, id.n=0 )
```

```

## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in plot.window(...): "id.n" is not a graphical parameter

```

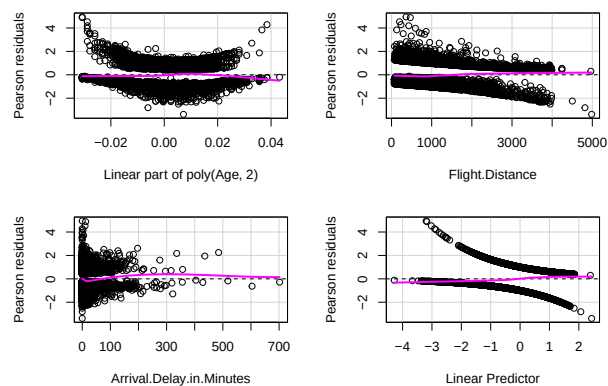
```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

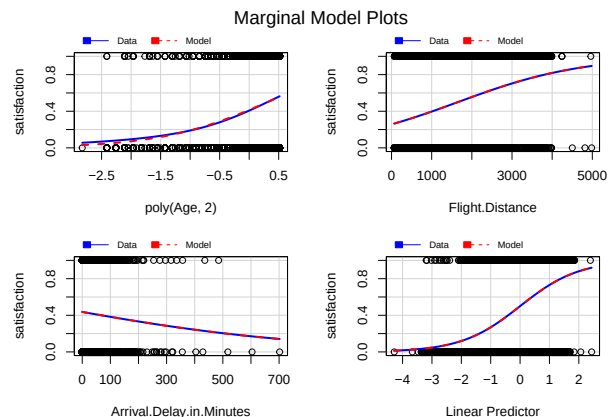
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```



```
##                               Test stat Pr(>|Test stat|)
## poly(Age, 2)
## Flight.Distance               0.6370           0.4248
## Arrival.Delay.in.Minutes      1.6834           0.1945
```

```
marginalModelPlots(m5)
```

```
## Warning in mmps(...): Splines and/or polynomials replaced by a fitted linear
## combination
```



```
#####
# Check for influential data and delete it. #
#####
influencePlot(m5)
```

```
##          StudRes          Hat          CookD
## 92    -0.3951640 0.024516487 0.0004128381
## 182    2.5401398 0.001041344 0.0049820099
## 1337   1.9289976 0.023058287 0.0244247031
## 1805   1.7947106 0.022646537 0.0179719530
## 1873   2.5492690 0.001022814 0.0050126003
## 4294  -0.8312199 0.053155849 0.0047076418
```

```
Boxplot(cooks.distance(m5), id=list(labels=rownames(train)))
```

```
## [1] "1337" "1805" "224" "4094" "1238" "428" "2160" "1873" "182" "4294"
```

```
cd <- cooks.distance(m5)
ss <- summary(cd); ss
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 5.295e-06 6.042e-05 9.316e-05 2.054e-04 1.789e-04 2.442e-02
```

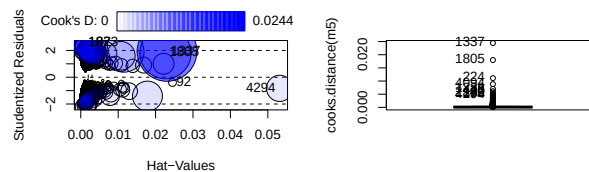
```
IQR <- ss[5] - ss[2]
severelowethresh<- ss[3] - 3*IQR;
severeuperthresh<-ss[5] + 3*IQR;
llrem <- which((cd < severelowethresh) | (cd > severeuperthresh));length(llrem)
```

```
## [1] 323
```

```
train <- train[-llrem,]
```

```
#retrain the model
```

```
m5 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes, family=binomial, data=train)
```



3.2 Include factor main effects to the best numeric model & Residual Analysis

Now that we have the best model only including numerical variables, we will include factor main effects to try to improve the model. We explain it step by step:

- 1- Adding the variable Gender is not statistically relevant, as shown by the AIC, Anova, and anova methods.
- 2- Including Class, Customer.Type, and Type.of.Travel significantly improves the model. The Anova function indicates that the net effect of the numerical variable Flight.Distance is no longer relevant, so we remove it. Adding Inflight.wifi.service and Departure.Arrival.time.convenient further enhances the model.
- 3- Although Ease.of.Online.booking slightly improves the AIC, its high VIF value suggests multicollinearity issues, so we exclude it.
- 4- Adding Gate.location and Food.and.drink improves the model, but Departure.Arrival.time.convenient becomes statistically insignificant, leading to its removal. Adding Online.boarding renders Arrival.Delay.in.Minutes and Food.and.drink insignificant, so they are also removed.
- 5- Including Seat.comfort, Inflight.entertainment, and On.board.service initially improves the model, but Inflight.entertainment has a high VIF and is excluded.
- 6- Adding Leg.room.service, Baggage.handling, and Checkin.service reveals that Food.and.drink is no longer significant and is removed. Including Inflight.service and Cleanliness provides minimal improvement relative to added complexity, so they are excluded.
- 7- Assessing self-defined factors, only DelayedArrival significantly contributes to the model; other factors are removed based on Anova tests and iterative analysis.
- 8- A step() operation confirms all included variables are properly justified.

The final model is:

```
satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel + Inflight.wifi.service + Gate.location + Online.boarding + Seat.comfort + On.board.service + Leg.room.service + Baggage.handling + Checkin.service + DelayedArrival
```

This model achieves an AIC value of 2235.28.

While the AIC suggests this model outperforms the numerical-only version, the anova test indicates the improvement in fit is statistically significant, implying the increased complexity is worth it given the model benefits.

Residual plots show outliers for the categorical variables, but this will be treated if they are influential in a bad way for the model.

Finally, we assess influential data removing it from the model.

```
factors <- names(which(sapply(train, is.numeric)==FALSE));factors
```

```
## [1] "Gender"
## [3] "Type.of.Travel"
## [5] "Inflight.wifi.service"
## [7] "Ease.of.Online.booking"
## [9] "Food.and.drink"
## [11] "Seat.comfort"
## [13] "On.board.service"
## [15] "Baggage.handling"
## [17] "Inflight.service"
## [19] "AgeGroup"
## [21] "DelayedArrival"

"Customer.Type"
"Class"
"Departure.Arrival.time.convenient"
"Gate.location"
"Online.boarding"
"Inflight.entertainment"
"Leg.room.service"
"Checkin.service"
"Cleanliness"
"f.FlightDist"
"DelayedDeparture"
```

```
#####
# Include Gender # not worth it
#####

m6 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes + Gender, family=binomial, data=train)
summary(m6)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Arrival.Delay.in.Minutes + Gender, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.406e+00  6.937e-02 -20.262  < 2e-16 ***
## poly(Age, 2)1     3.612e+01  2.940e+00  12.284  < 2e-16 ***
## poly(Age, 2)2    -6.325e+01  3.461e+00 -18.277  < 2e-16 ***
## Flight.Distance   9.162e-04  4.065e-05  22.542  < 2e-16 ***
## Arrival.Delay.in.Minutes -1.010e-02  1.563e-03  -6.465  1.02e-10 ***
## GenderMale       -4.891e-02  6.919e-02  -0.707    0.48
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 4968.0 on 4671 degrees of freedom
## AIC: 4980
##
## Number of Fisher Scoring iterations: 5
```

```
anova(m5,m6, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Gender
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4672      4968.5
## 2      4671      4968.0  1  0.49986   0.4796
```

```
Anova(m6, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      578.39  2  < 2.2e-16 ***
## Flight.Distance    629.08  1  < 2.2e-16 ***
## Arrival.Delay.in.Minutes  48.65  1  3.052e-12 ***
```

```
## Gender                0.50  1      0.4796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#####
# Include Class #
#####
m6 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes + Class, family=binomial, data=train)
summary(m6)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Arrival.Delay.in.Minutes + Class, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.111e-01  8.558e-02   1.298   0.194
## poly(Age, 2)1    3.181e+01  3.219e+00   9.880 < 2e-16 ***
## poly(Age, 2)2   -5.333e+01  3.750e+00 -14.221 < 2e-16 ***
## Flight.Distance    4.857e-04  4.551e-05  10.674 < 2e-16 ***
## Arrival.Delay.in.Minutes -1.119e-02  1.741e-03  -6.428  1.3e-10 ***
## ClassEco          -2.143e+00  8.544e-02 -25.078 < 2e-16 ***
## ClassEco Plus     -1.664e+00  1.495e-01 -11.126 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 4252.1  on 4670  degrees of freedom
## AIC: 4266.1
##
## Number of Fisher Scoring iterations: 5
```

```
Anova(m6, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      327.62  2 < 2.2e-16 ***
## Flight.Distance    121.41  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes  48.84  1 2.778e-12 ***
## Class             716.43  2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m5, m6, test="Chisq")
```

```
## Analysis of Deviance Table
```

```
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Class
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4672      4968.5
## 2      4670      4252.1  2   716.43 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#####
# Include Customer.Type #
#####
m7 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes + Class + Customer.Type,
          family=binomial, data=train)
summary(m7)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Arrival.Delay.in.Minutes + Class + Customer.Type, family = binomial,
##      data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -7.009e-01  1.169e-01  -5.995 2.03e-09 ***
## poly(Age, 2)1     2.087e+01  3.330e+00   6.267 3.68e-10 ***
## poly(Age, 2)2    -5.237e+01  3.710e+00 -14.115 < 2e-16 ***
## Flight.Distance   3.600e-04  4.730e-05   7.612 2.71e-14 ***
## Arrival.Delay.in.Minutes -1.142e-02  1.765e-03  -6.471 9.72e-11 ***
## ClassEco         -2.277e+00  8.939e-02 -25.476 < 2e-16 ***
## ClassEco Plus    -1.908e+00  1.529e-01 -12.477 < 2e-16 ***
## Customer.TypeLoyal Customer  1.256e+00  1.192e-01  10.534 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 4132.0  on 4669  degrees of freedom
## AIC: 4148
##
## Number of Fisher Scoring iterations: 5
```

```
Anova(m7, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      282.83  2 < 2.2e-16 ***
## Flight.Distance    59.93  1  9.808e-15 ***
```

```
## Arrival.Delay.in.Minutes    50.15  1  1.423e-12 ***
## Class                      768.63  2  < 2.2e-16 ***
## Customer.Type              120.12  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m6, m7, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Class
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Class + Customer.Type
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         4670      4252.1
## 2         4669      4132.0  1   120.12 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m7)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.083761  2      1.020313
## Flight.Distance                1.213401  1      1.101545
## Arrival.Delay.in.Minutes       1.013627  1      1.006791
## Class                         1.214938  2      1.049877
## Customer.Type                  1.176985  1      1.084889
```

```
par( mfrow=c(2,2))
residualPlot(m7, id.n=0)
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in box(...): "id.n" is not a graphical parameter
```

```
## Warning in title(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```

```
#####
# Include Type.of.Travel #
#####
```

```
m8 <- glm(satisfaction ~ poly(Age,2) + Flight.Distance +
          Arrival.Delay.in.Minutes + Class + Customer.Type
          + Type.of.Travel, family=binomial, data=train)
summary(m8)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Flight.Distance +
##      Arrival.Delay.in.Minutes + Class + Customer.Type + Type.of.Travel,
##      family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -8.965e-01  1.171e-01  -7.656 1.93e-14 ***
## poly(Age, 2)1       1.870e+01  3.606e+00   5.186 2.15e-07 ***
## poly(Age, 2)2      -4.245e+01  3.894e+00 -10.902 < 2e-16 ***
## Flight.Distance     2.949e-04  4.993e-05   5.906 3.50e-09 ***
## Arrival.Delay.in.Minutes -1.308e-02  1.800e-03  -7.264 3.77e-13 ***
## ClassEco           -1.325e+00  1.024e-01 -12.948 < 2e-16 ***
## ClassEco Plus      -1.433e+00  1.649e-01  -8.692 < 2e-16 ***
## Customer.TypeLoyal Customer  1.800e+00  1.179e-01  15.262 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.242e+00  1.262e-01 -17.763 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 3762.9  on 4668  degrees of freedom
## AIC: 3780.9
##
## Number of Fisher Scoring iterations: 5
```

```
Anova(m8, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      151.58  2 < 2.2e-16 ***
## Flight.Distance    35.88  1  2.094e-09 ***
## Arrival.Delay.in.Minutes  60.35  1  7.924e-15 ***
## Class             196.55  2 < 2.2e-16 ***
## Customer.Type      253.40  1 < 2.2e-16 ***
## Type.of.Travel     369.05  1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m7, m8, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Class + Customer.Type
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4669      4132.0
## 2      4668      3762.9  1   369.05 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m8)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      1.126429  2      1.030210
## Flight.Distance   1.242083  1      1.114488
## Arrival.Delay.in.Minutes 1.019044  1      1.009477
## Class             1.416275  2      1.090905
## Customer.Type     1.257266  1      1.121279
## Type.of.Travel    1.301429  1      1.140802
```

```
#####
# Remove Flight.Distance #
#####
m9 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
          + Class + Customer.Type
          + Type.of.Travel, family=binomial, data=train)
summary(m9)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel, family = binomial,
##      data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.599475   0.105675  -5.673 1.40e-08 ***
## poly(Age, 2)1    17.850597   3.562950   5.010 5.44e-07 ***
## poly(Age, 2)2   -41.321873   3.844395 -10.749 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.012765   0.001753  -7.281 3.31e-13 ***
## ClassEco        -1.535591   0.096922 -15.844 < 2e-16 ***
## ClassEco Plus   -1.686673   0.160433 -10.513 < 2e-16 ***
## Customer.TypeLoyal Customer  1.992816   0.114334  17.430 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.301188   0.125745 -18.301 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 3798.8 on 4669 degrees of freedom
## AIC: 3814.8
##
## Number of Fisher Scoring iterations: 5
```

```
Anova(m9, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2) 145.12 2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 60.56 1 7.129e-15 ***
## Class 304.82 2 < 2.2e-16 ***
## Customer.Type 334.75 1 < 2.2e-16 ***
## Type.of.Travel 393.10 1 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m9, m8, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel
## Model 2: satisfaction ~ poly(Age, 2) + Flight.Distance + Arrival.Delay.in.Minutes +
## Class + Customer.Type + Type.of.Travel
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1 4669 3798.8
## 2 4668 3762.9 1 35.884 2.094e-09 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m9)
```

```
##
## GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2) 1.115798 2 1.027771
## Arrival.Delay.in.Minutes 1.016513 1 1.008223
## Class 1.243152 2 1.055920
## Customer.Type 1.173882 1 1.083458
## Type.of.Travel 1.291654 1 1.136510
```

```
#####
# Include Inflight.wifi.service #
#####
m10 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
+ Class + Customer.Type + Type.of.Travel + Inflight.wifi.service
, family=binomial, data=train)
summary(m10)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service,
##      family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.091687   0.152871  -0.600  0.54866
## poly(Age, 2)1      23.880933   4.342065   5.500 3.80e-08 ***
## poly(Age, 2)2     -47.193908   4.800192  -9.832 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.010581  0.002086  -5.073 3.92e-07 ***
## ClassEco          -2.314107   0.133885 -17.284 < 2e-16 ***
## ClassEco Plus     -2.262773   0.206163 -10.976 < 2e-16 ***
## Customer.TypeLoyal Customer  2.522092  0.155833  16.185 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.281383  0.161065 -14.164 < 2e-16 ***
## Inflight.wifi.service.L      4.680500  0.266024  17.594 < 2e-16 ***
## Inflight.wifi.service.Q      2.821713  0.220177  12.816 < 2e-16 ***
## Inflight.wifi.service.C      0.451658  0.150368   3.004 0.00267 **
## Inflight.wifi.service^4     -0.162080  0.105884  -1.531 0.12583
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 2731.7  on 4665  degrees of freedom
## AIC: 2755.7
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m10, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      124.66  2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes  29.15  1 6.715e-08 ***
## Class             404.46  2 < 2.2e-16 ***
## Customer.Type      327.69  1 < 2.2e-16 ***
## Type.of.Travel     235.58  1 < 2.2e-16 ***
## Inflight.wifi.service 1067.12  4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m9, m10, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
##      Customer.Type + Type.of.Travel
## Model 2: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
```

```
##      Customer.Type + Type.of.Travel + Inflight.wifi.service
##      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         4669      3798.8
## 2         4665      2731.7  4   1067.1 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m10)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.108113  2         1.025997
## Arrival.Delay.in.Minutes      1.011149  1         1.005559
## Class                         1.514044  2         1.109263
## Customer.Type                 1.197370  1         1.094244
## Type.of.Travel                1.337844  1         1.156652
## Inflight.wifi.service         1.402522  4         1.043191
```

```
#####
# Include Departure.Arrival.time.convenient #
#####
```

```
m11 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
           + Class + Customer.Type + Type.of.Travel + Inflight.wifi.service
           + Departure.Arrival.time.convenient, family=binomial, data=train)
summary(m11)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      Departure.Arrival.time.convenient, family = binomial, data = train)
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -0.10989    0.15144  -0.726  0.46807
## poly(Age, 2)1                  22.49707    4.39597   5.118 3.09e-07 ***
## poly(Age, 2)2                 -46.13382    4.94381  -9.332 < 2e-16 ***
## Arrival.Delay.in.Minutes       -0.01085    0.00212  -5.119 3.07e-07 ***
## ClassEco                      -2.30759    0.13763 -16.766 < 2e-16 ***
## ClassEco Plus                 -2.22235    0.21345 -10.411 < 2e-16 ***
## Customer.TypeLoyal Customer    2.52583    0.15409  16.392 < 2e-16 ***
## Type.of.TravelPersonal Travel  -2.07480    0.16769 -12.373 < 2e-16 ***
## Inflight.wifi.service.L         5.15030    0.28172  18.282 < 2e-16 ***
## Inflight.wifi.service.Q         2.92050    0.23193  12.592 < 2e-16 ***
## Inflight.wifi.service.C         0.45181    0.16069   2.812  0.00493 **
## Inflight.wifi.service^4        -0.16587    0.11779  -1.408  0.15908
## Departure.Arrival.time.convenient.L -0.93105    0.13858  -6.719 1.83e-11 ***
## Departure.Arrival.time.convenient.Q -0.24331    0.13120  -1.855  0.06366 .
## Departure.Arrival.time.convenient.C  0.13207    0.12182   1.084  0.27829
## Departure.Arrival.time.convenient^4  0.08952    0.11722   0.764  0.44504
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 2678.5 on 4661 degrees of freedom
## AIC: 2710.5
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m11, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2) 111.30 2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 29.73 1 4.969e-08 ***
## Class 377.16 2 < 2.2e-16 ***
## Customer.Type 330.84 1 < 2.2e-16 ***
## Type.of.Travel 172.88 1 < 2.2e-16 ***
## Inflight.wifi.service 1076.69 4 < 2.2e-16 ***
## Departure.Arrival.time.convenient 53.19 4 7.779e-11 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m11)
```

```
##
## GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2) 1.117176 2 1.028088
## Arrival.Delay.in.Minutes 1.014511 1 1.007229
## Class 1.557666 2 1.117168
## Customer.Type 1.211000 1 1.100455
## Type.of.Travel 1.396079 1 1.181558
## Inflight.wifi.service 3.488404 4 1.169037
## Departure.Arrival.time.convenient 2.793443 4 1.137018
```

```
#####
# Include Ease.of.Online.booking # not worth it -> huge vif
#####

m11 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
+ Class + Customer.Type + Type.of.Travel + Inflight.wifi.service
+ Departure.Arrival.time.convenient + Ease.of.Online.booking
, family=binomial, data=train)
summary(m11)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
## Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
## Departure.Arrival.time.convenient + Ease.of.Online.booking,
## family = binomial, data = train)
##
```

```
## Coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.11145    0.15167  -0.735  0.46246
## poly(Age, 2)1      22.34606    4.41583   5.060 4.18e-07 ***
## poly(Age, 2)2     -45.83905    4.97373  -9.216 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.01080    0.00212  -5.096 3.47e-07 ***
## ClassEco          -2.28737    0.13756 -16.628 < 2e-16 ***
## ClassEco Plus     -2.19579    0.21335 -10.292 < 2e-16 ***
## Customer.TypeLoyal Customer 2.57671    0.15570  16.549 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.19394    0.17320 -12.667 < 2e-16 ***
## Inflight.wifi.service.L    5.39160    0.29843  18.067 < 2e-16 ***
## Inflight.wifi.service.Q    3.14120    0.24701  12.717 < 2e-16 ***
## Inflight.wifi.service.C    0.52386    0.17212   3.044 0.00234 **
## Inflight.wifi.service^4   -0.12044    0.12561  -0.959 0.33763
## Departure.Arrival.time.convenient.L -0.79093    0.16149  -4.898 9.70e-07 ***
## Departure.Arrival.time.convenient.Q -0.06299    0.15551  -0.405 0.68544
## Departure.Arrival.time.convenient.C  0.13785    0.14468   0.953 0.34068
## Departure.Arrival.time.convenient^4  0.17965    0.13913   1.291 0.19662
## Ease.of.Online.booking.L   -0.33373    0.17787  -1.876 0.06062 .
## Ease.of.Online.booking.Q   -0.39395    0.17156  -2.296 0.02166 *
## Ease.of.Online.booking.C   -0.06358    0.15589  -0.408 0.68339
## Ease.of.Online.booking^4   -0.16593    0.14242  -1.165 0.24399
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 2668.8  on 4657  degrees of freedom
## AIC: 2708.8
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m11, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##               LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      108.26  2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 29.62  1 5.257e-08 ***
## Class             367.63  2 < 2.2e-16 ***
## Customer.Type     338.21  1 < 2.2e-16 ***
## Type.of.Travel    181.44  1 < 2.2e-16 ***
## Inflight.wifi.service 917.45  4 < 2.2e-16 ***
## Departure.Arrival.time.convenient 28.95  4 7.985e-06 ***
## Ease.of.Online.booking  9.66  4  0.0465 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m11)
```

```
##               GVIF Df GVIF^(1/(2*Df))
```

```
## poly(Age, 2) 1.122901 2 1.029403
## Arrival.Delay.in.Minutes 1.014991 1 1.007468
## Class 1.560566 2 1.117688
## Customer.Type 1.237473 1 1.112418
## Type.of.Travel 1.498429 1 1.224103
## Inflight.wifi.service 6.205337 4 1.256307
## Departure.Arrival.time.convenient 10.464370 4 1.341109
## Ease.of.Online.booking 17.074812 4 1.425754
```

```
#####
# Include Gate.location #
#####
```

```
m11 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
+ Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
Departure.Arrival.time.convenient +
Gate.location ,
family=binomial, data=train)
summary(m11)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      Departure.Arrival.time.convenient + Gate.location, family = binomial,
##      data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.323541   0.163102  -1.984   0.0473 *
## poly(Age, 2)1      23.105301   4.478382   5.159 2.48e-07 ***
## poly(Age, 2)2     -47.078358   5.071846  -9.282 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.010855   0.002142  -5.068 4.03e-07 ***
## ClassEco          -2.225207   0.138595 -16.055 < 2e-16 ***
## ClassEco Plus     -2.087260   0.216493  -9.641 < 2e-16 ***
## Customer.TypeLoyal Customer 2.605869   0.163250  15.962 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.443132   0.183637 -13.304 < 2e-16 ***
## Inflight.wifi.service.L 5.177358   0.280573  18.453 < 2e-16 ***
## Inflight.wifi.service.Q 2.864527   0.231081  12.396 < 2e-16 ***
## Inflight.wifi.service.C 0.361641   0.161670   2.237  0.0253 *
## Inflight.wifi.service^4 -0.205808   0.120771  -1.704  0.0884 .
## Departure.Arrival.time.convenient.L -0.121557   0.178223  -0.682  0.4952
## Departure.Arrival.time.convenient.Q -0.102512   0.168284  -0.609  0.5424
## Departure.Arrival.time.convenient.C 0.027703   0.150483   0.184  0.8539
## Departure.Arrival.time.convenient^4 0.068525   0.145479   0.471  0.6376
## Gate.location.L    -1.446841   0.188867  -7.661 1.85e-14 ***
## Gate.location.Q     -0.388805   0.172729  -2.251  0.0244 *
## Gate.location.C      0.143689   0.154756   0.928  0.3532
## Gate.location^4      0.067302   0.136539   0.493  0.6221
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 2607.6 on 4657 degrees of freedom
## AIC: 2647.6
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m11, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2) 109.63 2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 29.05 1 7.061e-08 ***
## Class 331.44 2 < 2.2e-16 ***
## Customer.Type 321.75 1 < 2.2e-16 ***
## Type.of.Travel 207.99 1 < 2.2e-16 ***
## Inflight.wifi.service 1119.99 4 < 2.2e-16 ***
## Departure.Arrival.time.convenient 1.14 4 0.8883
## Gate.location 70.88 4 1.481e-14 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m10, m11, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service
## Model 2: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service +
## Departure.Arrival.time.convenient + Gate.location
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1 4665 2731.7
## 2 4657 2607.6 8 124.07 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m11)
```

```
## GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2) 1.123375 2 1.029511
## Arrival.Delay.in.Minutes 1.017213 1 1.008570
## Class 1.545138 2 1.114915
## Customer.Type 1.231702 1 1.109821
## Type.of.Travel 1.649303 1 1.284252
## Inflight.wifi.service 3.769994 4 1.180436
## Departure.Arrival.time.convenient 15.819232 4 1.412206
## Gate.location 11.115482 4 1.351267
```

```
m11 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
          + Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
          + Gate.location ,
          family=binomial, data=train)
summary(m11)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      +Gate.location, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.340637    0.162423  -2.097 0.035974 *
## poly(Age, 2)1      23.235876    4.468167   5.200 1.99e-07 ***
## poly(Age, 2)2     -47.321771    5.047586  -9.375 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.010837    0.002139  -5.067 4.05e-07 ***
## ClassEco          -2.223119    0.137628 -16.153 < 2e-16 ***
## ClassEco Plus     -2.081713    0.215562  -9.657 < 2e-16 ***
## Customer.TypeLoyal Customer  2.617500    0.162872  16.071 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.509013    0.171020 -14.671 < 2e-16 ***
## Inflight.wifi.service.L      5.146868    0.276428  18.619 < 2e-16 ***
## Inflight.wifi.service.Q      2.842106    0.226686  12.538 < 2e-16 ***
## Inflight.wifi.service.C      0.364380    0.157265   2.317 0.020504 *
## Inflight.wifi.service^4     -0.192830    0.116312  -1.658 0.097346 .
## Gate.location.L          -1.527756    0.152340 -10.029 < 2e-16 ***
## Gate.location.Q          -0.459381    0.137164  -3.349 0.000811 ***
## Gate.location.C           0.149672    0.128096   1.168 0.242632
## Gate.location^4           0.101996    0.112419   0.907 0.364257
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 2608.8  on 4661  degrees of freedom
## AIC: 2640.8
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m11, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      112.00  2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes   28.94  1 7.466e-08 ***
## Class             334.54  2 < 2.2e-16 ***
## Customer.Type      323.98  1 < 2.2e-16 ***
## Type.of.Travel     257.34  1 < 2.2e-16 ***
```

```
## Inflight.wifi.service      1186.69  4  < 2.2e-16 ***
## Gate.location              122.93  4  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m10, m11, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
##      Customer.Type + Type.of.Travel + Inflight.wifi.service
## Model 2: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
##      Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      +Gate.location
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         4665      2731.7
## 2         4661      2608.8  4   122.93 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m11)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.113519  2         1.027246
## Arrival.Delay.in.Minutes       1.016564  1         1.008248
## Class                          1.523725  2         1.111032
## Customer.Type                  1.213233  1         1.101469
## Type.of.Travel                 1.432385  1         1.196823
## Inflight.wifi.service          2.715906  4         1.133025
## Gate.location                  2.093610  4         1.096761
```

```
#####
# Include Food.and.drink      #
#####
```

```
m12 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
            + Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
            Gate.location + Food.and.drink, family=binomial, data=train)
summary(m12)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      Gate.location + Food.and.drink, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.384285   0.162709  -2.362 0.018187 *
## poly(Age, 2)1    22.966543   4.549545   5.048 4.46e-07 ***
## poly(Age, 2)2   -47.011962   5.081578  -9.251 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.010701   0.002174  -4.923 8.52e-07 ***
```

```
## ClassEco -2.127270 0.138932 -15.312 < 2e-16 ***
## ClassEco Plus -1.998687 0.218158 -9.162 < 2e-16 ***
## Customer.TypeLoyal Customer 2.584266 0.163162 15.839 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.541625 0.173175 -14.677 < 2e-16 ***
## Inflight.wifi.service.L 5.080829 0.279438 18.182 < 2e-16 ***
## Inflight.wifi.service.Q 2.908942 0.231136 12.585 < 2e-16 ***
## Inflight.wifi.service.C 0.405197 0.160837 2.519 0.011759 *
## Inflight.wifi.service^4 -0.180081 0.119240 -1.510 0.130984
## Gate.location.L -1.482182 0.154418 -9.598 < 2e-16 ***
## Gate.location.Q -0.479343 0.139068 -3.447 0.000567 ***
## Gate.location.C 0.155117 0.129459 1.198 0.230840
## Gate.location^4 0.078753 0.113274 0.695 0.486902
## Food.and.drink.L 0.782282 0.130283 6.004 1.92e-09 ***
## Food.and.drink.Q -0.084514 0.124498 -0.679 0.497239
## Food.and.drink.C 0.221340 0.113346 1.953 0.050845 .
## Food.and.drink^4 -0.085753 0.108159 -0.793 0.427869
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 2570.5 on 4657 degrees of freedom
## AIC: 2610.5
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m12, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2) 108.62 2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 27.35 1 1.700e-07 ***
## Class 296.07 2 < 2.2e-16 ***
## Customer.Type 311.53 1 < 2.2e-16 ***
## Type.of.Travel 256.71 1 < 2.2e-16 ***
## Inflight.wifi.service 1108.34 4 < 2.2e-16 ***
## Gate.location 113.70 4 < 2.2e-16 ***
## Food.and.drink 38.26 4 9.882e-08 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m11, m12, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service +
## +Gate.location
## Model 2: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service +
```

```
##      Gate.location + Food.and.drink
##      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4661      2608.8
## 2      4657      2570.5  4   38.265 9.882e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m12)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.128841  2          1.030761
## Arrival.Delay.in.Minutes      1.017875  1          1.008898
## Class                         1.535597  2          1.113190
## Customer.Type                 1.230765  1          1.109398
## Type.of.Travel                1.459557  1          1.208121
## Inflight.wifi.service         2.913875  4          1.143033
## Gate.location                 2.077519  4          1.095704
## Food.and.drink                1.166788  4          1.019469
```

```
#####
# Include Online.boarding , remove Food.and.drink #
#####
```

```
m13 <- glm(satisfaction ~ poly(Age,2) + Arrival.Delay.in.Minutes
+ Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
Gate.location + Food.and.drink + Online.boarding, family=binomial,
data=train)
Anova(m13, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      90.23  2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 25.40  1 4.653e-07 ***
## Class            152.77  2 < 2.2e-16 ***
## Customer.Type    194.59  1 < 2.2e-16 ***
## Type.of.Travel   269.60  1 < 2.2e-16 ***
## Inflight.wifi.service 495.28  4 < 2.2e-16 ***
## Gate.location     55.35  4 2.743e-11 ***
## Food.and.drink     4.19  4  0.3812
## Online.boarding    392.05  4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
m13 <- glm(satisfaction ~ poly(Age,2) +Arrival.Delay.in.Minutes + Class +
Customer.Type + Type.of.Travel
+ Inflight.wifi.service + Gate.location + Online.boarding,
family=binomial, data=train)
summary(m13)
```

```
##
```

```
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes +
##      Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
##      Gate.location + Online.boarding, family = binomial, data = train)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.501487   0.184228  -2.722  0.00649 **
## poly(Age, 2)1      14.088712   4.965474   2.837  0.00455 **
## poly(Age, 2)2     -47.725813   5.521702  -8.643 < 2e-16 ***
## Arrival.Delay.in.Minutes -0.011772  0.002467  -4.772 1.83e-06 ***
## ClassEco          -1.679747   0.147422 -11.394 < 2e-16 ***
## ClassEco Plus     -1.598539   0.232794  -6.867 6.57e-12 ***
## Customer.TypeLoyal Customer  2.264454  0.177272  12.774 < 2e-16 ***
## Type.of.TravelPersonal Travel -2.800094  0.186463 -15.017 < 2e-16 ***
## Inflight.wifi.service.L    4.059126  0.290076  13.993 < 2e-16 ***
## Inflight.wifi.service.Q    2.407718  0.243906   9.872 < 2e-16 ***
## Inflight.wifi.service.C    0.512311  0.175868   2.913  0.00358 **
## Inflight.wifi.service^4   -0.039723  0.134518  -0.295  0.76777
## Gate.location.L         -1.148754  0.162989  -7.048 1.81e-12 ***
## Gate.location.Q         -0.429838  0.148414  -2.896  0.00378 **
## Gate.location.C         -0.031649  0.137445  -0.230  0.81788
## Gate.location^4          0.037130  0.121234   0.306  0.75940
## Online.boarding.L        2.622712  0.184647  14.204 < 2e-16 ***
## Online.boarding.Q        1.568951  0.169183   9.274 < 2e-16 ***
## Online.boarding.C       -0.103998  0.149800  -0.694  0.48753
## Online.boarding^4       -0.231757  0.135909  -1.705  0.08815 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 2182.6  on 4657  degrees of freedom
## AIC: 2222.6
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m13, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)          91.10  2 < 2.2e-16 ***
## Arrival.Delay.in.Minutes  25.26  1 5.007e-07 ***
## Class                156.13  2 < 2.2e-16 ***
## Customer.Type         194.62  1 < 2.2e-16 ***
## Type.of.Travel        272.38  1 < 2.2e-16 ***
## Inflight.wifi.service  495.77  4 < 2.2e-16 ***
## Gate.location          56.85  4 1.328e-11 ***
## Online.boarding        426.13  4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m13)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      1.182628  2      1.042827
## Arrival.Delay.in.Minutes 1.017082  1      1.008505
## Class             1.526696  2      1.111573
## Customer.Type     1.265643  1      1.125008
## Type.of.Travel    1.574018  1      1.254599
## Inflight.wifi.service 3.236295  4      1.158127
## Gate.location     1.878873  4      1.082025
## Online.boarding   1.760556  4      1.073263
```

```
#####
# Include "Seat.comfort", "Inflight.entertainment", "On.board.service"#
#####
```

```
m14 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + + Arrival.Delay.in.Minutes +Inflight.wifi.service +
  Gate.location +
  + Inflight.entertainment +Online.boarding + Seat.comfort +
  On.board.service ,family=binomial,
  data=train)
summary(m14)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##      Type.of.Travel + +Arrival.Delay.in.Minutes + Inflight.wifi.service +
##      Gate.location + +Inflight.entertainment + Online.boarding +
##      Seat.comfort + On.board.service, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.880087   0.193744  -4.543 5.56e-06 ***
## poly(Age, 2)1    11.241928   5.346020   2.103 0.035478 *
## poly(Age, 2)2   -42.947973   5.694534  -7.542 4.63e-14 ***
## ClassEco        -1.398261   0.159960  -8.741 < 2e-16 ***
## ClassEco Plus   -1.213621   0.246303  -4.927 8.34e-07 ***
## Customer.TypeLoyal Customer  2.538998   0.188521  13.468 < 2e-16 ***
## Type.of.TravelPersonal Travel -3.170782   0.205196 -15.452 < 2e-16 ***
## Arrival.Delay.in.Minutes    -0.012941   0.002714  -4.769 1.85e-06 ***
## Inflight.wifi.service.L      3.781578   0.308195  12.270 < 2e-16 ***
## Inflight.wifi.service.Q      2.533925   0.262168   9.665 < 2e-16 ***
## Inflight.wifi.service.C      0.730638   0.192938   3.787 0.000153 ***
## Inflight.wifi.service^4     -0.248088   0.151619  -1.636 0.101784
## Gate.location.L            -0.864985   0.174866  -4.947 7.55e-07 ***
## Gate.location.Q            -0.519606   0.162429  -3.199 0.001379 **
## Gate.location.C            -0.115296   0.147195  -0.783 0.433458
## Gate.location^4             0.023914   0.129305   0.185 0.853277
## Inflight.entertainment.L     0.887740   0.215016   4.129 3.65e-05 ***
## Inflight.entertainment.Q    -0.481700   0.197122  -2.444 0.014539 *
## Inflight.entertainment.C    -0.340438   0.183788  -1.852 0.063977 .
## Inflight.entertainment^4     0.113031   0.170694   0.662 0.507855
```

```
## Online.boarding.L          2.678846    0.211093   12.690 < 2e-16 ***
## Online.boarding.Q          1.254715    0.186148    6.740 1.58e-11 ***
## Online.boarding.C          -0.100037    0.167921   -0.596 0.551351
## Online.boarding^4          -0.013391    0.154859   -0.086 0.931093
## Seat.comfort.L            -0.269503    0.199870   -1.348 0.177533
## Seat.comfort.Q             1.109925    0.179711    6.176 6.57e-10 ***
## Seat.comfort.C             0.283093    0.171043    1.655 0.097904 .
## Seat.comfort^4            -0.452566    0.161653   -2.800 0.005116 **
## On.board.service.L         1.511273    0.186655    8.097 5.65e-16 ***
## On.board.service.Q         0.587663    0.170173    3.453 0.000554 ***
## On.board.service.C         0.062796    0.165975    0.378 0.705173
## On.board.service^4         0.212171    0.150483    1.410 0.158558
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 1903.6 on 4645 degrees of freedom
## AIC: 1967.6
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m14, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##          LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      68.76  2 1.172e-15 ***
## Class              86.22  2 < 2.2e-16 ***
## Customer.Type     217.18  1 < 2.2e-16 ***
## Type.of.Travel    291.84  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 25.82  1 3.738e-07 ***
## Inflight.wifi.service 372.02  4 < 2.2e-16 ***
## Gate.location      30.75  4 3.447e-06 ***
## Inflight.entertainment 31.17  4 2.824e-06 ***
## Online.boarding    262.22  4 < 2.2e-16 ***
## Seat.comfort       55.17  4 2.999e-11 ***
## On.board.service   86.80  4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m13, m14, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service +
## Gate.location + Online.boarding
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
## +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
## +Inflight.entertainment + Online.boarding + Seat.comfort +
```

```
##      On.board.service
##      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         4657      2182.6
## 2         4645      1903.6 12   279.06 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m14) # Inflight entertainment huge VIF, remove it
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.242917 2         1.055870
## Class                         1.613608 2         1.127066
## Customer.Type                 1.420453 1         1.191828
## Type.of.Travel                1.824051 1         1.350574
## Arrival.Delay.in.Minutes      1.029317 1         1.014553
## Inflight.wifi.service         3.983605 4         1.188597
## Gate.location                 1.713327 4         1.069621
## Inflight.entertainment       10.682294 4         1.344569
## Online.boarding               2.729057 4         1.133709
## Seat.comfort                  5.293399 4         1.231592
## On.board.service              3.762373 4         1.180138
```

```
m14 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + +Arrival.Delay.in.Minutes +Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service ,family=binomial,
  data=train)
summary(m14)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##      Type.of.Travel + +Arrival.Delay.in.Minutes + Inflight.wifi.service +
##      Gate.location + Online.boarding + Seat.comfort + On.board.service,
##      family = binomial, data = train)
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -1.003840   0.193476  -5.188 2.12e-07 ***
## poly(Age, 2)1                   9.981703   5.239296   1.905 0.056759 .
## poly(Age, 2)2                  -43.123169   5.643336  -7.641 2.15e-14 ***
## ClassEco                       -1.357961   0.157481  -8.623 < 2e-16 ***
## ClassEco Plus                  -1.184476   0.240296  -4.929 8.26e-07 ***
## Customer.TypeLoyal Customer    2.682072   0.188937  14.196 < 2e-16 ***
## Type.of.TravelPersonal Travel  -3.295632   0.203606 -16.186 < 2e-16 ***
## Arrival.Delay.in.Minutes       -0.012943   0.002678  -4.833 1.34e-06 ***
## Inflight.wifi.service.L         3.960779   0.305965  12.945 < 2e-16 ***
## Inflight.wifi.service.Q         2.431953   0.259504   9.372 < 2e-16 ***
## Inflight.wifi.service.C         0.641693   0.187341   3.425 0.000614 ***
## Inflight.wifi.service^4        -0.227427   0.147156  -1.545 0.122230
## Gate.location.L                -0.912995   0.171355  -5.328 9.93e-08 ***
## Gate.location.Q                -0.442639   0.158529  -2.792 0.005236 **
## Gate.location.C                -0.110318   0.144773  -0.762 0.446059
```

```
## Gate.location^4          0.040905    0.128042    0.319 0.749374
## Online.boarding.L        2.651027    0.210119   12.617 < 2e-16 ***
## Online.boarding.Q        1.258782    0.184311    6.830 8.51e-12 ***
## Online.boarding.C       -0.071870    0.165825   -0.433 0.664719
## Online.boarding^4        0.003569    0.151465    0.024 0.981202
## Seat.comfort.L          0.240777    0.161823    1.488 0.136777
## Seat.comfort.Q           0.942582    0.155700    6.054 1.41e-09 ***
## Seat.comfort.C           0.134277    0.146046    0.919 0.357877
## Seat.comfort^4          -0.387493    0.143933   -2.692 0.007099 **
## On.board.service.L       1.875801    0.166305   11.279 < 2e-16 ***
## On.board.service.Q       0.461948    0.151282    3.054 0.002262 **
## On.board.service.C      -0.157384    0.142087   -1.108 0.268008
## On.board.service^4       0.279724    0.132007    2.119 0.034090 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 1934.7 on 4649 degrees of freedom
## AIC: 1990.7
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m14, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2)          70.69  2 4.472e-16 ***
## Class                 84.08  2 < 2.2e-16 ***
## Customer.Type        246.27  1 < 2.2e-16 ***
## Type.of.Travel       327.63  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 26.54  1 2.580e-07 ***
## Inflight.wifi.service 408.67  4 < 2.2e-16 ***
## Gate.location         33.03  4 1.178e-06 ***
## Online.boarding       259.54  4 < 2.2e-16 ***
## Seat.comfort          51.64  4 1.641e-10 ***
## On.board.service      169.84  4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m13, m14, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Arrival.Delay.in.Minutes + Class +
## Customer.Type + Type.of.Travel + Inflight.wifi.service +
## Gate.location + Online.boarding
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
## +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
## Online.boarding + Seat.comfort + On.board.service
```

```
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4657      2182.6
## 2      4649      1934.7  8   247.89 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m14)
```

```
##               GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      1.230087  2      1.053135
## Class             1.593236  2      1.123492
## Customer.Type     1.414629  1      1.189382
## Type.of.Travel    1.815840  1      1.347531
## Arrival.Delay.in.Minutes 1.027227  1      1.013522
## Inflight.wifi.service 3.511171  4      1.169988
## Gate.location     1.668553  4      1.066087
## Online.boarding    2.708029  4      1.132613
## Seat.comfort      1.871533  4      1.081495
## On.board.service  1.350078  4      1.038233
```

```
#####
# Include "Leg.room.service", "Baggage.handling", "Checkin.service"#
#####
```

```
m15 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + +Arrival.Delay.in.Minutes +Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
  Checkin.service,family=binomial,
  data=train)
```

```
summary(m15)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##   Type.of.Travel + +Arrival.Delay.in.Minutes + Inflight.wifi.service +
##   Gate.location + Online.boarding + Seat.comfort + On.board.service +
##   +Leg.room.service + Baggage.handling + Checkin.service, family = binomial,
##   data = train)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.393685   0.204396  -6.819 9.20e-12 ***
## poly(Age, 2)1     9.320205   5.572308   1.673 0.094407 .
## poly(Age, 2)2   -44.381083   5.967131  -7.438 1.03e-13 ***
## ClassEco        -1.104728   0.165282  -6.684 2.33e-11 ***
## ClassEco Plus   -0.943548   0.248279  -3.800 0.000144 ***
## Customer.TypeLoyal Customer  3.072748   0.204274  15.042 < 2e-16 ***
## Type.of.TravelPersonal Travel -3.725443   0.223385 -16.677 < 2e-16 ***
## Arrival.Delay.in.Minutes    -0.013614   0.002828  -4.813 1.48e-06 ***
## Inflight.wifi.service.L      4.027711   0.317458  12.687 < 2e-16 ***
## Inflight.wifi.service.Q      2.558914   0.268986   9.513 < 2e-16 ***
```

```

## Inflight.wifi.service.C      0.798335  0.198250  4.027 5.65e-05 ***
## Inflight.wifi.service^4     -0.041586  0.157196 -0.265 0.791355
## Gate.location.L             -0.774246  0.175437 -4.413 1.02e-05 ***
## Gate.location.Q             -0.460109  0.164623 -2.795 0.005191 **
## Gate.location.C             -0.161212  0.149626 -1.077 0.281285
## Gate.location^4             0.015188  0.132768  0.114 0.908923
## Online.boarding.L           2.638722  0.224069 11.776 < 2e-16 ***
## Online.boarding.Q           1.242344  0.194317  6.393 1.62e-10 ***
## Online.boarding.C           -0.114412  0.173464 -0.660 0.509527
## Online.boarding^4           0.032607  0.158206  0.206 0.836711
## Seat.comfort.L              0.136758  0.167931  0.814 0.415435
## Seat.comfort.Q              0.792893  0.162201  4.888 1.02e-06 ***
## Seat.comfort.C              0.169324  0.150686  1.124 0.261145
## Seat.comfort^4             -0.345386  0.147707 -2.338 0.019371 *
## On.board.service.L          1.020932  0.197613  5.166 2.39e-07 ***
## On.board.service.Q          0.246216  0.169424  1.453 0.146153
## On.board.service.C          -0.118992  0.167778 -0.709 0.478186
## On.board.service^4          0.413461  0.150501  2.747 0.006010 **
## Leg.room.service.L          0.914963  0.194808  4.697 2.64e-06 ***
## Leg.room.service.Q          -0.009840  0.175887 -0.056 0.955386
## Leg.room.service.C          -0.110342  0.150658 -0.732 0.463922
## Leg.room.service^4          -0.320994  0.140757 -2.280 0.022579 *
## Baggage.handling.L          0.671670  0.218541  3.073 0.002116 **
## Baggage.handling.Q          0.807629  0.189193  4.269 1.97e-05 ***
## Baggage.handling.C          0.104778  0.174365  0.601 0.547899
## Baggage.handling^4          -0.358968  0.161249 -2.226 0.026003 *
## Checkin.service.L           0.877137  0.167822  5.227 1.73e-07 ***
## Checkin.service.Q           0.247670  0.146386  1.692 0.090666 .
## Checkin.service.C           0.091870  0.152333  0.603 0.546450
## Checkin.service^4           0.306100  0.136263  2.246 0.024679 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 6355.4  on 4676  degrees of freedom
## Residual deviance: 1792.4  on 4637  degrees of freedom
## AIC: 1872.4
##
## Number of Fisher Scoring iterations: 7

```

```
Anova(m15, test= "LR")
```

```

## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##          LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      67.07  2  2.730e-15 ***
## Class              48.97  2  2.319e-11 ***
## Customer.Type      288.95  1 < 2.2e-16 ***
## Type.of.Travel     365.02  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 26.81  1  2.248e-07 ***
## Inflight.wifi.service 363.71  4 < 2.2e-16 ***
## Gate.location      24.20  4  7.287e-05 ***

```

```
## Online.boarding          226.99  4  < 2.2e-16 ***
## Seat.comfort             33.28  4  1.044e-06 ***
## On.board.service         35.93  4  2.998e-07 ***
## Leg.room.service         33.47  4  9.550e-07 ***
## Baggage.handling         48.93  4  6.027e-10 ***
## Checkin.service          34.55  4  5.747e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m14, m15, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##   Baggage.handling + Checkin.service
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4649      1934.7
## 2      4637      1792.4 12    142.32 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m15)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)    1.292180  2      1.066181
## Class           1.660309  2      1.135134
## Customer.Type   1.575759  1      1.255292
## Type.of.Travel  2.113838  1      1.453904
## Arrival.Delay.in.Minutes 1.035719  1      1.017703
## Inflight.wifi.service 4.384134  4      1.202916
## Gate.location   1.638155  4      1.063639
## Online.boarding  3.090156  4      1.151457
## Seat.comfort    1.885307  4      1.082487
## On.board.service 3.226081  4      1.157669
## Leg.room.service 2.242488  4      1.106219
## Baggage.handling 3.774640  4      1.180618
## Checkin.service  1.304123  4      1.033748
```

```
#####
# Include "Inflight.service", "Cleanliness"#
#####
```

```
m16 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + + Arrival.Delay.in.Minutes +Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
  Checkin.service + Inflight.service + Cleanliness ,family=binomial,
```

```
data=train)
```

```
summary(m16)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##     Type.of.Travel + +Arrival.Delay.in.Minutes + Inflight.wifi.service +
##     Gate.location + Online.boarding + Seat.comfort + On.board.service +
##     +Leg.room.service + Baggage.handling + Checkin.service +
##     Inflight.service + Cleanliness, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.432e+00  2.063e-01  -6.941 3.88e-12 ***
## poly(Age, 2)1       1.057e+01  5.656e+00   1.868 0.061733 .
## poly(Age, 2)2      -4.366e+01  5.980e+00  -7.300 2.87e-13 ***
## ClassEco           -1.108e+00  1.680e-01  -6.591 4.37e-11 ***
## ClassEco Plus      -9.469e-01  2.510e-01  -3.772 0.000162 ***
## Customer.TypeLoyal Customer  3.185e+00  2.092e-01  15.222 < 2e-16 ***
## Type.of.TravelPersonal Travel -3.817e+00  2.285e-01 -16.704 < 2e-16 ***
## Arrival.Delay.in.Minutes    -1.311e-02  2.879e-03  -4.555 5.23e-06 ***
## Inflight.wifi.service.L       4.069e+00  3.213e-01  12.661 < 2e-16 ***
## Inflight.wifi.service.Q       2.558e+00  2.728e-01   9.378 < 2e-16 ***
## Inflight.wifi.service.C       8.592e-01  2.022e-01   4.250 2.14e-05 ***
## Inflight.wifi.service^4      -4.965e-02  1.611e-01  -0.308 0.757854
## Gate.location.L            -8.034e-01  1.768e-01  -4.543 5.55e-06 ***
## Gate.location.Q            -5.308e-01  1.667e-01  -3.185 0.001449 **
## Gate.location.C            -2.141e-01  1.518e-01  -1.411 0.158375
## Gate.location^4             2.076e-02  1.346e-01   0.154 0.877401
## Online.boarding.L           2.517e+00  2.280e-01  11.039 < 2e-16 ***
## Online.boarding.Q           1.259e+00  1.978e-01   6.365 1.95e-10 ***
## Online.boarding.C          -1.132e-01  1.759e-01  -0.643 0.520086
## Online.boarding^4          -9.392e-04  1.604e-01  -0.006 0.995328
## Seat.comfort.L            -3.119e-01  2.033e-01  -1.534 0.125038
## Seat.comfort.Q             8.073e-01  1.838e-01   4.393 1.12e-05 ***
## Seat.comfort.C             2.942e-01  1.736e-01   1.695 0.090096 .
## Seat.comfort^4            -4.017e-01  1.650e-01  -2.435 0.014891 *
## On.board.service.L          9.281e-01  2.129e-01   4.359 1.31e-05 ***
## On.board.service.Q          1.389e-01  1.792e-01   0.775 0.438168
## On.board.service.C         -1.309e-01  1.818e-01  -0.720 0.471610
## On.board.service^4          5.237e-01  1.609e-01   3.256 0.001132 **
## Leg.room.service.L          9.068e-01  2.018e-01   4.494 6.98e-06 ***
## Leg.room.service.Q         -4.878e-02  1.808e-01  -0.270 0.787339
## Leg.room.service.C         -1.509e-01  1.543e-01  -0.978 0.328011
## Leg.room.service^4         -2.699e-01  1.449e-01  -1.863 0.062478 .
## Baggage.handling.L          6.431e-01  2.482e-01   2.591 0.009564 **
## Baggage.handling.Q          6.608e-01  2.039e-01   3.241 0.001193 **
## Baggage.handling.C          1.149e-01  1.869e-01   0.614 0.538894
## Baggage.handling^4         -2.464e-01  1.690e-01  -1.458 0.144723
## Checkin.service.L           8.291e-01  1.708e-01   4.855 1.20e-06 ***
## Checkin.service.Q           1.912e-01  1.492e-01   1.282 0.199863
## Checkin.service.C           1.378e-01  1.537e-01   0.896 0.370089
```

```
## Checkin.service^4          2.671e-01  1.382e-01  1.933 0.053254 .
## Inflight.service.L         3.168e-02  2.632e-01  0.120 0.904192
## Inflight.service.Q         4.367e-01  2.182e-01  2.002 0.045293 *
## Inflight.service.C         5.103e-02  1.969e-01  0.259 0.795478
## Inflight.service^4        -3.419e-01  1.748e-01 -1.956 0.050477 .
## Cleanliness.L              7.491e-01  1.962e-01  3.818 0.000135 ***
## Cleanliness.Q              -2.445e-02  1.745e-01 -0.140 0.888533
## Cleanliness.C              -2.328e-01  1.696e-01 -1.373 0.169798
## Cleanliness^4              1.753e-01  1.563e-01  1.122 0.262000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 1765.5 on 4629 degrees of freedom
## AIC: 1861.5
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m16, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2)          65.15  2 7.119e-15 ***
## Class                 47.67  2 4.455e-11 ***
## Customer.Type         298.66  1 < 2.2e-16 ***
## Type.of.Travel        369.98  1 < 2.2e-16 ***
## Arrival.Delay.in.Minutes 23.99  1 9.708e-07 ***
## Inflight.wifi.service  360.43  4 < 2.2e-16 ***
## Gate.location          27.23  4 1.787e-05 ***
## Online.boarding        207.09  4 < 2.2e-16 ***
## Seat.comfort           31.46  4 2.466e-06 ***
## On.board.service       28.10  4 1.193e-05 ***
## Leg.room.service       30.55  4 3.779e-06 ***
## Baggage.handling       30.69  4 3.549e-06 ***
## Checkin.service        29.25  4 6.950e-06 ***
## Inflight.service        9.49  4 0.049892 *
## Cleanliness            17.60  4 0.001475 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m15, m16, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + Leg.room.service +
##   Baggage.handling + Checkin.service
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
```

```
##      +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##      Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##      Baggage.handling + Checkin.service + Inflight.service + Cleanliness
##      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4637      1792.4
## 2      4629      1765.5  8    26.942 0.0007235 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m16)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)                   1.310020 2          1.069842
## Class                         1.699194 2          1.141723
## Customer.Type                 1.660990 1          1.288794
## Type.of.Travel                2.219450 1          1.489782
## Arrival.Delay.in.Minutes      1.042976 1          1.021262
## Inflight.wifi.service         4.641998 4          1.211541
## Gate.location                 1.678234 4          1.066858
## Online.boarding               3.271169 4          1.159679
## Seat.comfort                  5.291671 4          1.231542
## On.board.service              4.974749 4          1.222071
## Leg.room.service              2.549795 4          1.124121
## Baggage.handling              6.394500 4          1.261031
## Checkin.service               1.354348 4          1.038643
## Inflight.service              8.786882 4          1.312138
## Cleanliness                   4.498939 4          1.206810
```

```
#####
# Include "AgeGroup", "f.FlightDist", "DelayedArrival" "DelayedDeparture" #
#####

m16 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + +Arrival.Delay.in.Minutes +Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
  Checkin.service + Inflight.service + Cleanliness + AgeGroup +
  f.FlightDist + DelayedArrival +
  DelayedDeparture,family=binomial,
  data=train)

summary(m16)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##      Type.of.Travel + +Arrival.Delay.in.Minutes + Inflight.wifi.service +
##      Gate.location + Online.boarding + Seat.comfort + On.board.service +
##      +Leg.room.service + Baggage.handling + Checkin.service +
##      Inflight.service + Cleanliness + AgeGroup + f.FlightDist +
##      DelayedArrival + DelayedDeparture, family = binomial, data = train)
##
## Coefficients:
```

##	Estimate	Std. Error	z value	Pr(> z)	
## (Intercept)	-8.779345	1.471113	-5.968	2.40e-09	***
## poly(Age, 2)1	-25.647262	12.140923	-2.112	0.034647	*
## poly(Age, 2)2	-3.953005	9.760151	-0.405	0.685467	
## ClassEco	-1.138013	0.178737	-6.367	1.93e-10	***
## ClassEco Plus	-0.884751	0.261325	-3.386	0.000710	***
## Customer.TypeLoyal Customer	3.271766	0.223460	14.641	< 2e-16	***
## Type.of.TravelPersonal Travel	-3.730562	0.232862	-16.020	< 2e-16	***
## Arrival.Delay.in.Minutes	-0.006825	0.003454	-1.976	0.048137	*
## Inflight.wifi.service.L	4.200052	0.333926	12.578	< 2e-16	***
## Inflight.wifi.service.Q	2.549207	0.282268	9.031	< 2e-16	***
## Inflight.wifi.service.C	0.884614	0.208140	4.250	2.14e-05	***
## Inflight.wifi.service^4	-0.080950	0.163692	-0.495	0.620933	
## Gate.location.L	-0.812539	0.178782	-4.545	5.50e-06	***
## Gate.location.Q	-0.508575	0.169095	-3.008	0.002633	**
## Gate.location.C	-0.229172	0.154278	-1.485	0.137427	
## Gate.location^4	0.008238	0.136737	0.060	0.951960	
## Online.boarding.L	2.468227	0.231462	10.664	< 2e-16	***
## Online.boarding.Q	1.283590	0.202773	6.330	2.45e-10	***
## Online.boarding.C	-0.154671	0.179282	-0.863	0.388288	
## Online.boarding^4	0.021490	0.163502	0.131	0.895428	
## Seat.comfort.L	-0.298901	0.206528	-1.447	0.147822	
## Seat.comfort.Q	0.876559	0.186849	4.691	2.72e-06	***
## Seat.comfort.C	0.294299	0.176563	1.667	0.095550	.
## Seat.comfort^4	-0.426825	0.168241	-2.537	0.011181	*
## On.board.service.L	0.945143	0.216406	4.367	1.26e-05	***
## On.board.service.Q	0.165062	0.182162	0.906	0.364869	
## On.board.service.C	-0.123891	0.185955	-0.666	0.505258	
## On.board.service^4	0.467193	0.163341	2.860	0.004233	**
## Leg.room.service.L	0.820122	0.204401	4.012	6.01e-05	***
## Leg.room.service.Q	-0.023818	0.183547	-0.130	0.896754	
## Leg.room.service.C	-0.169364	0.156719	-1.081	0.279837	
## Leg.room.service^4	-0.222549	0.147534	-1.508	0.131438	
## Baggage.handling.L	0.539646	0.247284	2.182	0.029088	*
## Baggage.handling.Q	0.703393	0.205238	3.427	0.000610	***
## Baggage.handling.C	0.067385	0.188418	0.358	0.720617	
## Baggage.handling^4	-0.193967	0.171368	-1.132	0.257687	
## Checkin.service.L	0.864183	0.173646	4.977	6.47e-07	***
## Checkin.service.Q	0.229534	0.151618	1.514	0.130052	
## Checkin.service.C	0.169460	0.155911	1.087	0.277081	
## Checkin.service^4	0.270156	0.140300	1.926	0.054159	.
## Inflight.service.L	0.143369	0.265534	0.540	0.589246	
## Inflight.service.Q	0.392859	0.220274	1.783	0.074505	.
## Inflight.service.C	0.081752	0.200207	0.408	0.683027	
## Inflight.service^4	-0.366047	0.177809	-2.059	0.039527	*
## Cleanliness.L	0.734143	0.199570	3.679	0.000235	***
## Cleanliness.Q	-0.022969	0.177323	-0.130	0.896935	
## Cleanliness.C	-0.238318	0.171850	-1.387	0.165510	
## Cleanliness^4	0.212862	0.159129	1.338	0.181003	
## AgeGroup18-35	6.534680	1.426338	4.581	4.62e-06	***
## AgeGroup36-60	7.251485	1.501328	4.830	1.36e-06	***
## AgeGroupOver 60	6.164222	1.534691	4.017	5.90e-05	***
## f.FlightDistMedium Distance	0.264227	0.158644	1.666	0.095806	.
## f.FlightDistLong Distance	0.443073	0.175282	2.528	0.011479	*

```
## DelayedArrivalOK          0.513070  0.172537  2.974 0.002942 **
## DelayedDepartureOK        -0.034002  0.167304 -0.203 0.838954
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 1702.5 on 4622 degrees of freedom
## AIC: 1812.5
##
## Number of Fisher Scoring iterations: 9
```

```
Anova(m16, test= "LR") # remove agegroup, f.FlightDist + DelayedDeparture
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
## LR Chisq Df Pr(>Chisq)
## poly(Age, 2)          7.08  2  0.028993 *
## Class                 42.95  2  4.708e-10 ***
## Customer.Type         274.33  1  < 2.2e-16 ***
## Type.of.Travel        336.27  1  < 2.2e-16 ***
## Arrival.Delay.in.Minutes  4.27  1  0.038681 *
## Inflight.wifi.service  361.08  4  < 2.2e-16 ***
## Gate.location         26.33  4  2.713e-05 ***
## Online.boarding       193.84  4  < 2.2e-16 ***
## Seat.comfort          34.55  4  5.754e-07 ***
## On.board.service      26.24  4  2.832e-05 ***
## Leg.room.service      24.87  4  5.344e-05 ***
## Baggage.handling      26.88  4  2.106e-05 ***
## Checkin.service       31.71  4  2.195e-06 ***
## Inflight.service       9.77  4  0.044431 *
## Cleanliness           16.82  4  0.002091 **
## AgeGroup              44.72  3  1.059e-09 ***
## f.FlightDist           6.65  2  0.036017 *
## DelayedArrival         8.92  1  0.002827 **
## DelayedDeparture       0.04  1  0.838959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m15, m16, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##   Baggage.handling + Checkin.service
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##   Baggage.handling + Checkin.service + Inflight.service + Cleanliness +
```

```
##      AgeGroup + f.FlightDist + DelayedArrival + DelayedDeparture
##      Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      4637      1792.4
## 2      4622      1702.5 15    89.886 1.042e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m16)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      9.530472 2      1.757028
## Class             1.902392 2      1.174424
## Customer.Type     1.824474 1      1.350731
## Type.of.Travel    2.231754 1      1.493906
## Arrival.Delay.in.Minutes 1.606688 1      1.267552
## Inflight.wifi.service 5.027739 4      1.223691
## Gate.location     1.673464 4      1.066478
## Online.boarding    3.342360 4      1.162804
## Seat.comfort      5.360407 4      1.233530
## On.board.service  5.024379 4      1.223588
## Leg.room.service  2.587739 4      1.126199
## Baggage.handling  6.277885 4      1.258133
## Checkin.service   1.387539 4      1.041791
## Inflight.service  8.794293 4      1.312276
## Cleanliness       4.518382 4      1.207460
## AgeGroup          9.901946 3      1.465391
## f.FlightDist      1.387203 2      1.085263
## DelayedArrival    1.814816 1      1.347151
## DelayedDeparture  1.651093 1      1.284949
```

```
#delete Arrival delay in minutes
```

```
m16 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
  Checkin.service + Inflight.service + Cleanliness + DelayedArrival
, family=binomial,
  data=train)
summary(m16)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##      Type.of.Travel + Inflight.wifi.service + Gate.location +
##      Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##      Baggage.handling + Checkin.service + Inflight.service + Cleanliness +
##      DelayedArrival, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.955272   0.222910  -8.772  < 2e-16 ***
## poly(Age, 2)1    11.043132   5.642953   1.957  0.050350 .
```

```

## poly(Age, 2)2 -44.452681 6.020719 -7.383 1.54e-13 ***
## ClassEco -1.144362 0.168808 -6.779 1.21e-11 ***
## ClassEco Plus -0.965803 0.250313 -3.858 0.000114 ***
## Customer.TypeLoyal Customer 3.186575 0.209683 15.197 < 2e-16 ***
## Type.of.TravelPersonal Travel -3.829131 0.229441 -16.689 < 2e-16 ***
## Inflight.wifi.service.L 4.114520 0.320918 12.821 < 2e-16 ***
## Inflight.wifi.service.Q 2.542214 0.272492 9.330 < 2e-16 ***
## Inflight.wifi.service.C 0.882966 0.201953 4.372 1.23e-05 ***
## Inflight.wifi.service^4 -0.049480 0.161363 -0.307 0.759118
## Gate.location.L -0.802372 0.176391 -4.549 5.39e-06 ***
## Gate.location.Q -0.500254 0.166362 -3.007 0.002638 **
## Gate.location.C -0.220622 0.152027 -1.451 0.146725
## Gate.location^4 0.020117 0.134930 0.149 0.881482
## Online.boarding.L 2.477387 0.226802 10.923 < 2e-16 ***
## Online.boarding.Q 1.276907 0.198703 6.426 1.31e-10 ***
## Online.boarding.C -0.176988 0.175899 -1.006 0.314322
## Online.boarding^4 0.005703 0.161531 0.035 0.971833
## Seat.comfort.L -0.311736 0.205281 -1.519 0.128869
## Seat.comfort.Q 0.820492 0.185310 4.428 9.53e-06 ***
## Seat.comfort.C 0.293273 0.174518 1.680 0.092864 .
## Seat.comfort^4 -0.393960 0.165711 -2.377 0.017436 *
## On.board.service.L 0.957530 0.213230 4.491 7.10e-06 ***
## On.board.service.Q 0.136915 0.178729 0.766 0.443650
## On.board.service.C -0.138913 0.182543 -0.761 0.446663
## On.board.service^4 0.534734 0.161453 3.312 0.000926 ***
## Leg.room.service.L 0.862437 0.202198 4.265 2.00e-05 ***
## Leg.room.service.Q -0.039035 0.181104 -0.216 0.829346
## Leg.room.service.C -0.142231 0.154523 -0.920 0.357335
## Leg.room.service^4 -0.249075 0.144604 -1.722 0.084986 .
## Baggage.handling.L 0.629125 0.247656 2.540 0.011075 *
## Baggage.handling.Q 0.670527 0.203788 3.290 0.001001 **
## Baggage.handling.C 0.090243 0.187211 0.482 0.629780
## Baggage.handling^4 -0.240396 0.168862 -1.424 0.154555
## Checkin.service.L 0.830436 0.171655 4.838 1.31e-06 ***
## Checkin.service.Q 0.186089 0.149482 1.245 0.213171
## Checkin.service.C 0.163936 0.153955 1.065 0.286951
## Checkin.service^4 0.259850 0.138264 1.879 0.060193 .
## Inflight.service.L 0.036571 0.260878 0.140 0.888513
## Inflight.service.Q 0.439550 0.217411 2.022 0.043202 *
## Inflight.service.C 0.077046 0.196804 0.391 0.695441
## Inflight.service^4 -0.354852 0.174675 -2.032 0.042204 *
## Cleanliness.L 0.744730 0.197192 3.777 0.000159 ***
## Cleanliness.Q -0.044403 0.175399 -0.253 0.800151
## Cleanliness.C -0.214949 0.170114 -1.264 0.206389
## Cleanliness^4 0.181494 0.157254 1.154 0.248442
## DelayedArrivalOK 0.681169 0.129132 5.275 1.33e-07 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 6355.4 on 4676 degrees of freedom
## Residual deviance: 1761.0 on 4629 degrees of freedom
## AIC: 1857

```

```
##
## Number of Fisher Scoring iterations: 7
```

```
Anova(m16, test= "LR")
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##              LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      66.36  2  3.901e-15 ***
## Class             50.55  2  1.057e-11 ***
## Customer.Type     298.09  1  < 2.2e-16 ***
## Type.of.Travel    371.02  1  < 2.2e-16 ***
## Inflight.wifi.service 365.58  4  < 2.2e-16 ***
## Gate.location      26.41  4  2.616e-05 ***
## Online.boarding    205.96  4  < 2.2e-16 ***
## Seat.comfort       31.64  4  2.271e-06 ***
## On.board.service   29.42  4  6.409e-06 ***
## Leg.room.service   27.49  4  1.583e-05 ***
## Baggage.handling   30.07  4  4.742e-06 ***
## Checkin.service    29.30  4  6.806e-06 ***
## Inflight.service    10.03  4   0.03999 *
## Cleanliness        17.01  4   0.00192 **
## DelayedArrival      28.48  1  9.469e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m15, m16, test="Chisq")
```

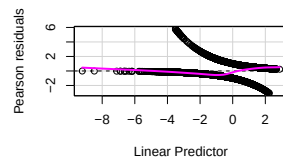
```
## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   +Arrival.Delay.in.Minutes + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + +Leg.room.service +
##   Baggage.handling + Checkin.service
## Model 2: satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + +Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4637      1792.4
## 2      4629      1761.0  8    31.436 0.0001175 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
vif(m16)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## poly(Age, 2)      1.311384  2      1.070120
## Class             1.707300  2      1.143082
## Customer.Type     1.659221  1      1.288108
## Type.of.Travel    2.221344  1      1.490417
```

```
## Inflight.wifi.service 4.628311 4 1.211094
## Gate.location 1.668195 4 1.066058
## Online.boarding 3.255883 4 1.159000
## Seat.comfort 5.411533 4 1.234995
## On.board.service 4.915225 4 1.220233
## Leg.room.service 2.530399 4 1.123049
## Baggage.handling 6.266514 4 1.257848
## Checkin.service 1.362195 4 1.039393
## Inflight.service 8.501936 4 1.306742
## Cleanliness 4.575561 4 1.209360
## DelayedArrival 1.048742 1 1.024081
```

```
#####
#Asses final model residuals#
#####
residualPlots( m16, id.n=0 )
```



```
## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded

## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

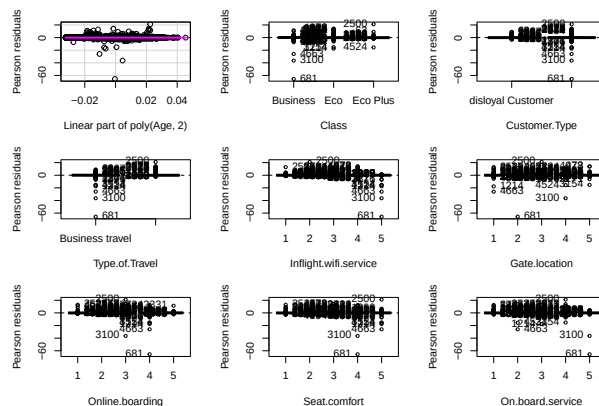
```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```



```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter

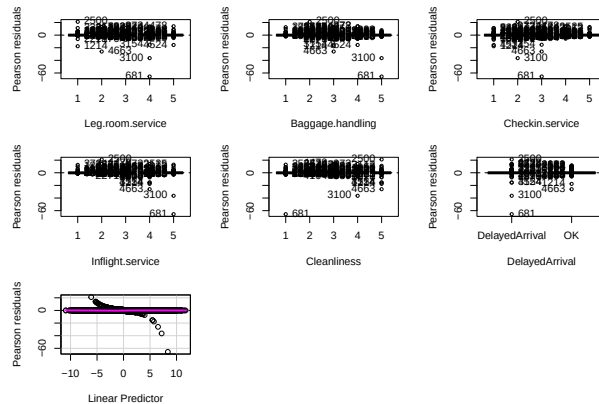
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter

## Warning in box(...): "id.n" is not a graphical parameter

## Warning in title(...): "id.n" is not a graphical parameter

## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter

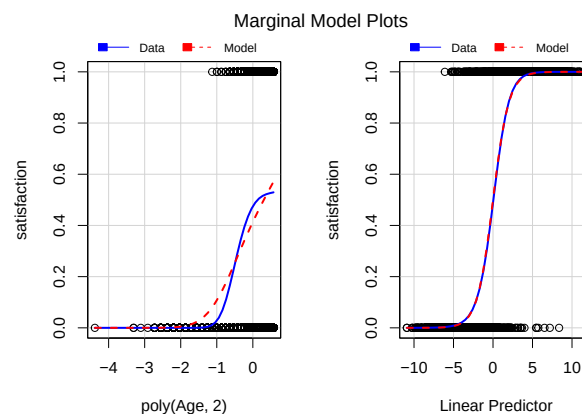
## Warning in residualPlots.default(model, ...): No possible lack-of-fit tests
```



```
marginalModelPlots(m16)
```

```
## Warning in mmpls(...): Splines and/or polynomials replaced by a fitted linear
## combination

## Warning in mmpls(...): Interactions and/or factors skipped
```



```
#####
# FINAL CHECK #
#####
m17 <- step(m16)
```

```
## Start: AIC=1856.98
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
## Inflight.wifi.service + Gate.location + Online.boarding +
## Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
## Checkin.service + Inflight.service + Cleanliness + DelayedArrival
##
##              Df Deviance    AIC
## <none>              1761.0 1857.0
## - Inflight.service    4   1771.0 1859.0
## - Cleanliness         4   1778.0 1866.0
## - Gate.location       4   1787.4 1875.4
## - Leg.room.service    4   1788.5 1876.5
## - Checkin.service     4   1790.3 1878.3
## - On.board.service    4   1790.4 1878.4
## - Baggage.handling    4   1791.0 1879.0
## - Seat.comfort        4   1792.6 1880.6
## - DelayedArrival      1   1789.5 1883.5
## - Class               2   1811.5 1903.5
## - poly(Age, 2)        2   1827.3 1919.3
## - Online.boarding     4   1967.0 2054.9
## - Customer.Type       1   2059.1 2153.1
## - Inflight.wifi.service 4   2126.6 2214.6
## - Type.of.Travel      1   2132.0 2226.0
```

```
#####
# Check for influential data and delete it. #
#####
influencePlot(m16)
```

```
##      StudRes      Hat      CookD
## 723  -4.135149 8.467405e-05 0.007593482
## 2007 -2.902368 7.846139e-03 0.008899881
## 3310 -3.839270 2.673427e-04 0.007402026
## 4076  1.044161 1.110213e-01 0.001918917
## 4675  1.002578 1.042375e-01 0.001614627
## 4830 -3.382426 1.775562e-03 0.009073907
```

```
Boxplot(cooks.distance(m5), id=list(labels=rownames(train)))
```

```
## [1] "768" "1014" "280" "2257" "2553" "3782" "3301" "1996" "605" "2790"
```

```
cd <- cooks.distance(m5)
ss <- summary(cd); ss
```

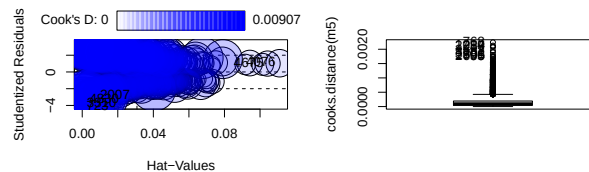
```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 1.100e-09 3.966e-05 9.424e-05 1.784e-04 1.943e-04 2.242e-03
```

```
IQR <- ss[5] - ss[2]
severelowethresh<- ss[3] - 3*IQR;
severeuperthresh<-ss[5] + 3*IQR;
llrem <- which((cd < severelowethresh) | (cd > severeuperthresh));length(llrem)
```

```
## [1] 292
```

```
train <- train[-llrem,]

m16 <- glm(satisfaction ~ poly(Age,2) + Class + Customer.Type +
  Type.of.Travel + Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
  Checkin.service + Inflight.service + Cleanliness + DelayedArrival
,family=binomial,
data=train)
```



3.3 Include factor main effects and interactions to the best numeric model & Residual Analysis

In this section, we add interactions to the model to see if we can improve it.

We first add interactions between all categorical values and the numerical variable. Then, using the step function, we go through a first cleaning of interaction in order to reduce the dimensionality and complexity of the model, obtaining model 19. By performing an anova test, we cannot reject the null hypothesis under a 99% confidence, and thus both models are equivalent, which makes us clearly keep the simplest one.

After removing those variables introducing multicollinearity, we obtain the final model. Residual and marginal model plots show a good model.

```
#####
# INCLUDE INTERACTIONS WITH ALL VARIABLES #
#####

m18 <- glm(satisfaction ~ poly(Age,2) *( Class + Customer.Type +
  Type.of.Travel + Inflight.wifi.service +
  Gate.location +Online.boarding + Seat.comfort +
  On.board.service + + Leg.room.service + Baggage.handling +
```

```

    Checkin.service + Inflight.service + Cleanliness + DelayedArrival)
, family=binomial,
data=train)

```

```
summary(m18)
```

```

##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) * (Class + Customer.Type +
##   Type.of.Travel + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + Leg.room.service +
##   Baggage.handling + Checkin.service + Inflight.service + Cleanliness +
##   DelayedArrival), family = binomial, data = train)
##
## Coefficients:
##
##               Estimate Std. Error z value
## (Intercept)      -2.826e+00  5.380e-01  -5.253
## poly(Age, 2)1       5.614e+01  3.591e+01   1.563
## poly(Age, 2)2      -5.482e+01  4.802e+01  -1.142
## ClassEco          -2.062e+00  2.790e-01  -7.391
## ClassEco Plus     -1.679e+00  4.510e-01  -3.723
## Customer.TypeLoyal Customer      4.323e+00  5.100e-01   8.477
## Type.of.TravelPersonal Travel    -4.061e+00  3.587e-01 -11.324
## Inflight.wifi.service.L          4.729e+00  5.952e-01   7.945
## Inflight.wifi.service.Q          3.677e+00  4.975e-01   7.391
## Inflight.wifi.service.C          1.133e+00  3.557e-01   3.184
## Inflight.wifi.service^4         -3.622e-01  2.806e-01  -1.291
## Gate.location.L                -8.303e-01  2.876e-01  -2.887
## Gate.location.Q                -9.268e-01  2.721e-01  -3.406
## Gate.location.C                -2.834e-01  2.347e-01  -1.207
## Gate.location^4                 2.597e-02  2.098e-01   0.124
## Online.boarding.L               3.020e+00  3.676e-01   8.214
## Online.boarding.Q               1.445e+00  3.233e-01   4.468
## Online.boarding.C               1.034e-01  3.091e-01   0.335
## Online.boarding^4               2.088e-01  2.779e-01   0.751
## Seat.comfort.L                 -4.933e-02  3.935e-01  -0.125
## Seat.comfort.Q                  8.553e-01  3.543e-01   2.414
## Seat.comfort.C                 -2.677e-02  3.338e-01  -0.080
## Seat.comfort^4                 -2.760e-01  2.913e-01  -0.948
## On.board.service.L              1.587e+00  4.137e-01   3.835
## On.board.service.Q              1.692e-01  3.244e-01   0.522
## On.board.service.C             -2.746e-01  3.141e-01  -0.874
## On.board.service^4              7.270e-01  2.806e-01   2.591
## Leg.room.service.L              6.742e-01  2.955e-01   2.281
## Leg.room.service.Q              2.910e-01  2.697e-01   1.079
## Leg.room.service.C             -4.931e-01  2.532e-01  -1.947
## Leg.room.service^4              4.501e-03  2.382e-01   0.019
## Baggage.handling.L              6.715e-01  4.488e-01   1.496
## Baggage.handling.Q              9.088e-01  3.741e-01   2.429
## Baggage.handling.C              3.341e-01  3.206e-01   1.042
## Baggage.handling^4             -3.068e-01  3.068e-01  -1.000
## Checkin.service.L              1.044e+00  2.901e-01   3.599
## Checkin.service.Q             -7.254e-02  2.489e-01  -0.291

```

## Checkin.service.C	2.360e-01	2.506e-01	0.942
## Checkin.service^4	5.013e-02	2.159e-01	0.232
## Inflight.service.L	-4.460e-01	4.804e-01	-0.928
## Inflight.service.Q	4.076e-01	3.685e-01	1.106
## Inflight.service.C	2.149e-01	3.312e-01	0.649
## Inflight.service^4	-5.034e-01	3.165e-01	-1.590
## Cleanliness.L	4.491e-01	3.884e-01	1.156
## Cleanliness.Q	-2.612e-02	3.417e-01	-0.076
## Cleanliness.C	-5.278e-01	3.319e-01	-1.590
## Cleanliness^4	4.039e-01	2.885e-01	1.400
## DelayedArrivalOK	9.951e-01	2.184e-01	4.556
## poly(Age, 2)1:ClassEco	-2.336e+01	2.041e+01	-1.144
## poly(Age, 2)2:ClassEco	-2.699e+01	2.661e+01	-1.014
## poly(Age, 2)1:ClassEco Plus	4.331e+00	3.421e+01	0.127
## poly(Age, 2)2:ClassEco Plus	-3.087e+01	4.353e+01	-0.709
## poly(Age, 2)1:Customer.TypeLoyal Customer	3.034e+01	3.439e+01	0.882
## poly(Age, 2)2:Customer.TypeLoyal Customer	1.144e-02	4.443e+01	0.000
## poly(Age, 2)1:Type.of.TravelPersonal Travel	1.146e+01	2.687e+01	0.427
## poly(Age, 2)2:Type.of.TravelPersonal Travel	-2.515e+01	3.213e+01	-0.783
## poly(Age, 2)1:Inflight.wifi.service.L	1.224e+02	4.156e+01	2.945
## poly(Age, 2)2:Inflight.wifi.service.L	4.699e+01	5.258e+01	0.894
## poly(Age, 2)1:Inflight.wifi.service.Q	3.762e+01	3.526e+01	1.067
## poly(Age, 2)2:Inflight.wifi.service.Q	1.189e+02	4.476e+01	2.655
## poly(Age, 2)1:Inflight.wifi.service.C	7.721e+01	2.573e+01	3.000
## poly(Age, 2)2:Inflight.wifi.service.C	6.084e+01	3.164e+01	1.923
## poly(Age, 2)1:Inflight.wifi.service^4	2.124e+01	2.038e+01	1.042
## poly(Age, 2)2:Inflight.wifi.service^4	-1.657e+01	2.730e+01	-0.607
## poly(Age, 2)1:Gate.location.L	1.531e+01	2.226e+01	0.688
## poly(Age, 2)2:Gate.location.L	4.595e+00	2.770e+01	0.166
## poly(Age, 2)1:Gate.location.Q	4.215e+01	2.110e+01	1.998
## poly(Age, 2)2:Gate.location.Q	-4.321e+01	2.532e+01	-1.706
## poly(Age, 2)1:Gate.location.C	-1.327e+01	1.796e+01	-0.739
## poly(Age, 2)2:Gate.location.C	-7.711e+00	2.231e+01	-0.346
## poly(Age, 2)1:Gate.location^4	7.733e+00	1.562e+01	0.495
## poly(Age, 2)2:Gate.location^4	-2.210e-01	2.031e+01	-0.011
## poly(Age, 2)1:Online.boarding.L	-1.002e+02	2.782e+01	-3.603
## poly(Age, 2)2:Online.boarding.L	5.714e+01	3.336e+01	1.712
## poly(Age, 2)1:Online.boarding.Q	1.796e+01	2.462e+01	0.730
## poly(Age, 2)2:Online.boarding.Q	-2.119e+00	3.007e+01	-0.070
## poly(Age, 2)1:Online.boarding.C	-1.600e+01	2.351e+01	-0.681
## poly(Age, 2)2:Online.boarding.C	-2.084e+01	2.857e+01	-0.729
## poly(Age, 2)1:Online.boarding^4	-3.493e+01	2.100e+01	-1.663
## poly(Age, 2)2:Online.boarding^4	3.035e+01	2.632e+01	1.153
## poly(Age, 2)1:Seat.comfort.L	-1.815e+01	2.916e+01	-0.622
## poly(Age, 2)2:Seat.comfort.L	5.436e+01	3.466e+01	1.568
## poly(Age, 2)1:Seat.comfort.Q	3.751e+01	2.664e+01	1.408
## poly(Age, 2)2:Seat.comfort.Q	-1.255e+01	3.052e+01	-0.411
## poly(Age, 2)1:Seat.comfort.C	1.702e+01	2.571e+01	0.662
## poly(Age, 2)2:Seat.comfort.C	-2.012e+01	2.892e+01	-0.696
## poly(Age, 2)1:Seat.comfort^4	-7.930e+00	2.214e+01	-0.358
## poly(Age, 2)2:Seat.comfort^4	-4.346e+00	2.549e+01	-0.170
## poly(Age, 2)1:On.board.service.L	-1.318e+01	2.856e+01	-0.462
## poly(Age, 2)2:On.board.service.L	6.631e+01	3.980e+01	1.666
## poly(Age, 2)1:On.board.service.Q	-7.934e+00	2.306e+01	-0.344

## poly(Age, 2)2:On.board.service.Q	-1.611e+01	3.123e+01	-0.516
## poly(Age, 2)1:On.board.service.C	5.122e+00	2.251e+01	0.228
## poly(Age, 2)2:On.board.service.C	-1.414e+01	3.013e+01	-0.469
## poly(Age, 2)1:On.board.service^4	-1.746e+01	2.007e+01	-0.870
## poly(Age, 2)2:On.board.service^4	2.198e+01	2.784e+01	0.789
## poly(Age, 2)1:Leg.room.service.L	2.683e+01	2.199e+01	1.220
## poly(Age, 2)2:Leg.room.service.L	-2.889e+01	2.846e+01	-1.015
## poly(Age, 2)1:Leg.room.service.Q	1.587e+01	1.963e+01	0.809
## poly(Age, 2)2:Leg.room.service.Q	3.478e+01	2.609e+01	1.333
## poly(Age, 2)1:Leg.room.service.C	-2.413e+01	1.880e+01	-1.284
## poly(Age, 2)2:Leg.room.service.C	-4.240e+01	2.384e+01	-1.778
## poly(Age, 2)1:Leg.room.service^4	-1.212e+01	1.739e+01	-0.697
## poly(Age, 2)2:Leg.room.service^4	7.302e+00	2.364e+01	0.309
## poly(Age, 2)1:Baggage.handling.L	-1.220e+01	3.342e+01	-0.365
## poly(Age, 2)2:Baggage.handling.L	3.970e+01	4.147e+01	0.957
## poly(Age, 2)1:Baggage.handling.Q	-9.540e+00	2.727e+01	-0.350
## poly(Age, 2)2:Baggage.handling.Q	7.729e+00	3.517e+01	0.220
## poly(Age, 2)1:Baggage.handling.C	1.172e+01	2.316e+01	0.506
## poly(Age, 2)2:Baggage.handling.C	5.697e+01	3.168e+01	1.798
## poly(Age, 2)1:Baggage.handling^4	1.103e+01	2.187e+01	0.504
## poly(Age, 2)2:Baggage.handling^4	-1.298e+01	2.947e+01	-0.440
## poly(Age, 2)1:Checkin.service.L	8.051e+00	2.175e+01	0.370
## poly(Age, 2)2:Checkin.service.L	1.441e+01	2.792e+01	0.516
## poly(Age, 2)1:Checkin.service.Q	1.827e+00	1.846e+01	0.099
## poly(Age, 2)2:Checkin.service.Q	-2.736e+01	2.390e+01	-1.145
## poly(Age, 2)1:Checkin.service.C	1.096e+01	1.896e+01	0.578
## poly(Age, 2)2:Checkin.service.C	-9.824e-01	2.390e+01	-0.041
## poly(Age, 2)1:Checkin.service^4	-8.048e+00	1.652e+01	-0.487
## poly(Age, 2)2:Checkin.service^4	-3.413e+01	2.070e+01	-1.649
## poly(Age, 2)1:Inflight.service.L	3.294e+01	3.479e+01	0.947
## poly(Age, 2)2:Inflight.service.L	-5.303e+01	4.335e+01	-1.223
## poly(Age, 2)1:Inflight.service.Q	-6.002e+01	2.678e+01	-2.241
## poly(Age, 2)2:Inflight.service.Q	2.920e+00	3.437e+01	0.085
## poly(Age, 2)1:Inflight.service.C	6.409e+00	2.385e+01	0.269
## poly(Age, 2)2:Inflight.service.C	-3.331e+00	3.085e+01	-0.108
## poly(Age, 2)1:Inflight.service^4	3.210e+01	2.222e+01	1.445
## poly(Age, 2)2:Inflight.service^4	-3.439e+00	3.051e+01	-0.113
## poly(Age, 2)1:Cleanliness.L	3.159e+01	2.934e+01	1.077
## poly(Age, 2)2:Cleanliness.L	-4.871e+01	3.281e+01	-1.485
## poly(Age, 2)1:Cleanliness.Q	5.797e+01	2.606e+01	2.225
## poly(Age, 2)2:Cleanliness.Q	2.137e+01	2.818e+01	0.758
## poly(Age, 2)1:Cleanliness.C	-2.314e+00	2.592e+01	-0.089
## poly(Age, 2)2:Cleanliness.C	-3.350e+01	2.780e+01	-1.205
## poly(Age, 2)1:Cleanliness^4	-9.767e+00	2.214e+01	-0.441
## poly(Age, 2)2:Cleanliness^4	1.950e+01	2.504e+01	0.779
## poly(Age, 2)1:DelayedArrivalOK	-1.933e+01	1.621e+01	-1.193
## poly(Age, 2)2:DelayedArrivalOK	6.973e-01	2.106e+01	0.033
## Pr(> z)			
## (Intercept)	1.50e-07	***	
## poly(Age, 2)1	0.118010		
## poly(Age, 2)2	0.253651		
## ClassEco	1.45e-13	***	
## ClassEco Plus	0.000197	***	
## Customer.TypeLoyal Customer	< 2e-16	***	

```

## Type.of.TravelPersonal Travel < 2e-16 ***
## Inflight.wifi.service.L 1.94e-15 ***
## Inflight.wifi.service.Q 1.46e-13 ***
## Inflight.wifi.service.C 0.001450 **
## Inflight.wifi.service^4 0.196720
## Gate.location.L 0.003888 **
## Gate.location.Q 0.000659 ***
## Gate.location.C 0.227296
## Gate.location^4 0.901475
## Online.boarding.L < 2e-16 ***
## Online.boarding.Q 7.88e-06 ***
## Online.boarding.C 0.737998
## Online.boarding^4 0.452426
## Seat.comfort.L 0.900221
## Seat.comfort.Q 0.015769 *
## Seat.comfort.C 0.936077
## Seat.comfort^4 0.343382
## On.board.service.L 0.000125 ***
## On.board.service.Q 0.601896
## On.board.service.C 0.381995
## On.board.service^4 0.009583 **
## Leg.room.service.L 0.022547 *
## Leg.room.service.Q 0.280609
## Leg.room.service.C 0.051524 .
## Leg.room.service^4 0.984926
## Baggage.handling.L 0.134612
## Baggage.handling.Q 0.015143 *
## Baggage.handling.C 0.297303
## Baggage.handling^4 0.317177
## Checkin.service.L 0.000320 ***
## Checkin.service.Q 0.770754
## Checkin.service.C 0.346318
## Checkin.service^4 0.816362
## Inflight.service.L 0.353245
## Inflight.service.Q 0.268716
## Inflight.service.C 0.516476
## Inflight.service^4 0.111765
## Cleanliness.L 0.247588
## Cleanliness.Q 0.939083
## Cleanliness.C 0.111748
## Cleanliness^4 0.161530
## DelayedArrivalOK 5.21e-06 ***
## poly(Age, 2)1:ClassEco 0.252540
## poly(Age, 2)2:ClassEco 0.310531
## poly(Age, 2)1:ClassEco Plus 0.899271
## poly(Age, 2)2:ClassEco Plus 0.478205
## poly(Age, 2)1:Customer.TypeLoyal Customer 0.377575
## poly(Age, 2)2:Customer.TypeLoyal Customer 0.999795
## poly(Age, 2)1:Type.of.TravelPersonal Travel 0.669708
## poly(Age, 2)2:Type.of.TravelPersonal Travel 0.433787
## poly(Age, 2)1:Inflight.wifi.service.L 0.003226 **
## poly(Age, 2)2:Inflight.wifi.service.L 0.371583
## poly(Age, 2)1:Inflight.wifi.service.Q 0.286095
## poly(Age, 2)2:Inflight.wifi.service.Q 0.007921 **

```

## poly(Age, 2)1:Inflight.wifi.service.C	0.002696 **
## poly(Age, 2)2:Inflight.wifi.service.C	0.054491 .
## poly(Age, 2)1:Inflight.wifi.service^4	0.297256
## poly(Age, 2)2:Inflight.wifi.service^4	0.543873
## poly(Age, 2)1:Gate.location.L	0.491546
## poly(Age, 2)2:Gate.location.L	0.868240
## poly(Age, 2)1:Gate.location.Q	0.045738 *
## poly(Age, 2)2:Gate.location.Q	0.087925 .
## poly(Age, 2)1:Gate.location.C	0.460051
## poly(Age, 2)2:Gate.location.C	0.729633
## poly(Age, 2)1:Gate.location^4	0.620493
## poly(Age, 2)2:Gate.location^4	0.991318
## poly(Age, 2)1:Online.boarding.L	0.000314 ***
## poly(Age, 2)2:Online.boarding.L	0.086808 .
## poly(Age, 2)1:Online.boarding.Q	0.465657
## poly(Age, 2)2:Online.boarding.Q	0.943818
## poly(Age, 2)1:Online.boarding.C	0.496056
## poly(Age, 2)2:Online.boarding.C	0.465812
## poly(Age, 2)1:Online.boarding^4	0.096280 .
## poly(Age, 2)2:Online.boarding^4	0.248856
## poly(Age, 2)1:Seat.comfort.L	0.533708
## poly(Age, 2)2:Seat.comfort.L	0.116810
## poly(Age, 2)1:Seat.comfort.Q	0.159203
## poly(Age, 2)2:Seat.comfort.Q	0.680812
## poly(Age, 2)1:Seat.comfort.C	0.508060
## poly(Age, 2)2:Seat.comfort.C	0.486562
## poly(Age, 2)1:Seat.comfort^4	0.720234
## poly(Age, 2)2:Seat.comfort^4	0.864632
## poly(Age, 2)1:On.board.service.L	0.644425
## poly(Age, 2)2:On.board.service.L	0.095719 .
## poly(Age, 2)1:On.board.service.Q	0.730856
## poly(Age, 2)2:On.board.service.Q	0.605954
## poly(Age, 2)1:On.board.service.C	0.819990
## poly(Age, 2)2:On.board.service.C	0.638841
## poly(Age, 2)1:On.board.service^4	0.384082
## poly(Age, 2)2:On.board.service^4	0.429849
## poly(Age, 2)1:Leg.room.service.L	0.222383
## poly(Age, 2)2:Leg.room.service.L	0.310091
## poly(Age, 2)1:Leg.room.service.Q	0.418700
## poly(Age, 2)2:Leg.room.service.Q	0.182543
## poly(Age, 2)1:Leg.room.service.C	0.199292
## poly(Age, 2)2:Leg.room.service.C	0.075353 .
## poly(Age, 2)1:Leg.room.service^4	0.485950
## poly(Age, 2)2:Leg.room.service^4	0.757433
## poly(Age, 2)1:Baggage.handling.L	0.715063
## poly(Age, 2)2:Baggage.handling.L	0.338487
## poly(Age, 2)1:Baggage.handling.Q	0.726433
## poly(Age, 2)2:Baggage.handling.Q	0.826071
## poly(Age, 2)1:Baggage.handling.C	0.612946
## poly(Age, 2)2:Baggage.handling.C	0.072182 .
## poly(Age, 2)1:Baggage.handling^4	0.613945
## poly(Age, 2)2:Baggage.handling^4	0.659644
## poly(Age, 2)1:Checkin.service.L	0.711222
## poly(Age, 2)2:Checkin.service.L	0.605790

```

## poly(Age, 2)1:Checkin.service.Q          0.921151
## poly(Age, 2)2:Checkin.service.Q          0.252320
## poly(Age, 2)1:Checkin.service.C          0.563139
## poly(Age, 2)2:Checkin.service.C          0.967208
## poly(Age, 2)1:Checkin.service^4          0.626089
## poly(Age, 2)2:Checkin.service^4          0.099187 .
## poly(Age, 2)1:Inflight.service.L          0.343840
## poly(Age, 2)2:Inflight.service.L          0.221251
## poly(Age, 2)1:Inflight.service.Q          0.025008 *
## poly(Age, 2)2:Inflight.service.Q          0.932297
## poly(Age, 2)1:Inflight.service.C          0.788120
## poly(Age, 2)2:Inflight.service.C          0.914007
## poly(Age, 2)1:Inflight.service^4          0.148495
## poly(Age, 2)2:Inflight.service^4          0.910256
## poly(Age, 2)1:Cleanliness.L          0.281592
## poly(Age, 2)2:Cleanliness.L          0.137584
## poly(Age, 2)1:Cleanliness.Q          0.026104 *
## poly(Age, 2)2:Cleanliness.Q          0.448245
## poly(Age, 2)1:Cleanliness.C          0.928859
## poly(Age, 2)2:Cleanliness.C          0.228283
## poly(Age, 2)1:Cleanliness^4          0.659041
## poly(Age, 2)2:Cleanliness^4          0.436060
## poly(Age, 2)1:DelayedArrivalOK          0.233026
## poly(Age, 2)2:DelayedArrivalOK          0.973583
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5907.1  on 4384  degrees of freedom
## Residual deviance: 1284.9  on 4247  degrees of freedom
## AIC: 1560.9
##
## Number of Fisher Scoring iterations: 9

```

```
Anova(m18, test= "LR")
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```

## Analysis of Deviance Table (Type II tests)
##
## Response: satisfaction
##
##      LR Chisq Df Pr(>Chisq)
## poly(Age, 2)      120.845  2 < 2.2e-16 ***
## Class              89.726  2 < 2.2e-16 ***
## Customer.Type      308.180  1 < 2.2e-16 ***
## Type.of.Travel     260.295  1 < 2.2e-16 ***
## Inflight.wifi.service 290.100  4 < 2.2e-16 ***
## Gate.location       22.638  4 0.0001496 ***
## Online.boarding     142.186  4 < 2.2e-16 ***
## Seat.comfort        27.190  4 1.820e-05 ***
## On.board.service    26.339  4 2.704e-05 ***
## Leg.room.service    18.659  4 0.0009169 ***

```

```
## Baggage.handling          20.768  4  0.0003521 ***
## Checkin.service           23.587  4  9.665e-05 ***
## Inflight.service          9.762  4  0.0446379 *
## Cleanliness               15.857  4  0.0032171 **
## DelayedArrival            36.689  1  1.386e-09 ***
## poly(Age, 2):Class         3.477  4  0.4813389
## poly(Age, 2):Customer.Type 0.854  2  0.6523670
## poly(Age, 2):Type.of.Travel 0.654  2  0.7211443
## poly(Age, 2):Inflight.wifi.service 25.746  8  0.0011606 **
## poly(Age, 2):Gate.location 7.543  8  0.4793512
## poly(Age, 2):Online.boarding 17.848  8  0.0223990 *
## poly(Age, 2):Seat.comfort 7.147  8  0.5208273
## poly(Age, 2):On.board.service 4.536  8  0.8058105
## poly(Age, 2):Leg.room.service 11.122  8  0.1948875
## poly(Age, 2):Baggage.handling 7.647  8  0.4687167
## poly(Age, 2):Checkin.service 6.800  8  0.5583915
## poly(Age, 2):Inflight.service 10.385  8  0.2390430
## poly(Age, 2):Cleanliness 13.487  8  0.0961645 .
## poly(Age, 2):DelayedArrival 1.611  2  0.4467557
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#####
# Remove not significant interactions using step #
#####
```

```
m19 <- step(m18)
```

```
## Start: AIC=1560.86
## satisfaction ~ poly(Age, 2) * (Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival)
##
##               Df Deviance    AIC
## - poly(Age, 2):On.board.service      8  1289.4 1549.4
## - poly(Age, 2):Checkin.service       8  1291.7 1551.7
## - poly(Age, 2):Seat.comfort           8  1292.0 1552.0
## - poly(Age, 2):Gate.location          8  1292.4 1552.4
## - poly(Age, 2):Baggage.handling       8  1292.5 1552.5
## - poly(Age, 2):Inflight.service       8  1295.2 1555.2
## - poly(Age, 2):Leg.room.service       8  1296.0 1556.0
## - poly(Age, 2):Class                  4  1288.3 1556.3
## - poly(Age, 2):Type.of.Travel         2  1285.5 1557.5
## - poly(Age, 2):Customer.Type          2  1285.7 1557.7
## - poly(Age, 2):Cleanliness            8  1298.3 1558.3
## - poly(Age, 2):DelayedArrival         2  1286.5 1558.5
## <none>                                1284.9 1560.9
## - poly(Age, 2):Online.boarding        8  1302.7 1562.7
## - poly(Age, 2):Inflight.wifi.service  8  1310.6 1570.6
##
## Step: AIC=1549.4
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
```

```

##      Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##      Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##      poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##      2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##      poly(Age, 2):Gate.location + poly(Age, 2):Online.boarding +
##      poly(Age, 2):Seat.comfort + poly(Age, 2):Leg.room.service +
##      poly(Age, 2):Baggage.handling + poly(Age, 2):Checkin.service +
##      poly(Age, 2):Inflight.service + poly(Age, 2):Cleanliness +
##      poly(Age, 2):DelayedArrival
##
##
##                                     Df Deviance    AIC
## - poly(Age, 2):Checkin.service      8   1295.7 1539.7
## - poly(Age, 2):Gate.location         8   1296.2 1540.2
## - poly(Age, 2):Seat.comfort          8   1296.8 1540.8
## - poly(Age, 2):Baggage.handling      8   1297.2 1541.2
## - poly(Age, 2):Inflight.service      8   1300.3 1544.3
## - poly(Age, 2):Leg.room.service      8   1300.3 1544.3
## - poly(Age, 2):Type.of.Travel        2   1289.5 1545.5
## - poly(Age, 2):Class                 4   1293.8 1545.8
## - poly(Age, 2):Cleanliness            8   1302.6 1546.6
## - poly(Age, 2):DelayedArrival         2   1290.9 1546.9
## - poly(Age, 2):Customer.Type          2   1291.4 1547.4
## <none>                                1289.4 1549.4
## - poly(Age, 2):Online.boarding        8   1305.8 1549.8
## - poly(Age, 2):Inflight.wifi.service  8   1315.7 1559.7
## - On.board.service                   4   1315.7 1567.7
##
## Step:  AIC=1539.72
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##      Inflight.wifi.service + Gate.location + Online.boarding +
##      Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##      Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##      poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##      2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##      poly(Age, 2):Gate.location + poly(Age, 2):Online.boarding +
##      poly(Age, 2):Seat.comfort + poly(Age, 2):Leg.room.service +
##      poly(Age, 2):Baggage.handling + poly(Age, 2):Inflight.service +
##      poly(Age, 2):Cleanliness + poly(Age, 2):DelayedArrival
##
##
##                                     Df Deviance    AIC
## - poly(Age, 2):Gate.location         8   1302.2 1530.2
## - poly(Age, 2):Seat.comfort          8   1303.0 1531.0
## - poly(Age, 2):Baggage.handling      8   1304.8 1532.8
## - poly(Age, 2):Leg.room.service      8   1306.3 1534.3
## - poly(Age, 2):Inflight.service      8   1306.6 1534.6
## - poly(Age, 2):Type.of.Travel        2   1295.8 1535.8
## - poly(Age, 2):Class                 4   1300.0 1536.0
## - poly(Age, 2):Cleanliness            8   1308.1 1536.1
## - poly(Age, 2):Customer.Type          2   1297.5 1537.5
## - poly(Age, 2):DelayedArrival         2   1297.5 1537.5
## <none>                                1295.7 1539.7
## - poly(Age, 2):Online.boarding        8   1312.0 1540.0
## - poly(Age, 2):Inflight.wifi.service  8   1320.8 1548.8
## - Checkin.service                    4   1319.7 1555.7

```

```

## - On.board.service          4    1323.1 1559.1
##
## Step: AIC=1530.21
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##   2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Seat.comfort +
##   poly(Age, 2):Leg.room.service + poly(Age, 2):Baggage.handling +
##   poly(Age, 2):Inflight.service + poly(Age, 2):Cleanliness +
##   poly(Age, 2):DelayedArrival
##
##
##              Df Deviance    AIC
## - poly(Age, 2):Seat.comfort      8    1309.6 1521.6
## - poly(Age, 2):Baggage.handling   8    1310.4 1522.4
## - poly(Age, 2):Inflight.service   8    1311.8 1523.8
## - poly(Age, 2):Leg.room.service   8    1313.0 1525.0
## - poly(Age, 2):Cleanliness        8    1314.1 1526.1
## - poly(Age, 2):Type.of.Travel     2    1302.2 1526.2
## - poly(Age, 2):Class              4    1307.0 1527.0
## - poly(Age, 2):DelayedArrival     2    1304.1 1528.1
## - poly(Age, 2):Customer.Type      2    1305.4 1529.4
## - poly(Age, 2):Online.boarding    8    1318.0 1530.0
## <none>                          1302.2 1530.2
## - poly(Age, 2):Inflight.wifi.service 8    1326.0 1538.0
## - Gate.location                   4    1324.5 1544.5
## - Checkin.service                 4    1326.5 1546.5
## - On.board.service                4    1328.7 1548.7
##
## Step: AIC=1521.56
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##   2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Leg.room.service +
##   poly(Age, 2):Baggage.handling + poly(Age, 2):Inflight.service +
##   poly(Age, 2):Cleanliness + poly(Age, 2):DelayedArrival
##
##
##              Df Deviance    AIC
## - poly(Age, 2):Baggage.handling   8    1318.7 1514.7
## - poly(Age, 2):Inflight.service   8    1320.2 1516.2
## - poly(Age, 2):Leg.room.service   8    1321.5 1517.5
## - poly(Age, 2):Type.of.Travel     2    1309.6 1517.6
## - poly(Age, 2):Class              4    1313.7 1517.7
## - poly(Age, 2):DelayedArrival     2    1311.1 1519.1
## - poly(Age, 2):Customer.Type      2    1312.0 1520.0
## <none>                          1309.6 1521.6
## - poly(Age, 2):Online.boarding    8    1327.4 1523.4
## - poly(Age, 2):Cleanliness        8    1332.5 1528.5
## - poly(Age, 2):Inflight.wifi.service 8    1333.4 1529.4

```

```

## - Gate.location          4    1331.8 1535.8
## - Checkin.service       4    1333.4 1537.4
## - On.board.service      4    1336.8 1540.8
## - Seat.comfort          4    1337.1 1541.1
##
## Step: AIC=1514.7
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##   2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Leg.room.service +
##   poly(Age, 2):Inflight.service + poly(Age, 2):Cleanliness +
##   poly(Age, 2):DelayedArrival
##
##
## Df Deviance    AIC
## - poly(Age, 2):Leg.room.service      8    1329.3 1509.3
## - poly(Age, 2):Class                  4    1322.1 1510.1
## - poly(Age, 2):Inflight.service       8    1330.9 1510.9
## - poly(Age, 2):Type.of.Travel         2    1319.0 1511.0
## - poly(Age, 2):DelayedArrival         2    1319.9 1511.9
## - poly(Age, 2):Customer.Type          2    1322.7 1514.7
## <none>                                1318.7 1514.7
## - poly(Age, 2):Online.boarding        8    1338.0 1518.0
## - poly(Age, 2):Cleanliness            8    1340.7 1520.7
## - poly(Age, 2):Inflight.wifi.service  8    1344.3 1524.3
## - Baggage.handling                    4    1339.3 1527.3
## - Gate.location                      4    1340.5 1528.5
## - Checkin.service                    4    1342.2 1530.2
## - Seat.comfort                      4    1345.6 1533.6
## - On.board.service                   4    1347.2 1535.2
##
## Step: AIC=1509.27
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Class + poly(Age, 2):Customer.Type + poly(Age,
##   2):Type.of.Travel + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Inflight.service +
##   poly(Age, 2):Cleanliness + poly(Age, 2):DelayedArrival
##
##
## Df Deviance    AIC
## - poly(Age, 2):Class                  4    1332.1 1504.1
## - poly(Age, 2):Type.of.Travel         2    1329.5 1505.5
## - poly(Age, 2):Inflight.service       8    1341.8 1505.8
## - poly(Age, 2):DelayedArrival         2    1330.7 1506.7
## - poly(Age, 2):Customer.Type          2    1332.8 1508.8
## <none>                                1329.3 1509.3
## - poly(Age, 2):Online.boarding        8    1348.8 1512.8
## - poly(Age, 2):Cleanliness            8    1350.8 1514.8
## - poly(Age, 2):Inflight.wifi.service  8    1353.0 1517.0
## - Leg.room.service                   4    1345.8 1517.8

```

```

## - Baggage.handling          4    1350.6 1522.6
## - Gate.location             4    1351.7 1523.7
## - Checkin.service           4    1353.7 1525.7
## - On.board.service          4    1357.0 1529.0
## - Seat.comfort              4    1357.8 1529.8
##
## Step: AIC=1504.09
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Customer.Type + poly(Age, 2):Type.of.Travel +
##   poly(Age, 2):Inflight.wifi.service + poly(Age, 2):Online.boarding +
##   poly(Age, 2):Inflight.service + poly(Age, 2):Cleanliness +
##   poly(Age, 2):DelayedArrival
##
##                                     Df Deviance    AIC
## - poly(Age, 2):Inflight.service      8    1344.2 1500.2
## - poly(Age, 2):Type.of.Travel        2    1332.7 1500.7
## - poly(Age, 2):DelayedArrival         2    1333.7 1501.7
## - poly(Age, 2):Customer.Type          2    1336.0 1504.0
## <none>                                1332.1 1504.1
## - poly(Age, 2):Online.boarding        8    1350.8 1506.8
## - poly(Age, 2):Cleanliness            8    1352.5 1508.5
## - poly(Age, 2):Inflight.wifi.service  8    1355.4 1511.4
## - Leg.room.service                   4    1348.5 1512.5
## - Baggage.handling                   4    1353.2 1517.2
## - Gate.location                      4    1355.2 1519.2
## - Checkin.service                    4    1357.1 1521.1
## - On.board.service                   4    1360.2 1524.2
## - Seat.comfort                       4    1360.7 1524.7
## - Class                              2    1420.2 1588.2
##
## Step: AIC=1500.18
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Customer.Type + poly(Age, 2):Type.of.Travel +
##   poly(Age, 2):Inflight.wifi.service + poly(Age, 2):Online.boarding +
##   poly(Age, 2):Cleanliness + poly(Age, 2):DelayedArrival
##
##                                     Df Deviance    AIC
## - poly(Age, 2):Type.of.Travel        2    1345.9 1497.9
## - poly(Age, 2):DelayedArrival         2    1346.5 1498.5
## <none>                                1344.2 1500.2
## - poly(Age, 2):Customer.Type          2    1348.9 1500.9
## - Inflight.service                   4    1353.1 1501.1
## - poly(Age, 2):Online.boarding        8    1364.5 1504.5
## - poly(Age, 2):Cleanliness            8    1366.2 1506.2
## - poly(Age, 2):Inflight.wifi.service  8    1368.8 1508.8
## - Leg.room.service                   4    1361.6 1509.6
## - Baggage.handling                   4    1363.4 1511.4
## - Gate.location                      4    1368.0 1516.0

```

```

## - Checkin.service          4    1368.8 1516.8
## - Seat.comfort             4    1375.6 1523.6
## - On.board.service         4    1378.4 1526.4
## - Class                    2    1428.1 1580.1
##
## Step: AIC=1497.91
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Customer.Type + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Cleanliness +
##   poly(Age, 2):DelayedArrival
##
##                                     Df Deviance    AIC
## - poly(Age, 2):DelayedArrival      2    1348.3 1496.3
## - poly(Age, 2):Customer.Type       2    1349.7 1497.7
## <none>                             1345.9 1497.9
## - Inflight.service                 4    1355.0 1499.0
## - poly(Age, 2):Cleanliness         8    1367.0 1503.0
## - poly(Age, 2):Online.boarding     8    1368.0 1504.0
## - poly(Age, 2):Inflight.wifi.service 8    1369.4 1505.4
## - Leg.room.service                 4    1364.3 1508.3
## - Baggage.handling                 4    1365.1 1509.1
## - Gate.location                    4    1370.1 1514.1
## - Checkin.service                  4    1371.5 1515.5
## - Seat.comfort                     4    1376.5 1520.5
## - On.board.service                  4    1380.5 1524.5
## - Class                            2    1429.7 1577.7
## - Type.of.Travel                   1    1629.1 1779.1
##
## Step: AIC=1496.3
## satisfaction ~ poly(Age, 2) + Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival +
##   poly(Age, 2):Customer.Type + poly(Age, 2):Inflight.wifi.service +
##   poly(Age, 2):Online.boarding + poly(Age, 2):Cleanliness
##
##                                     Df Deviance    AIC
## <none>                             1348.3 1496.3
## - poly(Age, 2):Customer.Type       2    1352.3 1496.3
## - Inflight.service                 4    1357.3 1497.3
## - poly(Age, 2):Cleanliness         8    1369.0 1501.0
## - poly(Age, 2):Online.boarding     8    1369.9 1501.9
## - poly(Age, 2):Inflight.wifi.service 8    1371.7 1503.7
## - Leg.room.service                 4    1367.1 1507.1
## - Baggage.handling                 4    1367.3 1507.3
## - Gate.location                    4    1372.3 1512.3
## - Checkin.service                  4    1374.1 1514.1
## - Seat.comfort                     4    1379.0 1519.0
## - On.board.service                  4    1382.8 1522.8
## - DelayedArrival                   1    1384.2 1530.2
## - Class                            2    1432.2 1576.2

```

```
## - Type.of.Travel 1 1633.9 1779.9
```

```
m19 <- glm(satisfaction ~poly(Age,2)*(Customer.Type + Online.boarding +  
      Leg.room.service + Checkin.service + Cleanliness) + Class +  
      Customer.Type + Type.of.Travel + Inflight.wifi.service +  
      Gate.location + Online.boarding + Seat.comfort + On.board.service +  
      Leg.room.service + Baggage.handling + Checkin.service +  
      Inflight.service + Cleanliness + DelayedArrival ,  
      family=binomial, data=train)  
summary(m19)
```

```
##  
## Call:  
## glm(formula = satisfaction ~ poly(Age, 2) * (Customer.Type +  
##   Online.boarding + Leg.room.service + Checkin.service + Cleanliness) +  
##   Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +  
##   Gate.location + Online.boarding + Seat.comfort + On.board.service +  
##   Leg.room.service + Baggage.handling + Checkin.service + Inflight.service +  
##   Cleanliness + DelayedArrival, family = binomial, data = train)  
##  
## Coefficients:  
##  
## Estimate Std. Error z value Pr(>|z|)  
## (Intercept) -2.72015 0.37990 -7.160 8.06e-13  
## poly(Age, 2)1 18.06442 24.79651 0.729 0.466304  
## poly(Age, 2)2 -53.37216 31.03617 -1.720 0.085491  
## Customer.TypeLoyal Customer 3.85766 0.37764 10.215 < 2e-16  
## Online.boarding.L 2.70920 0.31432 8.619 < 2e-16  
## Online.boarding.Q 1.44646 0.29258 4.944 7.66e-07  
## Online.boarding.C -0.21118 0.28642 -0.737 0.460940  
## Online.boarding^4 0.25388 0.25513 0.995 0.319690  
## Leg.room.service.L 0.78750 0.25887 3.042 0.002350  
## Leg.room.service.Q -0.02573 0.23309 -0.110 0.912110  
## Leg.room.service.C -0.11453 0.21147 -0.542 0.588096  
## Leg.room.service^4 -0.02771 0.19121 -0.145 0.884780  
## Checkin.service.L 1.03742 0.24505 4.233 2.30e-05  
## Checkin.service.Q 0.12804 0.20879 0.613 0.539728  
## Checkin.service.C 0.17429 0.21251 0.820 0.412118  
## Checkin.service^4 0.16089 0.18686 0.861 0.389229  
## Cleanliness.L 0.54653 0.27734 1.971 0.048769  
## Cleanliness.Q -0.26666 0.25737 -1.036 0.300157  
## Cleanliness.C -0.47450 0.24890 -1.906 0.056605  
## Cleanliness^4 0.27608 0.22693 1.217 0.223744  
## ClassEco -1.74402 0.20360 -8.566 < 2e-16  
## ClassEco Plus -1.32163 0.28048 -4.712 2.45e-06  
## Type.of.TravelPersonal Travel -3.75347 0.26256 -14.296 < 2e-16  
## Inflight.wifi.service.L 4.15968 0.37644 11.050 < 2e-16  
## Inflight.wifi.service.Q 2.75138 0.31580 8.712 < 2e-16  
## Inflight.wifi.service.C 0.78403 0.23501 3.336 0.000849  
## Inflight.wifi.service^4 -0.19857 0.18971 -1.047 0.295249  
## Gate.location.L -0.80090 0.20134 -3.978 6.95e-05  
## Gate.location.Q -0.63259 0.19347 -3.270 0.001076  
## Gate.location.C -0.22932 0.17698 -1.296 0.195068  
## Gate.location^4 -0.02328 0.15653 -0.149 0.881752  
## Seat.comfort.L -0.27210 0.25328 -1.074 0.282681
```

## Seat.comfort.Q	1.00042	0.22861	4.376	1.21e-05
## Seat.comfort.C	0.19515	0.21655	0.901	0.367510
## Seat.comfort^4	-0.29838	0.19816	-1.506	0.132130
## On.board.service.L	1.13409	0.24803	4.572	4.82e-06
## On.board.service.Q	0.15902	0.21009	0.757	0.449087
## On.board.service.C	-0.22319	0.21125	-1.057	0.290730
## On.board.service^4	0.67804	0.18720	3.622	0.000292
## Baggage.handling.L	0.63901	0.28750	2.223	0.026239
## Baggage.handling.Q	0.60200	0.23398	2.573	0.010087
## Baggage.handling.C	0.10213	0.21283	0.480	0.631309
## Baggage.handling^4	-0.27792	0.19506	-1.425	0.154220
## Inflight.service.L	-0.22768	0.30247	-0.753	0.451607
## Inflight.service.Q	0.38686	0.25771	1.501	0.133326
## Inflight.service.C	0.22024	0.22662	0.972	0.331125
## Inflight.service^4	-0.43539	0.20029	-2.174	0.029723
## DelayedArrivalOK	0.93184	0.15284	6.097	1.08e-09
## poly(Age, 2)1:Customer.TypeLoyal Customer	25.71803	24.98614	1.029	0.303343
## poly(Age, 2)2:Customer.TypeLoyal Customer	-31.48552	31.27932	-1.007	0.314131
## poly(Age, 2)1:Online.boarding.L	-69.00111	22.64906	-3.047	0.002315
## poly(Age, 2)2:Online.boarding.L	18.78431	26.27764	0.715	0.474708
## poly(Age, 2)1:Online.boarding.Q	4.75453	21.93792	0.217	0.828422
## poly(Age, 2)2:Online.boarding.Q	9.94080	25.21722	0.394	0.693429
## poly(Age, 2)1:Online.boarding.C	6.96773	21.02493	0.331	0.740340
## poly(Age, 2)2:Online.boarding.C	-39.76247	25.93934	-1.533	0.125300
## poly(Age, 2)1:Online.boarding^4	-16.26768	18.75565	-0.867	0.385751
## poly(Age, 2)2:Online.boarding^4	28.07422	23.23520	1.208	0.226946
## poly(Age, 2)1:Leg.room.service.L	36.76717	18.33087	2.006	0.044883
## poly(Age, 2)2:Leg.room.service.L	-26.53661	22.26983	-1.192	0.233420
## poly(Age, 2)1:Leg.room.service.Q	3.51338	17.31032	0.203	0.839163
## poly(Age, 2)2:Leg.room.service.Q	14.36624	21.24473	0.676	0.498897
## poly(Age, 2)1:Leg.room.service.C	-14.16780	16.09715	-0.880	0.378782
## poly(Age, 2)2:Leg.room.service.C	-9.99990	19.15164	-0.522	0.601570
## poly(Age, 2)1:Leg.room.service^4	-10.12383	14.11421	-0.717	0.473202
## poly(Age, 2)2:Leg.room.service^4	8.29999	17.20452	0.482	0.629500
## poly(Age, 2)1:Checkin.service.L	-7.10293	17.94027	-0.396	0.692163
## poly(Age, 2)2:Checkin.service.L	18.44831	22.36888	0.825	0.409524
## poly(Age, 2)1:Checkin.service.Q	-5.91836	15.75143	-0.376	0.707114
## poly(Age, 2)2:Checkin.service.Q	-10.01289	19.56553	-0.512	0.608818
## poly(Age, 2)1:Checkin.service.C	8.02938	16.36843	0.491	0.623751
## poly(Age, 2)2:Checkin.service.C	-0.83122	19.64363	-0.042	0.966248
## poly(Age, 2)1:Checkin.service^4	-15.37239	14.39829	-1.068	0.285676
## poly(Age, 2)2:Checkin.service^4	-25.22680	17.57199	-1.436	0.151109
## poly(Age, 2)1:Cleanness.L	16.28127	18.32043	0.889	0.374167
## poly(Age, 2)2:Cleanness.L	-22.09598	20.77725	-1.063	0.287569
## poly(Age, 2)1:Cleanness.Q	54.67137	17.79760	3.072	0.002127
## poly(Age, 2)2:Cleanness.Q	5.37006	19.79541	0.271	0.786177
## poly(Age, 2)1:Cleanness.C	9.37589	17.69849	0.530	0.596281
## poly(Age, 2)2:Cleanness.C	-25.50366	20.07107	-1.271	0.203847
## poly(Age, 2)1:Cleanness^4	-14.73710	16.26276	-0.906	0.364837
## poly(Age, 2)2:Cleanness^4	12.68645	19.34565	0.656	0.511967
##				
## (Intercept)	***			
## poly(Age, 2)1	.			
## poly(Age, 2)2	.			

```

## Customer.TypeLoyal Customer      ***
## Online.boarding.L                 ***
## Online.boarding.Q                 ***
## Online.boarding.C
## Online.boarding^4
## Leg.room.service.L                **
## Leg.room.service.Q
## Leg.room.service.C
## Leg.room.service^4
## Checkin.service.L                 ***
## Checkin.service.Q
## Checkin.service.C
## Checkin.service^4
## Cleanliness.L                     *
## Cleanliness.Q
## Cleanliness.C                      .
## Cleanliness^4
## ClassEco                          ***
## ClassEco Plus                     ***
## Type.of.TravelPersonal Travel     ***
## Inflight.wifi.service.L           ***
## Inflight.wifi.service.Q           ***
## Inflight.wifi.service.C           ***
## Inflight.wifi.service^4
## Gate.location.L                   ***
## Gate.location.Q                   **
## Gate.location.C
## Gate.location^4
## Seat.comfort.L                    ***
## Seat.comfort.Q                    ***
## Seat.comfort.C
## Seat.comfort^4
## On.board.service.L                ***
## On.board.service.Q
## On.board.service.C
## On.board.service^4                ***
## Baggage.handling.L                *
## Baggage.handling.Q                *
## Baggage.handling.C
## Baggage.handling^4
## Inflight.service.L
## Inflight.service.Q
## Inflight.service.C
## Inflight.service^4                *
## DelayedArrivalOK                  ***
## poly(Age, 2)1:Customer.TypeLoyal Customer
## poly(Age, 2)2:Customer.TypeLoyal Customer
## poly(Age, 2)1:Online.boarding.L   **
## poly(Age, 2)2:Online.boarding.L
## poly(Age, 2)1:Online.boarding.Q
## poly(Age, 2)2:Online.boarding.Q
## poly(Age, 2)1:Online.boarding.C
## poly(Age, 2)2:Online.boarding.C
## poly(Age, 2)1:Online.boarding^4

```

```

## poly(Age, 2)2:Online.boarding^4
## poly(Age, 2)1:Leg.room.service.L          *
## poly(Age, 2)2:Leg.room.service.L
## poly(Age, 2)1:Leg.room.service.Q
## poly(Age, 2)2:Leg.room.service.Q
## poly(Age, 2)1:Leg.room.service.C
## poly(Age, 2)2:Leg.room.service.C
## poly(Age, 2)1:Leg.room.service^4
## poly(Age, 2)2:Leg.room.service^4
## poly(Age, 2)1:Checkin.service.L
## poly(Age, 2)2:Checkin.service.L
## poly(Age, 2)1:Checkin.service.Q
## poly(Age, 2)2:Checkin.service.Q
## poly(Age, 2)1:Checkin.service.C
## poly(Age, 2)2:Checkin.service.C
## poly(Age, 2)1:Checkin.service^4
## poly(Age, 2)2:Checkin.service^4
## poly(Age, 2)1:Ccleanliness.L
## poly(Age, 2)2:Ccleanliness.L
## poly(Age, 2)1:Ccleanliness.Q              **
## poly(Age, 2)2:Ccleanliness.Q
## poly(Age, 2)1:Ccleanliness.C
## poly(Age, 2)2:Ccleanliness.C
## poly(Age, 2)1:Ccleanliness^4
## poly(Age, 2)2:Ccleanliness^4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5907.1  on 4384  degrees of freedom
## Residual deviance: 1358.3  on 4303  degrees of freedom
## AIC: 1522.3
##
## Number of Fisher Scoring iterations: 8

```

```
anova(m19, m18, test="Chisq")
```

```

## Analysis of Deviance Table
##
## Model 1: satisfaction ~ poly(Age, 2) * (Customer.Type + Online.boarding +
##   Leg.room.service + Checkin.service + Cleanliness) + Class +
##   Customer.Type + Type.of.Travel + Inflight.wifi.service +
##   Gate.location + Online.boarding + Seat.comfort + On.board.service +
##   Leg.room.service + Baggage.handling + Checkin.service + Inflight.service +
##   Cleanliness + DelayedArrival
## Model 2: satisfaction ~ poly(Age, 2) * (Class + Customer.Type + Type.of.Travel +
##   Inflight.wifi.service + Gate.location + Online.boarding +
##   Seat.comfort + On.board.service + Leg.room.service + Baggage.handling +
##   Checkin.service + Inflight.service + Cleanliness + DelayedArrival)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      4303      1358.2
## 2      4247      1284.9 56      73.39 0.05937 .
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#####
# Remove too big vif values #
#####
vif(m19)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

##	GVIF	Df	GVIF ^{1/(2*Df)}
## poly(Age, 2)	259.667778	2	4.014251
## Customer.Type	3.400300	1	1.843990
## Online.boarding	21.947290	4	1.471203
## Leg.room.service	8.745371	4	1.311361
## Checkin.service	6.085175	4	1.253240
## Cleanliness	26.146314	4	1.503752
## Class	1.945491	2	1.181020
## Type.of.Travel	2.386572	1	1.544853
## Inflight.wifi.service	6.227862	4	1.256876
## Gate.location	1.920833	4	1.085016
## Seat.comfort	8.358370	4	1.303963
## On.board.service	5.896860	4	1.248325
## Baggage.handling	7.803606	4	1.292817
## Inflight.service	10.392048	4	1.339947
## DelayedArrival	1.111801	1	1.054420
## poly(Age, 2):Customer.Type	205.599521	2	3.786653
## poly(Age, 2):Online.boarding	59.590918	8	1.291067
## poly(Age, 2):Leg.room.service	14.360641	8	1.181200
## poly(Age, 2):Checkin.service	11.971831	8	1.167845
## poly(Age, 2):Cleanliness	26.288649	8	1.226692

```
m20 <- glm(satisfaction ~poly(Age,2)*(Leg.room.service + Checkin.service + Cleanliness) +
           Class + Customer.Type + Type.of.Travel + Inflight.wifi.service +
           Gate.location + Online.boarding + Seat.comfort + On.board.service+
           Leg.room.service + Baggage.handling +
           Inflight.service + Cleanliness + DelayedArrival ,
           family=binomial, data=train)
vif(m20)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

##	GVIF	Df	GVIF ^{1/(2*Df)}
## poly(Age, 2)	3.168983	2	1.334228
## Leg.room.service	8.602438	4	1.308663
## Checkin.service	5.738399	4	1.244082
## Cleanliness	19.276353	4	1.447532
## Class	1.892793	2	1.172940
## Customer.Type	1.660864	1	1.288745
## Type.of.Travel	2.344522	1	1.531183
## Inflight.wifi.service	5.874237	4	1.247725

## Gate.location	1.858550	4	1.080555
## Online.boarding	3.856216	4	1.183778
## Seat.comfort	7.916674	4	1.295143
## On.board.service	5.706173	4	1.243206
## Baggage.handling	7.328415	4	1.282703
## Inflight.service	9.798727	4	1.330136
## DelayedArrival	1.099368	1	1.048508
## poly(Age, 2):Leg.room.service	13.091164	8	1.174387
## poly(Age, 2):Checkin.service	9.513667	8	1.151189
## poly(Age, 2):Cleanliness	11.881539	8	1.167292

```
summary(m20)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) * (Leg.room.service +
##   Checkin.service + Cleanliness) + Class + Customer.Type +
##   Type.of.Travel + Inflight.wifi.service + Gate.location +
##   Online.boarding + Seat.comfort + On.board.service + Leg.room.service +
##   Baggage.handling + Inflight.service + Cleanliness + DelayedArrival,
##   family = binomial, data = train)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.755144   0.276541  -9.963 < 2e-16 ***
## poly(Age, 2)1    30.249877   8.363211   3.617 0.000298 ***
## poly(Age, 2)2   -72.724896  10.121222  -7.185 6.70e-13 ***
## Leg.room.service.L    0.795078   0.258407   3.077 0.002092 **
## Leg.room.service.Q    0.023950   0.230150   0.104 0.917119
## Leg.room.service.C   -0.088273   0.210951  -0.418 0.675618
## Leg.room.service^4  -0.066293   0.190125  -0.349 0.727330
## Checkin.service.L    1.087078   0.239244   4.544 5.52e-06 ***
## Checkin.service.Q    0.120788   0.206753   0.584 0.559078
## Checkin.service.C    0.163329   0.210732   0.775 0.438308
## Checkin.service^4    0.177925   0.186330   0.955 0.339634
## Cleanliness.L        0.738769   0.269564   2.741 0.006133 **
## Cleanliness.Q       -0.216072   0.239144  -0.904 0.366248
## Cleanliness.C       -0.543689   0.241960  -2.247 0.024638 *
## Cleanliness^4        0.336756   0.220448   1.528 0.126613
## ClassEco            -1.686286   0.200919  -8.393 < 2e-16 ***
## ClassEco Plus      -1.283334   0.279907  -4.585 4.54e-06 ***
## Customer.TypeLoyal Customer    3.953650   0.264029  14.974 < 2e-16 ***
## Type.of.TravelPersonal Travel  -3.761449   0.260188 -14.457 < 2e-16 ***
## Inflight.wifi.service.L    4.181347   0.368595  11.344 < 2e-16 ***
## Inflight.wifi.service.Q    2.738750   0.310498   8.821 < 2e-16 ***
## Inflight.wifi.service.C    0.771917   0.230229   3.353 0.000800 ***
## Inflight.wifi.service^4   -0.206420   0.187803  -1.099 0.271711
## Gate.location.L       -0.811493   0.201291  -4.031 5.54e-05 ***
## Gate.location.Q       -0.648746   0.192519  -3.370 0.000752 ***
## Gate.location.C       -0.232085   0.175552  -1.322 0.186158
## Gate.location^4       -0.034305   0.154966  -0.221 0.824806
## Online.boarding.L      2.425163   0.263532   9.203 < 2e-16 ***
## Online.boarding.Q      1.328715   0.230272   5.770 7.92e-09 ***
## Online.boarding.C      0.033281   0.204624   0.163 0.870799
```

```

## Online.boarding^4          0.005969    0.187316    0.032 0.974577
## Seat.comfort.L            -0.343042    0.248543   -1.380 0.167521
## Seat.comfort.Q             0.995458    0.225129    4.422 9.79e-06 ***
## Seat.comfort.C             0.243475    0.214836    1.133 0.257085
## Seat.comfort^4            -0.309405    0.198520   -1.559 0.119101
## On.board.service.L         1.123775    0.245553    4.577 4.73e-06 ***
## On.board.service.Q         0.119551    0.208982    0.572 0.567281
## On.board.service.C        -0.234712    0.209671   -1.119 0.262957
## On.board.service^4         0.664802    0.184982    3.594 0.000326 ***
## Baggage.handling.L         0.636400    0.285513    2.229 0.025816 *
## Baggage.handling.Q         0.598513    0.231193    2.589 0.009631 **
## Baggage.handling.C         0.100296    0.211118    0.475 0.634737
## Baggage.handling^4        -0.273663    0.192694   -1.420 0.155550
## Inflight.service.L        -0.255340    0.299477   -0.853 0.393870
## Inflight.service.Q         0.414894    0.254105    1.633 0.102518
## Inflight.service.C         0.221987    0.225062    0.986 0.323968
## Inflight.service^4        -0.421183    0.198252   -2.124 0.033630 *
## DelayedArrivalOK          0.925412    0.151272    6.118 9.50e-10 ***
## poly(Age, 2)1:Leg.room.service.L 36.666446 18.416314    1.991 0.046483 *
## poly(Age, 2)2:Leg.room.service.L -27.531885 22.538593   -1.222 0.221880
## poly(Age, 2)1:Leg.room.service.Q  2.731337 17.112434    0.160 0.873187
## poly(Age, 2)2:Leg.room.service.Q 17.936032 21.105150    0.850 0.395413
## poly(Age, 2)1:Leg.room.service.C -15.246911 16.059009   -0.949 0.342402
## poly(Age, 2)2:Leg.room.service.C -4.666284 19.115274   -0.244 0.807143
## poly(Age, 2)1:Leg.room.service^4 -9.250724 14.067390   -0.658 0.510795
## poly(Age, 2)2:Leg.room.service^4  6.121276 17.074581    0.359 0.719967
## poly(Age, 2)1:Checkin.service.L -15.330248 17.275140   -0.887 0.374855
## poly(Age, 2)2:Checkin.service.L  26.915727 21.140256    1.273 0.202948
## poly(Age, 2)1:Checkin.service.Q  -7.990897 15.723766   -0.508 0.611310
## poly(Age, 2)2:Checkin.service.Q -10.183397 19.323285   -0.527 0.598193
## poly(Age, 2)1:Checkin.service.C   6.923472 16.401627    0.422 0.672937
## poly(Age, 2)2:Checkin.service.C  -2.808824 19.883014   -0.141 0.887659
## poly(Age, 2)1:Checkin.service^4 -18.188732 14.382785   -1.265 0.206008
## poly(Age, 2)2:Checkin.service^4 -19.920490 17.679584   -1.127 0.259848
## poly(Age, 2)1:Cleanliness.L      -7.398027 17.017736   -0.435 0.663762
## poly(Age, 2)2:Cleanliness.L     -14.163550 20.167705   -0.702 0.482499
## poly(Age, 2)1:Cleanliness.Q       53.184130 16.186469    3.286 0.001017 **
## poly(Age, 2)2:Cleanliness.Q       10.850908 18.706843    0.580 0.561881
## poly(Age, 2)1:Cleanliness.C       15.053133 16.604839    0.907 0.364644
## poly(Age, 2)2:Cleanliness.C     -31.050833 19.905527   -1.560 0.118781
## poly(Age, 2)1:Cleanliness^4     -21.620386 15.293055   -1.414 0.157439
## poly(Age, 2)2:Cleanliness^4      16.159378 18.861787    0.857 0.391596
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 5907.1  on 4384  degrees of freedom
## Residual deviance: 1374.0  on 4313  degrees of freedom
## AIC: 1518
##
## Number of Fisher Scoring iterations: 8

```

Now, we proceed to remove unusual and influent data. In order to do it, we remove the cook's distance

outliers iteratively until we iterate a certain number of times. This way, we remove those points which bias our model incorrectly.

```
#####
# Check for influential data and delete it. #
#####
# Set the maximum number of iterations
max_iterations <- 3

for (i in 1:max_iterations) {
  # Generate the influence plot
  influencePlot(m20)

  # Identify influential data points
  cd <- cooks.distance(m20)
  ss <- summary(cd); ss
  IQR <- ss[5] - ss[2]
  severelowethresh<- ss[3] - 3*IQR;
  severeuperthresh<-ss[5] + 3*IQR;
  llrem <- which((cd < severelowethresh) | (cd > severeuperthresh));length(llrem)

  # Break the loop if no influential data points are found
  if (length(llrem) == 0) {
    message("No more influential points to remove. Stopping iterations.")
    break
  }

  # Remove the identified data points
  train <- train[-llrem, ]

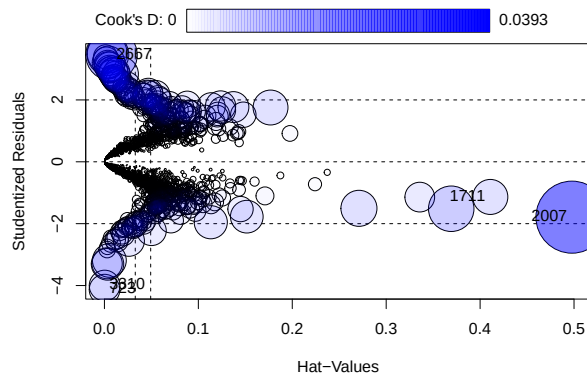
  # Refit the model with the updated data
  m20 <- glm(satisfaction ~ poly(Age, 2) + Class + Customer.Type +
             Type.of.Travel + Inflight.wifi.service + Gate.location +
             Online.boarding + Seat.comfort + On.board.service +
             Leg.room.service + Baggage.handling + Checkin.service +
             Inflight.service + Cleanliness + DelayedArrival,
             family = binomial, data = train)

  message("Iteration ", i, " completed. Model refitted.")
}
```

```
## Warning: glm.fit: algorithm did not converge
```

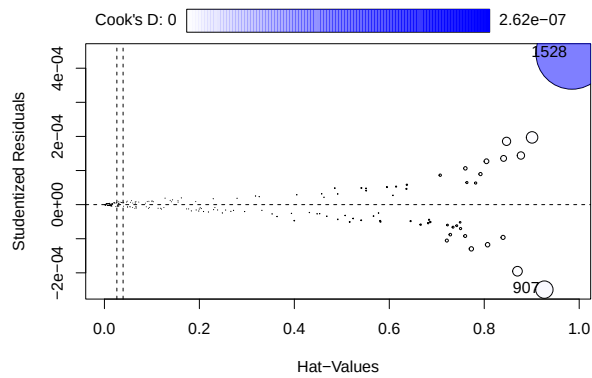
```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Iteration 1 completed. Model refitted.
```



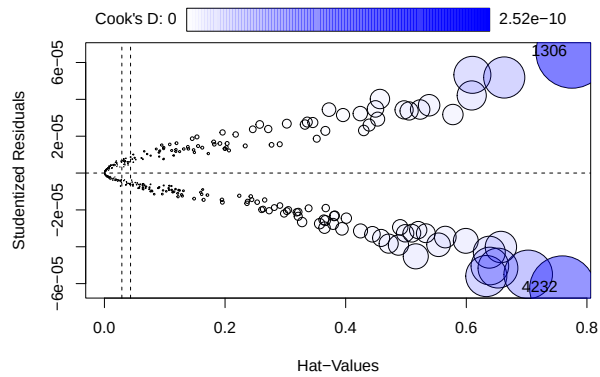
```
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

## Iteration 2 completed. Model refitted.
```



```
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

## Iteration 3 completed. Model refitted.
```



```
#####  
#Asses final model residuals#  
#####
```

```
residualPlots( m20, id.n=0 )
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a  
## graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a  
## graphical parameter
```

```
## Warning in box(...): "id.n" is not a graphical parameter
```

```
## Warning in title(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a  
## graphical parameter
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

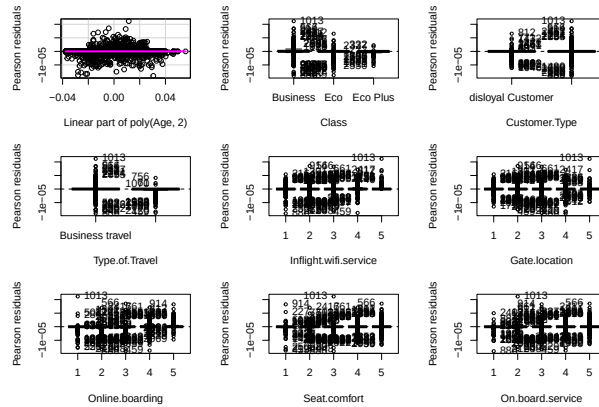
```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :  
## Duplicated argument id.n = 0 is disregarded
```



```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in (function (z, notch = FALSE, width = NULL, varwidth = FALSE, :
## Duplicated argument id.n = 0 is disregarded
```

```
## Warning in plot.window(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "id.n" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

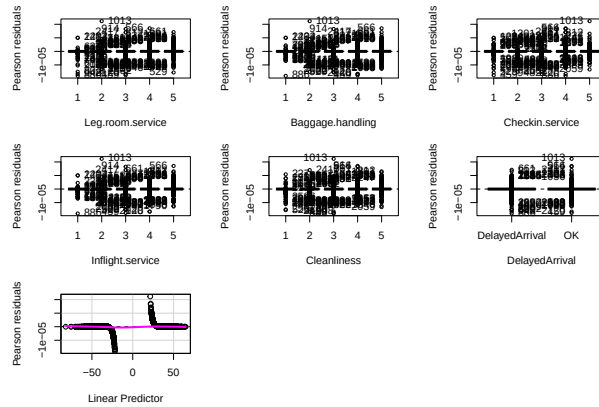
```
## Warning in axis(side = side, at = at, labels = labels, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in box(...): "id.n" is not a graphical parameter
```

```
## Warning in title(...): "id.n" is not a graphical parameter
```

```
## Warning in plot.xy(xy.coords(x, y), type = type, ...): "id.n" is not a
## graphical parameter
```

```
## Warning in residualPlots.default(model, ...): No possible lack-of-fit tests
```



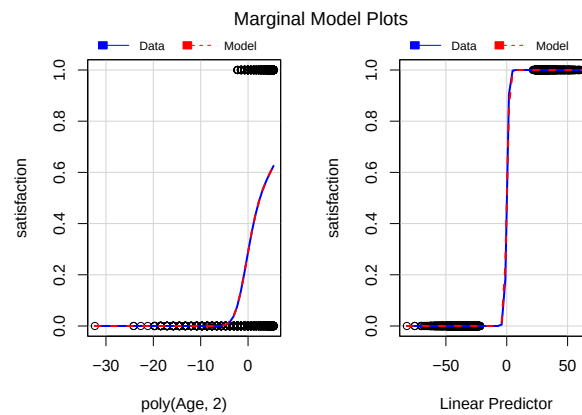
```
marginalModelPlots(m20)
```

```
## Warning in mmps(...): Splines and/or polynomials replaced by a fitted linear
## combination
```

```
## Warning: glm.fit: algorithm did not converge
```

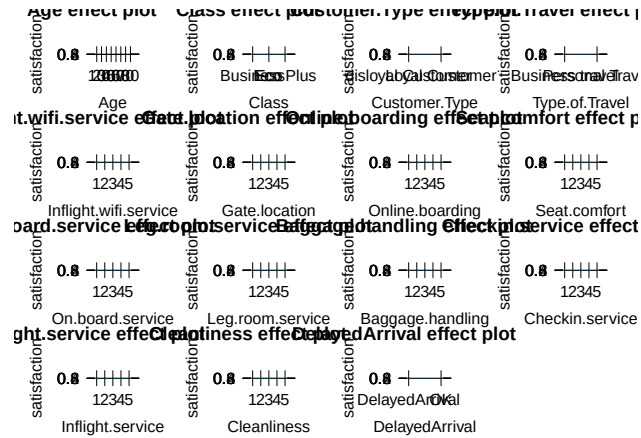
```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning in mmps(...): Interactions and/or factors skipped
```



3.4 Effects study

```
plot(allEffects(m20))
```



3.5 Final model interpretation

The final model that obtained after a long model discovery is the following, which has a pseudo-coefficient of determination (McFadden) of 0.80.

```
#Summary of the parameters
summary(m20)
```

```
##
## Call:
## glm(formula = satisfaction ~ poly(Age, 2) + Class + Customer.Type +
##      Type.of.Travel + Inflight.wifi.service + Gate.location +
##      Online.boarding + Seat.comfort + On.board.service + Leg.room.service +
##      Baggage.handling + Checkin.service + Inflight.service + Cleanliness +
##      DelayedArrival, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -2.049e+01  1.804e+04  -0.001    0.999
## poly(Age, 2)1       8.493e+01  3.606e+05   0.000    1.000
## poly(Age, 2)2      -2.962e+02  3.994e+05  -0.001    0.999
## ClassEco           -9.930e+00  1.596e+04  -0.001    1.000
## ClassEco Plus     -1.011e+01  2.902e+04   0.000    1.000
## Customer.TypeLoyal Customer  2.582e+01  1.623e+04   0.002    0.999
## Type.of.TravelPersonal Travel -2.396e+01  1.938e+04  -0.001    0.999
## Inflight.wifi.service.L      2.201e+01  1.900e+04   0.001    0.999
## Inflight.wifi.service.Q      1.578e+01  1.760e+04   0.001    0.999
## Inflight.wifi.service.C      4.414e+00  1.580e+04   0.000    1.000
## Inflight.wifi.service^4      1.564e-01  1.319e+04   0.000    1.000
## Gate.location.L             -6.375e+00  2.033e+04   0.000    1.000
## Gate.location.Q             -6.177e+00  1.640e+04   0.000    1.000
## Gate.location.C             -1.935e+00  1.412e+04   0.000    1.000
## Gate.location^4             -4.371e-01  1.174e+04   0.000    1.000
## Online.boarding.L           1.319e+01  1.714e+04   0.001    0.999
## Online.boarding.Q           6.164e+00  1.530e+04   0.000    1.000
## Online.boarding.C           5.680e-01  1.483e+04   0.000    1.000
## Online.boarding^4          -1.353e+00  1.347e+04   0.000    1.000
## Seat.comfort.L             4.963e-01  1.714e+04   0.000    1.000
```

```
## Seat.comfort.Q          6.429e+00  1.649e+04  0.000  1.000
## Seat.comfort.C          3.310e-02  1.708e+04  0.000  1.000
## Seat.comfort^4         -2.476e+00  1.686e+04  0.000  1.000
## On.board.service.L      8.553e+00  1.958e+04  0.000  1.000
## On.board.service.Q     -3.457e-01  1.592e+04  0.000  1.000
## On.board.service.C     -1.542e+00  1.535e+04  0.000  1.000
## On.board.service^4      2.000e+00  1.296e+04  0.000  1.000
## Leg.room.service.L      5.295e+00  1.876e+04  0.000  1.000
## Leg.room.service.Q     -6.137e-01  1.665e+04  0.000  1.000
## Leg.room.service.C     -6.086e-01  1.341e+04  0.000  1.000
## Leg.room.service^4     -1.312e+00  1.356e+04  0.000  1.000
## Baggage.handling.L      9.831e-01  2.946e+04  0.000  1.000
## Baggage.handling.Q      5.107e+00  2.372e+04  0.000  1.000
## Baggage.handling.C     -5.043e-01  2.035e+04  0.000  1.000
## Baggage.handling^4     -2.420e+00  1.557e+04  0.000  1.000
## Checkin.service.L       3.695e+00  1.530e+04  0.000  1.000
## Checkin.service.Q       1.988e+00  1.420e+04  0.000  1.000
## Checkin.service.C       1.212e+00  1.532e+04  0.000  1.000
## Checkin.service^4       1.181e+00  1.301e+04  0.000  1.000
## Inflight.service.L      9.942e-01  2.913e+04  0.000  1.000
## Inflight.service.Q     -5.330e-01  2.400e+04  0.000  1.000
## Inflight.service.C      1.644e+00  2.018e+04  0.000  1.000
## Inflight.service^4     -2.303e+00  1.468e+04  0.000  1.000
## Cleanliness.L           4.832e+00  1.946e+04  0.000  1.000
## Cleanliness.Q          -1.507e+00  1.578e+04  0.000  1.000
## Cleanliness.C          -2.170e+00  1.723e+04  0.000  1.000
## Cleanliness^4           1.279e+00  1.527e+04  0.000  1.000
## DelayedArrivalOK        5.397e+00  1.178e+04  0.000  1.000
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 3.5353e+03  on 2694  degrees of freedom
## Residual deviance: 3.2902e-08  on 2647  degrees of freedom
## AIC: 96
##
## Number of Fisher Scoring iterations: 25
```

```
#Pseudo coefficient of determination R^2
pseudoR2 <- pR2(m20)["McFadden"];
```

```
## fitting null model for pseudo-r2
```

```
pseudoR2
```

```
## McFadden
##      1
```

```
#Residual analysis and influent data
residualPlots(m20)
```



```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.3644 1.0000 1.0000
```

```
#Classification threshold
predicted_class <- ifelse(predicted_prob > 0.5, 1, 0)
table(predicted_class)
```

```
## predicted_class
##      0      1
## 1713  982
```

```
# Confusion Matrix
actual_class <- train$satisfaction
table(actual_class)
```

```
## actual_class
##      0      1
## 1713  982
```

```
conf_matrix <- confusionMatrix((predicted_class), (actual_class));conf_matrix
```

```
##      [,1] [,2]
## [1,] 1713  0
## [2,]  0 982
```

```
# Confusion matrix components
TN <- 2725
FP <- 114
FN <- 154
TP <- 1887
```

```
# Calculations
accuracy <- (TP + TN) / (TP + TN + FP + FN)
precision <- TP / (TP + FP)
recall <- TP / (TP + FN)
f1_score <- 2 * (precision * recall) / (precision + recall)
```

```
# Print results
cat("Accuracy:", accuracy, "\n")
```

```
## Accuracy: 0.945082
```

```
cat("Precision:", precision, "\n")
```

```
## Precision: 0.9430285
```

```
cat("Recall:", recall, "\n")
```

```
## Recall: 0.9245468
```

```
cat("F1 Score:", f1_score, "\n")
```

```
## F1 Score: 0.9336962
```

```
# AUC and ROC Curve
```

```
roc_curve <- roc(actual_class, predicted_prob)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc_value <- auc(roc_curve)
```

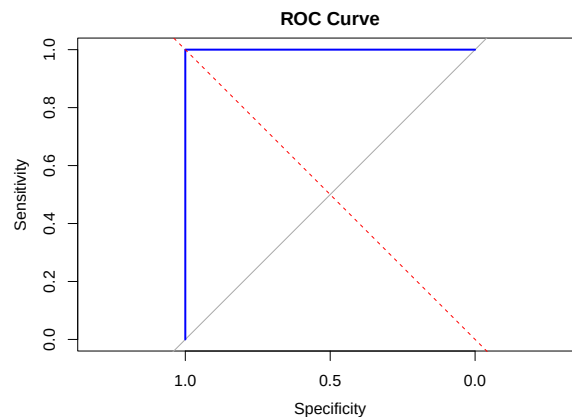
```
cat("AUC:", auc_value, "\n")
```

```
## AUC: 1
```

```
# Plot the ROC Curve
```

```
plot(roc_curve, col = "blue", main = "ROC Curve", lwd = 2)
```

```
abline(a = 0, b = 1, lty = 2, col = "red") # Add diagonal reference line
```



3.5 Model evaluation

Evaluation of the aforementioned model with unseen data will be performed using various categorical data criteria such as Precision, Recall, Accuracy, F1-score and the AUC/ROC curve.

First, from the confusion matrix we see that the model miss classifies 151 out of 2000 observations, which corresponds to a relatively low error rate (7.55%). These are 48 satisfied clients which are set to neutral or dissatisfied, and 103 which are neutral or dissatisfied and are classified as satisfied.

In the test set, the model obtains:

- Accuracy: 0.9245
- Precision: 0.94482
- Recall: 0.8886
- F1 Score: 0.9158
- AUC: 0.9644

```

#Evaluation of the model using train data (Confusion matrix and AUC)
# Get predicted probabilities and classifications
predicted_prob <- predict(m20, newdata = test, type = "response")
summary(predicted_prob)

##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000619 0.4273985 1.0000000 1.0000000

#Classification threshold
predicted_class <- ifelse(predicted_prob > 0.5, 1, 0)
table(predicted_class)

## predicted_class
##      0      1
## 1146   854

# Confusion Matrix
actual_class <- test$satisfaction
table(actual_class)

## actual_class
##      0      1
## 1075   925

conf_matrix <- confusionMatrix((predicted_class), (actual_class));conf_matrix

##      [,1] [,2]
## [1,] 1022   53
## [2,]  124  801

# Confusion matrix components
TN <- 1028
FP <- 48
FN <- 103
TP <- 822

# Calculations
accuracy <- (TP + TN) / (TP + TN + FP + FN)
precision <- TP / (TP + FP)
recall <- TP / (TP + FN)
f1_score <- 2 * (precision * recall) / (precision + recall)

# Print results
cat("Accuracy:", accuracy, "\n")

## Accuracy: 0.9245377

cat("Precision:", precision, "\n")

## Precision: 0.9448276

```

```
cat("Recall:", recall, "\n")
```

```
## Recall: 0.8886486
```

```
cat("F1 Score:", f1_score, "\n")
```

```
## F1 Score: 0.9158774
```

```
# AUC and ROC Curve
```

```
roc_curve <- roc(actual_class, predicted_prob)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
auc_value <- auc(roc_curve)
```

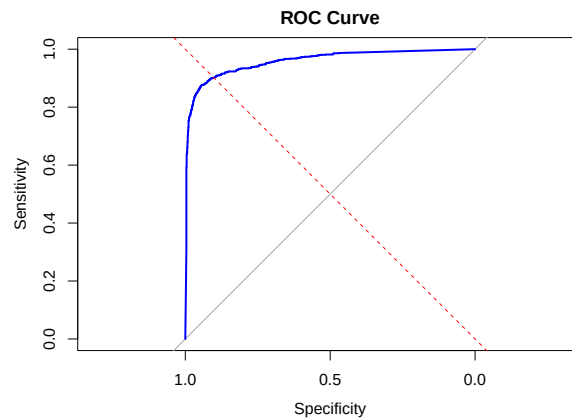
```
cat("AUC:", auc_value, "\n")
```

```
## AUC: 0.9586987
```

```
# Plot the ROC Curve
```

```
plot(roc_curve, col = "blue", main = "ROC Curve", lwd = 2)
```

```
abline(a = 0, b = 1, lty = 2, col = "red") # Add diagonal reference line
```



```
# Hosmer-Lemeshow Goodness of Fit (GOF) Test
```

```
library(ResourceSelection)
```

```
## ResourceSelection 0.3-6    2023-06-27
```

```
?hoslem.test
```

```
## starting httpd help server ...
```

```
## done
```

```
#hoslem.test(y = test$satisfaction, x = predicted_prob, g = 2)
```

4 Conclusion and Further Work

In conclusion, several insights can be extracted after the completion of this work:

Exploratory Data Analysis: Exploratory data analysis is fundamental to any data-driven project. It has allowed us to understand the structure and hidden patterns in data such as the most influential factors affecting the target variable customer satisfaction, such as customer type, travel class, delay status...

Model interpretability: Iterative residual and influence analysis has ensured a robust model almost free from influential outliers and multicollinearity issues. The addition of factors and some interactions between variables has been proved to greatly improve the model goodness of fit and predictive performance. Lastly, it is apparent that the final model is also rich in simplicity and explainability due to the lack of over complicated polynomials and expressions, only the raw variables and some interactions between age and factors are used for the model (Cleanliness, Customer Type, Online.Onboarding, Leg room, Inflight service).

Data Preparation: The dataset opposed some challenges during the data preparation/cleaning phase given the number of outliers and missing values in some variables. However, after performing the data quality report and imputation it can clearly be concluded that the latent quality of this data is astonishingly high, and that is the main rationale behind the performance of our final model.

Model Performance: Firstly, it's clear that the goodness of fit for the train data is great given the low residual deviance, AIC and high McFadden Pseudo R^2 . Not only that but also the quality of the predictive performance on unseen data is astonishingly good, after empirically measuring Recall, Precision, Accuracy, F1-score and AUC of at least 88%. This points out the fact that high quality data produces high quality models, therefore validating the garbage-in garbage-out old saying.

This project still has some potential areas of improvement such as:

1. **Advanced feature engineering:** Exploring additional derived variables such as time-related features or complex interactions driven by business knowledge.
2. **Data Integration:** Incorporating external data such as weather, ticket pricing, flight routes... for a more in-depth and holistic analysis and increased predictive power.
3. **Advanced Modelling:** Trying unsupervised clustering methods or advanced classification models like neural networks or decision trees could enhance predictive abilities, analysis depth or improve interpretability in the case of the latter.